DevOrSysAdminScripts

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

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Current number of commits: 862
Current git SHA1: 02bdefe1a70225151c24b7d3620cad21bbb2b724
Code lines: 12008 total lines, 11048 not empty lines, 960 empty lines.

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1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_dependencies_notes.exec.php
│   ├── build_dependencies_notes.libr.php
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spelling_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── dependencies_data.json
│   ├── dependencies_notes.txt
│   ├── DevOrSysAdminScripts.html.tpl
```

```
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpcs_ruleset.xml
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── repository_data.txt.tpl
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── .gitignore
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 01234\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456\
│   ├──
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
```

```
│    │ an_see_that_it_works.txt
│    ├── funny.sh
│    ├── spampty_data.txt
│    ├── string_escaping.html
│    ├── string_escaping.xhtml
│    └──
│      yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│      that_you_can_see_that_it_works.txt
├── README.html
├── README.md
├── README.md.tpl
├── string_escaping.libr.php
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 83 files

[ 12K Apr  2 09:49]  ./
├── [4.0K Apr  2 09:47]  build_and_checks_dependencies/
│    ├── [4.4K Apr  1 19:39]  build_and_checks.exec.sh.tpl*
│    ├── [1.2K Mar 26 12:30]  build_dependencies_notes.exec.php*
│    ├── [6.9K Mar 26 12:37]  build_dependencies_notes.libr.php
│    ├── [4.2K Apr  1 20:41]  build_md_from_printable_md.exec.sh*
│    ├── [3.3K Apr  1 20:43]  check_shell_scripts_beginnings.libr.sh
│    ├── [5.4K Apr  1 20:47]  check_shell_scripts_indentation.libr.sh
│    ├── [3.8K Apr  1 21:56]  check_URLs.libr.sh
│    ├── [ 32K Apr  2 09:30]  common_build_and_checks.exec.sh*
│    ├── [ 27K Apr  2 09:29]  common_build_and_checks.exec.sh.tpl
│    ├── [ 18K Apr  2 04:13]  common_options.libr.sh
│    ├── [6.2K Apr  2 04:19]  comparisons.libr.sh
│    ├── [ 22K Apr  2 04:33]  create_PDF.exec.sh*
│    ├── [3.7K Apr  2 04:36]  generate_from_template.libr.sh
│    ├── [2.0K Mar 24 20:52]  get_common_text_glob_patterns.libr.sh
│    ├── [3.3K Apr  2 04:39]  grammar_and_spelling_check.libr.sh
│    ├── [4.0K Apr  1 20:35]  licenses_templates/
│    │    ├── [ 14K Apr  1 20:35]  build_licenses_templates.exec.sh*
│    │    ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│    │    ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│    │    └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│    ├── [6.2K Apr  2 04:41]  lines_counts.libr.sh
│    ├── [ 15K Apr  2 04:51]  lines_filters.libr.sh
│    ├── [3.0K Apr  1 14:56]  lines_maps.libr.sh
│    ├── [2.8K Apr  2 05:50]  overwrite_if_not_equal.libr.sh
│    ├── [2.5K Apr  1 15:50]  python_black_complement.libr.sh
│    ├── [3.2K Apr  2 04:58]  python_isort_complement.libr.sh
```

```
│   ├── [2.1K Apr  2 05:19]  repository_data.libr.sh
│   ├── [4.2K Apr  2 09:18]  shell_checks_complement.libr.sh
│   ├── [2.0K Mar 26 12:37]  split_score.exec.php
│   ├── [3.2K Mar 26 23:11]  split_score.libr.php
│   ├── [ 48K Apr  2 05:30]  strings_functions.libr.sh
│   ├── [4.4K Apr  2 05:35]  too_long_code_lines.libr.sh
│   ├── [4.3K Apr  2 05:45]  update_common_build_and_checks.exec.sh*
│   └── [6.0K Apr  2 05:49]  update_or_check_files_names_listing.exec.sh*
├── [4.6K Apr  2 09:30]  build_and_checks.exec.sh*
├── [4.0K Apr  2 09:49]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [4.9K Mar 26 23:19]  dependencies_data.json
│   ├── [4.1K Mar 26 23:20]  dependencies_notes.txt
│   ├── [5.7K Mar 26 23:16]  DevOrSysAdminScripts.html.tpl
│   ├── [ 35K Apr  2 09:48]  DevOrSysAdminScripts.tex
│   ├── [4.2K Mar 26 23:16]  DevOrSysAdminScripts.tex.tpl
│   ├── [5.5K Mar 26 22:08]  files_names_listing.txt
│   ├── [  23 Mar 24 19:33]  .gitignore
│   ├── [1.4K Mar 26 23:16]  licenses_notes.txt
│   ├── [ 13K Mar 26 22:59]  phpcs_ruleset.xml
│   ├── [2.2K Mar 26 12:12]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.5K Mar 22 15:51]  repository_data.txt
│   ├── [ 786 Mar 22 15:39]  repository_data.txt.tpl
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 166 Mar 24 19:33]  shellcheck.ini
│   └── [3.6K Apr  2 09:16]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [479K Apr  2 09:48]  DevOrSysAdminScripts.html
├── [412K Apr  2 09:48]  DevOrSysAdminScripts.pdf
├── [4.0K Apr  1 19:35]  git_helpers/
│   ├── [3.8K Apr  1 19:16]  git_checks.libr.sh
│   ├── [2.1K Apr  1 19:21]  post-commit*
│   ├── [ 15K Apr  1 19:35]  post_commit_renaming_comment.libr.sh
│   └── [1.8K Apr  1 19:23]  pre-commit*
├── [   8 Mar 26 10:37]  .gitignore
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [1.1K Mar 31 19:59]  LiensUtiles.txt
├── [4.0K Mar 18 17:27]  miscellaneous/
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890
│   │ 0123\
│   ├── [   0 Mar  4 10:03]
```

```
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 01234\
    │── [   0 Mar  4 10:01]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 012345
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 012345\
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 0123456
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 0123456\
    │── [   0 Mar  4 09:13]
    │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
    │ an_see_that_it_works.txt
    │── [1.3K Mar  9 15:15]  funny.sh
    │── [  94 Mar  3 19:56]  spampty_data.txt
    │── [ 442 Mar 18 16:50]  string_escaping.html
    │── [ 654 Mar 18 17:27]  string_escaping.xhtml
    └── [   0 Mar  4 09:13]
      yet_another_long_long_long_long_long_long_long_long_long_filename_so_
      that_you_can_see_that_it_works.txt
├── [8.1K Mar 22 21:29]  README.html
├── [4.3K Mar 22 21:29]  README.md
├── [4.3K Mar 22 21:29]  README.md.tpl
├── [7.0K Mar 26 22:16]  string_escaping.libr.php
├── [ 16K Mar 26 20:11]  unicode.libr.php
├── [5.4K Apr  2 09:39]  upgrade_build_and_checks.exec.sh*
└── [2.5K Apr  1 19:06]  wget_sha512.libr.sh

6 directories, 83 files
```

2 Notes on licenses and dependencies

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:

basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;

- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, phpdoc, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/,
https://savannah.gnu.org/projects/findutils/.

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/,
https://savannah.gnu.org/projects/grep/.

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/,
https://savannah.gnu.org/projects/sed/.

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/,
https://github.com/koalaman/shellcheck.

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/,
https://savannah.gnu.org/projects/wget/.

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/,
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/,
https://github.com/util-linux/util-linux official?.

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,

https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.

phpdoc (phpDocumentor) can be found at the following URLs:
https://phpdoc.org/,
https://docs.phpdoc.org/,
https://github.com/phpdocumentor/phpdocumentor.

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;)).

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer,
https://packagist.org/packages/squizlabs/php_codesniffer.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

python3 can be found at the following URLs:
https://www.python.org/,
https://docs.python.org/3/.

If you see something that I missed regarding my dependencies, please tell me/email me :).

3 Listing of files

3.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
```

```
if [[ "$*" == *--verbose* ]]; then
  printf "%s %s\n" "$0" "$*"
  LFBFL_verbose="--verbose"
fi
readonly LFBFL_verbose

local LFBFL_check_download=""
if [[ "$*" == *--check-download=keep* ]]; then
  LFBFL_check_download="--check-download=keep"
elif [[ "$*" == *--check-download=y* ]]; then
  LFBFL_check_download="--check-download=y"
fi
readonly LFBFL_check_download

source ./wget_sha512.libr.sh

LFBFL_subdir="build_and_checks_dependencies"
mkdir --parents "${LFBFL_subdir}/licenses_templates"
local LFBFL_variables_directory
LFBFL_variables_directory="${LFBFL_work_directory}/"
LFBFL_variables_directory+="build_and_checks_variables"
readonly LFBFL_variables_directory
mkdir --parents -- "${LFBFL_variables_directory}/temp"
if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
  printf "temp/\nupgrade_venvs_ts\n"\
    > "${LFBFL_variables_directory}/.gitignore"
fi

local LFBFL_dependencies_URL
LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
readonly LFBFL_dependencies_URL

declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
local LFBFL_script_download_URL
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
LFBFL_script_download_URL+="${LFBFL_common_file_name}"
readonly LFBFL_script_download_URL
declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
@sha512_common_build_and_checks.exec.sh@
wget_sha512 "${LFBFL_file_path}"\
  "${LFBFL_script_download_URL}"\
  "${LFBFL_correct_sha512}"\
  "${LFBFL_verbose}"\
  "${LFBFL_check_download}"
chmod +x "${LFBFL_file_path}"
```

```

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.2 build_and_checks_dependencies/build_dependencies_notes.exec.php

```
#!/usr/bin/php
<?php

/**
This file is part of DevOrSysAdminScripts library.

```

```
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@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once 'string_escaping.libr.php';
require_once 'build_dependencies_notes.libr.php';

build_dependencies_notes($argv[1]);
?>
```

3.3 build_and_checks_dependencies/build_dependencies_notes.libr.php

```
<?php

/**
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```

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2026-03-26T11:59+01:00: This file was renamed from
"build_dependencies_notes.exec.php"
to
"build_dependencies_notes.libr.php".

@category Library
@package DevOrSysAdminScripts
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*/

declare(strict_types=1);
declare(encoding='UTF-8');

// require_once 'string_escaping.libr.php';



/**
This function looks at the dependencies data found in a work directory,
and generate three files: a txt file, a TeX file (.sub),
and an HTML file (.sub) with the data in a nicer format.

@param string $s_work_directory The directory where the work must be done.

@return void
*/
```

```
function build_dependencies_notes(string $s_work_directory) : void {
  $s_subdir1 = $s_work_directory.'/build_and_checks_variables';
  $s_subdir2 = $s_subdir1.'/temp';
  $s_data_file_path = $s_subdir1.'/dependencies_data.json';
  $s_data = file_get_contents($s_data_file_path);
  $arr_data = json_decode($s_data, true);
  $s_result_content1 = '';
  $s_result_content2 = '';
  $s_result_content3 = '';

  foreach($arr_data as $arr_dependency){
    $s_URL_word = 'URL';
    if(count($arr_dependency['URLs']) > 1){
      $s_URL_word .= 's';
    }

    $s_result_content3 .= "<p>\n";

    $s_dependency1 = $arr_dependency['dependency_name'];
    $s_dependency2 = string_escaping\TeX\escape_text($s_dependency1);
    $s_dependency3 = string_escaping\HTML\escape_text($s_dependency1);

    $s_sentence_end = ' can be found at the following '.$s_URL_word.':';
    $s_result_content1 .= $s_dependency1.$s_sentence_end."\n";
    $s_result_content2 .= $s_dependency2.$s_sentence_end."\\newline\n";
    $s_result_content3 .= $s_dependency3.$s_sentence_end."<br>\n";

    for(
      $i = 0, $i_max = count($arr_dependency['URLs']) - 1;
      $i <= $i_max;
      ++$i
    ){
      $mixed_URL = $arr_dependency['URLs'][$i];
      $s_comment = '';
      $s_URL = '';
      if(is_array($mixed_URL)){
        $s_URL = $mixed_URL['URL'];
        $s_comment = ' '.$mixed_URL['comment'];
      }
      else{
        $s_URL = $mixed_URL;
      }

      $s_result_content1 .= $s_URL.$s_comment."\n";

      if(strlen($s_URL) > 60){
```

```
        $s_result_content2 .= (
          "{\\small\n"
          .'  \url{'.$s_URL."}\n"
          .'}'
        );
      }
      else{
        $s_result_content2 .= '\url{'.$s_URL.'}';
      }
      $s_result_content2 .= string_escaping\TeX\escape_text($s_comment);
      if($i === $i_max){
        $s_result_content2 .= ".\n";
      }
      else{
        $s_result_content2 .= ",\\newline\n";
      }

      $s_result_content3 .= (
        "<a\n"
        .'href="'.string_escaping\HTML\escape_attribute($s_URL)."\"\n"
        ."><span style=\"font-family:monospace\"\n"
        .'>'.string_escaping\HTML\escape_text($s_URL)."</span\n"
        .'></a>'
      );
      $s_result_content3 .= string_escaping\HTML\escape_text($s_comment);
      if($i === $i_max){
        $s_result_content3 .= ".\n";
      }
      else{
        $s_result_content3 .= ",<br>\n";
      }
    }//end for urls

    $s_result_content1 .= "\n";
    $s_result_content2 .= "\n";
    $s_result_content3 .= "</p>\n\n";
  }//end foreach($arr_data as $arr_dependency)

  // Add a "footer".
  $s_closing_sentence = (
    "If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
  );
  $s_result_content1 .= $s_closing_sentence;
  $s_result_content2 .= $s_closing_sentence;
  $s_result_content3 .= "<p>\n".$s_closing_sentence."</p>\n";
```

```
// We overwrite results if needed.
$s_result_path1 = $s_subdir1.'/dependencies_notes.txt';
$s_existing_result_content1 = '';
if(file_exists($s_result_path1)){
  $s_existing_result_content1 = file_get_contents($s_result_path1);
}
if($s_result_content1 !== $s_existing_result_content1){
  file_put_contents($s_result_path1, $s_result_content1);
}
/*
Example of generated content of dependencies_notes.txt:
------------------------------------------------------------------------
more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?
------------------------------------------------------------------------
*/
$s_result_path2 = $s_subdir2.'/dependencies_notes.tex.sub';
$s_existing_result_content2 = '';
if(file_exists($s_result_path2)){
  $s_existing_result_content2 = file_get_contents($s_result_path2);
}
if($s_result_content2 !== $s_existing_result_content2){
  file_put_contents($s_result_path2, $s_result_content2);
}
/*
Example of generated content of dependencies_notes.tex.sub:
------------------------------------------------------------------------
more of util-linux can be found at the following URLs:\newline
\url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
{\small
  \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
},\newline
\url{https://github.com/util-linux/util-linux} official?.
------------------------------------------------------------------------
*/
$s_result_path3 = $s_subdir2.'/dependencies_notes.html.sub';
$s_existing_result_content3 = '';
if(file_exists($s_result_path3)){
  $s_existing_result_content3 = file_get_contents($s_result_path3);
}
if($s_result_content3 !== $s_existing_result_content3){
  file_put_contents($s_result_path3, $s_result_content3);
}
```

```
  /*
  Example of generated content of dependencies_notes.html.sub:
  <p>
  more of util-linux can be found at the following URLs:<br>
  <a
  href="https://www.kernel.org/pub/linux/utils/util-linux/"
  ><span style="font-family:monospace"
  >https://www.kernel.org/pub/linux/utils/util-linux/</span
  ></a>,<br>
  <a
  href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
  ><span style="font-family:monospace"
  >https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
  ></a>,<br>
  <a
  href="https://github.com/util-linux/util-linux"
  ><span style="font-family:monospace"
  >https://github.com/util-linux/util-linux</span
  ></a> official?.
  </p>
  */
}//end build_dependencies_notes()
?>
```

3.4 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
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```
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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name
  get_some_option LFBFL_base_name --base-name README 1 "$@"
  if [[ LFBFL_i_verbose -eq 1 && "${LFBFL_base_name}" != "README" ]]; then
    printf "Searching md file: %s.\n" "${LFBFL_base_name}"
  fi

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
    sed --regexp-extended --null-data\
```

```
      --expression="${LFBFL_sed_expression}"\
      --expression="${LFBFL_sed_expression}"\
      --expression="${LFBFL_sed_expression}"\
      --expression="${LFBFL_sed_expression}"\
      "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
    # Unfortunately, we cannot return early, since we want to propose
    # HTML file generation with pandoc from .md file,
    # if the file .md.tpl doesn't exist.
  fi

  if [[ ! -f "${LFBFL_base_name}.md" ]]; then
    printf "No file %s.md\n" "${LFBFL_base_name}"
    return 1
  fi

  pandoc --from=markdown --to=html --standalone\
    --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
    --output="${LFBFL_base_name}.html.temp"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed --in-place --regexp-extended\
    --expression='s~(  <meta .*) />~\1>~g'\
    --expression='s~<html .*>~<html lang="en">~'\
    --expression='/    code\{white-space: pre-wrap;\}/d'\
    "${LFBFL_base_name}.html.temp"
  sed --in-place --regexp-extended --null-data\
    --expression='s~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
    --expression="${LFBFL_prewrap}"\
    "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
```

```
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

3.5 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # $1=file_path
  #
  # declare -r LFBFL_file_path=$(basename -- "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
```

```
  #   return 0
  # fi
  # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head --lines=1 -- "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || printf "%s:File has wrong shell script beginning.\n" "$1"
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  local LFBFL_head
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_beginnings: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
```

```
      && printf -- "%s:Checking shell script beginning.\n"\
          "${LFBFL_file_path}"
    LFBFL_head=$(head --lines=1 -- "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || printf -- "%s:File has wrong shell script beginning.\n"\
          "${LFBFL_file_path}"
  done
}
```

3.6 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_one_shell_script_indentation(){
  # $1=file_path
  declare -i LFBFL_i_file_with_error=0
  local LFBFL_some_line
  local LFBFL_previous_line="workeduntilsomeasshole"
```

```
36    # created a command named "workeduntilsomeasshole"
37    # and called it without neither indentation, nor parameters
38    # in a .sh script XD.
39    # Manual justification of the lines L-4 and L-3 was a serendipity.
40    # I just tried a joke and was lucky ;) :).
41    local LFBFL_line_end
42    local LFBFL_previous_line_end
43    declare -i LFBFL_i_line_length
44    declare -i LFBFL_i_line_end_length
45    declare -i LFBFL_i_line_start_length
46    local LFBFL_spacing
47    declare -i LFBFL_i_offset
48    local LFBFL_substring1
49    local LFBFL_substring2
50    declare -r LFBFL_pattern='^ *-e '
51
52    # This grep would have been enough without some sed commands.
53    declare -r LFBFL_s_lines=$(
54      grep --before-context=1 --extended-regexp --line-number\
55        --regexp='^(  )* ([^ ]|$)' --with-filename -- "$1"
56    )
57    if [[ -z "${LFBFL_s_lines}" ]]; then
58      return
59    fi
60
61    declare -a LFBFL_arr_lines
62    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
63    readonly LFBFL_arr_lines
64
65    for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
66      if [[ "${LFBFL_some_line}" == "--" ]]; then
67        # When switching between code fragments, we reinit.
68        LFBFL_previous_line="workeduntilsomeasshole"
69        continue
70      fi
71      # Note how it's funny that this loop always work because we check
72      # shell scripts beginning in another script and thus we always
73      # have a line before.
74      if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]]; then
75        LFBFL_previous_line="${LFBFL_some_line}"
76        continue
77      fi
78      # Remove the name of the file and then the line number
79      LFBFL_line_end=${LFBFL_some_line#*:}
80      LFBFL_line_end=${LFBFL_line_end#*:}
81      # But for the previous line '-' or ':' is used instead of ':'
```

```
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_i_line_length=${#LFBFL_some_line}
    LFBFL_i_line_end_length=${#LFBFL_line_end}
    LFBFL_i_line_start_length=$((
      LFBFL_i_line_length - LFBFL_i_line_end_length
    ))
    LFBFL_i_offset=$((LFBFL_i_line_start_length - 2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_i_offset += 1));;
      :) ((LFBFL_i_offset += 1));;
      [0-9]) ((LFBFL_i_offset += 2));;
      *) printf ":-(\n";;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_i_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        printf -- "%s\n" "${LFBFL_some_line}"
        LFBFL_i_file_with_error=1
      fi
    else
      printf -- "%s\n" "${LFBFL_some_line}"
      LFBFL_i_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_i_file_with_error -eq 1 ]]; then
    printf -- "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
```

```
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_indentation: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf -- "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

3.7 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# that it will be useful,
```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )

  local LFBFL_pattern
  local LFBFL_base_file_name
  local LFBFL_substitution
```

```
  local LFBFL_substitution2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "check_URLs: Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep --with-filename -- 'http:'\
      | grep --invert-match "| xargs grep --with-filename -- 'htt"\
      | grep --invert-match --perl-regexp\
        "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"\
      | grep --invert-match 'http://www.w3.org/1999/xhtml'\
      | grep --invert-match\
        'http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd'
    # Second grep after xargs grep just above will remove false positives
    # from substitutions that fit on one line.

    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]] && printf "check_URLs: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf -- "check_URLs: Handling the file: %s.\n"\
          "${LFBFL_file_path}"
      fi
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep --quiet "${LFBFL_substitution}" -- "${LFBFL_file_path}";
        then
          sed --in-place\
          --expression="s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            -- "${LFBFL_file_path}"
        fi
      done
```

```
    done
  done
}
```

3.8 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
```

```
41  local LFBFL_work_directory_option="--work-directory="
42  LFBFL_work_directory_option+="${LFBFL_work_directory}"
43  declare -r LFBFL_dependencies_URL="$2"
44  local LFBFL_verbose=""
45  declare -i LFBFL_i_verbose=0
46  if [[ "$*" == *--verbose* ]]; then
47    printf -- "%s %s\n" "$0" "$*"
48    LFBFL_verbose="--verbose"
49    LFBFL_i_verbose=1
50  fi
51  readonly LFBFL_verbose
52  readonly LFBFL_i_verbose
53  local LFBFL_check_download=""
54  if [[ "$*" == *--check-download=keep* ]]; then
55    LFBFL_check_download="--check-download=keep"
56  elif [[ "$*" == *--check-download=y* ]]; then
57    LFBFL_check_download="--check-download=y"
58  fi
59  readonly LFBFL_check_download
60
61  declare -a LFBFL_some_common_options=(
62    "${LFBFL_verbose}"
63    # "$--work-directory={LFBFL_work_directory}" doesn't work
64    "${LFBFL_work_directory_option}"
65  )
66
67  source ./wget_sha512.libr.sh
68
69  local LFBFL_subdir="build_and_checks_dependencies"
70  declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
71  declare -r LFBFL_subdir3="build_and_checks_variables"
72  local LFBFL_file_name
73  local LFBFL_script_download_URL
74  local LFBFL_file_path
75  local LFBFL_correct_sha512
76
77  wrapped_wget_sha512(){
78    wget_sha512 "${LFBFL_file_path}"\
79      "${LFBFL_script_download_URL}"\
80      "${LFBFL_correct_sha512}"\
81      "${LFBFL_verbose}"\
82      "${LFBFL_check_download}"
83  }
84
85  LFBFL_file_name="build_dependencies_notes.exec.php"
86  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='624368d161351d7a261fbe3165d51907f46394c1b0e79'
LFBFL_correct_sha512+='b980ec11173c324b2c5531cf2a2b60b2dc62edcaa9c8'
LFBFL_correct_sha512+='71dca55065f8802df4275420836252786f9606b'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="build_dependencies_notes.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2152594899ac7ebd1bf56f8cb84b8b68f2e45d56d2942'
LFBFL_correct_sha512+='e6c6ed520f84c130012009be712e510f61c06f135937'
LFBFL_correct_sha512+='dc908b0db213071bae736cc3e18bca22b10bd02'
wrapped_wget_sha512

LFBFL_file_name="build_md_from_printable_md.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='1daf2d9ccdda7822ea9612888e792564ae50804613872'
LFBFL_correct_sha512+='918de7096c7c085f31eb28d3b63bad9a4d2e6ff0f7bc'
LFBFL_correct_sha512+='66672cd2cbeb13532c37225662684c47da11ef3'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='b44f28fc0bde88ea9876ba0f7fd18002624e3437d5def'
LFBFL_correct_sha512+='e96d3c98ac8a33aeda2837f7c166bbb6da6593fca989'
LFBFL_correct_sha512+='baee7e8759b758234b23b0855e48793e1134adb'
wrapped_wget_sha512

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='366fd5322d549cd9019dc6988436a0f7ea04c3264c662'
LFBFL_correct_sha512+='9f13782ec5daff983c533aef50885383855a80b134e0'
LFBFL_correct_sha512+='8604f0bf1e40f4ec6dc27040032b524ae073c30'
wrapped_wget_sha512

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='90c92977df7085553d163e12d8bb905d4d610299367a7'
LFBFL_correct_sha512+='10fb9cf68fb3f9cd4e3fca94bbe3762c2e23a52a581d'
LFBFL_correct_sha512+='1b76a8c4ed0a17067f6af40268f2a223f3582d0'
```

```
wrapped_wget_sha512

LFBFL_file_name="common_options.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='3cbda63f2337a8006b68a25113f8d05b35ee92ea5a04b'
LFBFL_correct_sha512+='07aaa5209bc943888114b5b97c26ec360b7a3c3c613a'
LFBFL_correct_sha512+='d66510257ffa6b58a1ff35318d2ef96327cefbd'
wrapped_wget_sha512

LFBFL_file_name="comparisons.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='8fbc6f65ad2fef3b96c4e4838bfee52205a71c2a88dfb'
LFBFL_correct_sha512+='77f3afa347e3fcc70d0b0cffcd0c96c945e4b0e858f8'
LFBFL_correct_sha512+='7389c8260c37e1645c47aa91c0c38e6999749ff'
wrapped_wget_sha512

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='229485706ebe003d3b6d36aeab2c93511473f928e1317'
LFBFL_correct_sha512+='fef66133152378ab7b8ced1119d1eea54c0ceadfb7c1'
LFBFL_correct_sha512+='8b838cce296ea65b76d86ca63365254d7b4df74'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='0c932bfc3ceafa2ac3bdd46610ff4ecbe0ec378b957b7'
LFBFL_correct_sha512+='3ca7972a9689f63983ec9cc1e6392826e1d057e26403'
LFBFL_correct_sha512+='86940368175c5ae8f3bfe890063142c8abfd0c6'
wrapped_wget_sha512

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='d38b894f63ac0db1eab4447a765b723642e12e3443aee'
LFBFL_correct_sha512+='01bd08e44bd2971bf11d3f0b98d2932bcaaeab4f30b7'
LFBFL_correct_sha512+='9f7da666e02a0fb1dada8a348ad866a104e4d53'
wrapped_wget_sha512

LFBFL_file_name="grammar_and_spelling_check.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
179     LFBFL_correct_sha512='27df239b4976fdf0299f1628d32bcec25783f4a5dda5f'
180     LFBFL_correct_sha512+='48e80e8ac342ec88f01bfc88e7cb26e45b904a9121ea'
181     LFBFL_correct_sha512+='1ed57ad5952e0d4bc504126092bc297631efecb'
182     wrapped_wget_sha512
183
184     # /licenses_templates/ ---------------------------------------------
185     local LFBFL_dependencies_URL2
186     LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
187     readonly LFBFL_dependencies_URL2
188     LFBFL_file_name="build_licenses_templates.exec.sh"
189     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
190     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
191     LFBFL_correct_sha512='e1295d0fb03b5433b17e833d63e7347c9ce50e4647c6e'
192     LFBFL_correct_sha512+='ccf068582c3ebc0555acb32af2d76f8f56211b43f2e0'
193     LFBFL_correct_sha512+='047f391b50a9dcabff914cedc13f2cd0ebb6ae2'
194     wrapped_wget_sha512
195     chmod +x "./${LFBFL_file_path}"
196
197     LFBFL_file_name="license_file_header_AGPLv3+.tpl"
198     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
199     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
200     LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
201     LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
202     LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
203     wrapped_wget_sha512
204
205     LFBFL_file_name="license_file_header_GPLv3+.tpl"
206     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
207     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
208     LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
209     LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
210     LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
211     wrapped_wget_sha512
212
213     LFBFL_file_name="license_file_header_LGPLv3+.tpl"
214     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
215     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
216     LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
217     LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
218     LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
219     wrapped_wget_sha512
220     # ------------------------------------------------------------------
221
222     LFBFL_file_name="lines_counts.libr.sh"
223     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
224     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
LFBFL_correct_sha512='6bea1ee84fcb5271b913ce7b5d1cac2d7ecdba86966a3'
LFBFL_correct_sha512+='999df462aae833e87eff9d1405e874da2618fa9540ae'
LFBFL_correct_sha512+='e684af6d26c3c74e70b8442c0cd9ec70dbb81c2'
wrapped_wget_sha512

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='1190e2cdb408f0ec4a497c74096bec6fc11972e01c36f'
LFBFL_correct_sha512+='d050ea2aee8b3bd9f44483576fc1ba1a6a2e81d00b17'
LFBFL_correct_sha512+='95eca0c606eb7cb74e469da9951d761fad99def'
wrapped_wget_sha512

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='45df460ebb31fd5f2f2a62a70f03d6ab9fd3bfd8d6e6f'
LFBFL_correct_sha512+='e45650e339f5b6bad5557a8c8d3a830d8539ba5442df'
LFBFL_correct_sha512+='68945da69270219159ceea45b14b395793fc025'
wrapped_wget_sha512

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='f75d5232cba494dd9c1a53111dc10b78414ec180bf293'
LFBFL_correct_sha512+='6114bdc8c415b41d0c3dd05b31ad9cf66affa1080922'
LFBFL_correct_sha512+='cc2d9fce18a4cdcef8feee8816ee100196fc190'
wrapped_wget_sha512

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='60d40634f255ce3e0e80e4899b451fdfa4b1fcc5f1c10'
LFBFL_correct_sha512+='18b86edb31c86f212f8176b8d666323d600d35a86c04'
LFBFL_correct_sha512+='1f8e4259486484605a22c509d5de673c8688499'
wrapped_wget_sha512

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='bc2e4c884bf79afb2a39bb0d63576e92f256c45904dd0'
LFBFL_correct_sha512+='01941cccf84ca08590d376abc29d7153eaee58dc4957'
LFBFL_correct_sha512+='4f8e2860b1e14177ec71c931f283477d4cfded8'
wrapped_wget_sha512

LFBFL_file_name="repository_data.libr.sh"
```

```
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='6f9355a7aa72d5fa4ca702700290555c1922e3928b8a2'
LFBFL_correct_sha512+='2a48f1d103d5a6a6ad568da0969d1d3eaab974fdec81'
LFBFL_correct_sha512+='8a7fcf2c094984e4ad35f7774c9942ba7cb39a3'
wrapped_wget_sha512

LFBFL_file_name="shell_checks_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='7da795615505504037acbf355efc0a74bb5ce6ca6fbd3'
LFBFL_correct_sha512+='68d69efabb07d2da95707357b0330bb94aebcafd7b40'
LFBFL_correct_sha512+='b179739406b63a9a142872d6513be0e40e796cf'
wrapped_wget_sha512

LFBFL_file_name="split_score.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='0eaca0166e71a1992f897c7f87e77da3babd297cd85c6'
LFBFL_correct_sha512+='e4251ae920f15b4cbd9c3d4a7e2ae21eaf74b366a0e7'
LFBFL_correct_sha512+='681977dff58901c6d40b3e46f38e0cbec86be54'
wrapped_wget_sha512

LFBFL_file_name="split_score.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='47b667d83125fa17babd46201ae8a8b7f7abeef123276'
LFBFL_correct_sha512+='40e7ed87faeacac997455207c4ab4df2734649676960'
LFBFL_correct_sha512+='1efcadaf35338ba2bec2410e01065ff7ed871f7'
wrapped_wget_sha512

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='eaddaa296834038d726d1309cce1524e019130a85eec9'
LFBFL_correct_sha512+='cabc7eab23670fc3f09d9023a5ee075f80cdf2a60fb8'
LFBFL_correct_sha512+='0d59aec6af9c96bf9aee7a7520c03bba084cb80'
wrapped_wget_sha512

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='c1ba12766290ea450a67f1d87764fe212b34bdbccfa11'
LFBFL_correct_sha512+='c968978c07603701575301baeb69f3638edf9f84ee76'
LFBFL_correct_sha512+='d36c266af12b930b40e36692d5e6f740f0a1b66'
wrapped_wget_sha512
```

```
LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='b0455e0c75a645e858489e4bf6ec6fc488cee605e48d0'
LFBFL_correct_sha512+='862d87bda2d9778d80595ab2778907b4d2816903d78a'
LFBFL_correct_sha512+='f17c047ca5b8e12223b6f41079c75aadb7ed28c'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spelling_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=shell_checks_complement.libr.sh
source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

enhanced_set_bash_option extglob
```

```
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

declare -a LFBFL_return_traps_stack
local LFBFL_previous_return_trap
init_return_trap

enhanced_set_shell_option pipefail --trap-unset

declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
keep_options LFBFL_some_common_options2 --verbose

printf "Building license headers\n"
"./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
  "${LFBFL_some_common_options[@]}"

printf "Building README.md\n"
"./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
  --work-directory="${LFBFL_work_directory}"\
  --base-name=README\
  "${LFBFL_verbose}"

printf "Building other MarkDown files\n"
local LFBFL_some_directory
declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
local LFBFL_s_files_paths
LFBFL_s_files_paths=$(
  find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
  | relevant_find
)
declare -a LFBFL_arr_files_paths
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    printf -- "Found template %s.\n" "${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
    LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
    LFBFL_file_name=${LFBFL_file_name%.md.tpl}
    "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
      --work-directory="${LFBFL_some_directory}"\
      --base-name="${LFBFL_file_name}"\
      "${LFBFL_verbose}"
```

```
    done
  fi

  printf "Building dependencies_notes\n"
  # https://github.com/php/php-src/issues/21538
  local LFBFL_annoying_warning
  LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
  LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
  LFBFL_annoying_warning=" by settings in"
  readonly LFBFL_annoying_warning
  php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
    -- "${LFBFL_work_directory}"\
    2>&1\
    | grep --invert-match "${LFBFL_annoying_warning}"

  declare -i LFBFL_i_directory_changed
  pushd_to_work_directory
  LFBFL_i_directory_changed=$?
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_path}" max_line_length\
    --result-variable-prefix=LFBFL_i_

  LFBFL_some_common_options+=(
    "--max-line-length=${LFBFL_i_max_line_length}"
  )
  LFBFL_some_common_options2+=(
    "--max-line-length=${LFBFL_i_max_line_length}"
  )

  declare -i LFBFL_i_upgrade_venvs=0
  local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
  readonly LFBFL_upgrade_venvs_ts_file
  declare -i LFBFL_i_upgrade_venvs_ts
  declare -i LFBFL_i_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
    "${LFBFL_some_common_options2[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_i_upgrade_venvs_ts=$(
```

```
      stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
    ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      LFBFL_i_upgrade_venvs=1
    fi
  else
    LFBFL_i_upgrade_venvs=1
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    local LFBFL_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Analyzing too long lines\n"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_some_common_options2[@]}"

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»..."
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
    "[^${LFBFL_usual_characters}]" .\
    | relevant_grep

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_path}"\
    "${LFBFL_some_common_options2[@]}"

  printf "Running shellcheck\n"
```

```
LFBFL_s_files_paths=$(
  find . -type f -name "*.sh"\
  | relevant_find
)
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
      -- "${LFBFL_file_path}"
  done
fi
printf "Running shell_checks_complement\n"
shell_checks_complement "${LFBFL_some_common_options2[@]}"

printf "Analyzing shell scripts beginnings\n"
check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
  | relevant_grep

printf "Analyzing shell scripts indentation\n"
check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
  | relevant_grep

printf -- "---Python---\n"

maybe_deactivate(){
  if command -v deactivate
  then
    deactivate
  fi
}

source_without_return_trap(){
  declare -i LFBFL_i_no_return_traps=1
  # shellcheck disable=SC1090,SC1091
  source -- "$1"
}

printf "Running isort\n"
local LFBFL_isort_venv=""
grep_variable "${LFBFL_data_file_path}" isort_venv\
  --result-variable-prefix=LFBFL_\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_isort_venv}" ]]; then
  maybe_deactivate
  source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
fi
```

```
547   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
548     pip install --upgrade isort
549   fi
550   isort --profile=black .
551   if [[ -n "${LFBFL_isort_venv}" ]]; then
552     deactivate
553   fi
554   printf "Running python_isort_complement\n"
555   python_isort_complement "${LFBFL_some_common_options2[@]}"
556
557   printf "Running black\n"
558   # First, we update the configuration file with max_line_length.
559   local LFBFL_update_max_length="s/^line-length = [0-9]*$"
560   LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
561   sed --regexp-extended --expression="${LFBFL_update_max_length}"\
562     "${LFBFL_subdir3}/black.toml"\
563     > "${LFBFL_subdir3}/black.toml.temp"
564   overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
565     "${LFBFL_subdir3}/black.toml.temp"
566
567   local LFBFL_black_venv=""
568   grep_variable "${LFBFL_data_file_path}" black_venv\
569     --result-variable-prefix=LFBFL_\
570     --replace-line-returns-by=""
571   if [[ -n "${LFBFL_black_venv}" ]]; then
572     maybe_deactivate
573     source_without_return_trap "${LFBFL_black_venv}/bin/activate"
574   fi
575   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
576     pip install --upgrade black
577   fi
578   black --config="${LFBFL_subdir3}/black.toml" .
579   if [[ -n "${LFBFL_black_venv}" ]]; then
580     deactivate
581   fi
582   printf "Running python_black_complement\n"
583   python_black_complement "${LFBFL_some_common_options2[@]}"
584
585   printf "Probing if mypy should be runned\n"
586   local LFBFL_mypy_venv=""
587   grep_variable "${LFBFL_data_file_path}" mypy_venv\
588     --result-variable-prefix=LFBFL_\
589     --replace-line-returns-by=""
590   if [[ -n "${LFBFL_mypy_venv}" ]]; then
591     maybe_deactivate
592     source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
```

```
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
        mypy -- "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_path}" bandit_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
    --recursive .
  # Saving new baseline in temp if necessary.
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
```

```
    --format=json\
    --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
    --recursive .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_path}" pylint_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_path}" ruff_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config="${LFBFL_subdir3}/ruff.toml"
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHP native linter\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.php"\
```

```
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      php --syntax-check -f="${LFBFL_file_path}"
    done
  fi

  printf "Running PHP_CodeSniffer\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require squizlabs/php_codesniffer
  fi
  phpcs --report=code\
    --standard=build_and_checks_variables/phpcs_ruleset.xml

  printf "Running PHPMD\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
  local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

  local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
  LFBFL_phpmd_rulesets+="design,naming,unusedcode"
  readonly LFBFL_phpmd_rulesets

  phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  # Saving new baseline in temp if necessary.
  rm "${LFBFL_temp_phpmd_baseline}"
  phpmd --color --generate-baseline\
    --baseline-file="${LFBFL_temp_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  sed --in-place --expression='s/" file="/"\n    file="/g'\
    "${LFBFL_temp_phpmd_baseline}"
  sed --in-place --regexp-extended\
    --expression='s~(file=".*)/>~\1\n  />~g'\
    "${LFBFL_temp_phpmd_baseline}"
  diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"

  printf "Running phpDocumentor\n"
  # shellcheck disable=SC2312
```

```
  docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
    --result-variable-prefix=LFBFL_
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd -- "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
    "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_path}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
```

```
fi

declare -i LFBFL_i_vnu_jar_ts
local LFBFL_vnu_jar_path=""
grep_variable "${LFBFL_data_file_path}" vnu_jar_path\
  --result-variable-prefix=LFBFL_\
  --replace-line-returns-by=""
if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
  LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
  LFBFL_i_current_ts=$(date +%s)
  ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
  ((
    LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
  ))
  if [[ LFBFL_i_current_ts -gt 0 ]]; then
    printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
    printf "Go get the latest version here:\n"
    printf "https://github.com/validator/validator/releases\n"
  fi
  # vnu has false positives for xhtml
  local LFBFL_use_vnu_for_xhtml=""
  grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml\
    --result-variable-prefix=LFBFL_
  local LFBFL_find_regexp
  if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
    LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
  else
    LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
  fi
  declare -r LFBFL_files_for_Nu=$(
    find . -iregex "${LFBFL_find_regexp}"\
    | relevant_find
  )
  if [[ -n "${LFBFL_files_for_Nu}" ]]; then
    declare -a LFBFL_files_for_Nu2
    mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
    readonly LFBFL_files_for_Nu2
    java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
      --verbose -- "${LFBFL_files_for_Nu2[@]}"
  fi
fi
# ------------------------------------------------------------------

if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  touch -- "${LFBFL_upgrade_venvs_ts_file}"
fi
```

```
}

common_build_and_checks "$@"
```

3.9 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
```

```
41    local LFBFL_work_directory_option="--work-directory="
42    LFBFL_work_directory_option+="${LFBFL_work_directory}"
43    declare -r LFBFL_dependencies_URL="$2"
44    local LFBFL_verbose=""
45    declare -i LFBFL_i_verbose=0
46    if [[ "$*" == *--verbose* ]]; then
47      printf -- "%s %s\n" "$0" "$*"
48      LFBFL_verbose="--verbose"
49      LFBFL_i_verbose=1
50    fi
51    readonly LFBFL_verbose
52    readonly LFBFL_i_verbose
53    local LFBFL_check_download=""
54    if [[ "$*" == *--check-download=keep* ]]; then
55      LFBFL_check_download="--check-download=keep"
56    elif [[ "$*" == *--check-download=y* ]]; then
57      LFBFL_check_download="--check-download=y"
58    fi
59    readonly LFBFL_check_download
60
61    declare -a LFBFL_some_common_options=(
62      "${LFBFL_verbose}"
63      # "$--work-directory={LFBFL_work_directory}" doesn't work
64      "${LFBFL_work_directory_option}"
65    )
66
67    source ./wget_sha512.libr.sh
68
69    local LFBFL_subdir="build_and_checks_dependencies"
70    declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
71    declare -r LFBFL_subdir3="build_and_checks_variables"
72    local LFBFL_file_name
73    local LFBFL_script_download_URL
74    local LFBFL_file_path
75    local LFBFL_correct_sha512
76
77    wrapped_wget_sha512(){
78      wget_sha512 "${LFBFL_file_path}"\
79        "${LFBFL_script_download_URL}"\
80        "${LFBFL_correct_sha512}"\
81        "${LFBFL_verbose}"\
82        "${LFBFL_check_download}"
83    }
84
85    LFBFL_file_name="build_dependencies_notes.exec.php"
86    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_dependencies_notes.exec.php@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="build_dependencies_notes.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_dependencies_notes.libr.php@
wrapped_wget_sha512

LFBFL_file_name="build_md_from_printable_md.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_build_md_from_printable_md.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_beginnings.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_indentation.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_URLs.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="common_options.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_common_options.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="comparisons.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_comparisons.libr.sh@
```

```
wrapped_wget_sha512

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_create_PDF.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_generate_from_template.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_get_common_text_glob_patterns.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="grammar_and_spelling_check.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_grammar_and_spelling_check.libr.sh@
wrapped_wget_sha512

# /licenses_templates/ ---------------------------------------------
local LFBFL_dependencies_URL2
LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
readonly LFBFL_dependencies_URL2
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_build_licenses_templates.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_AGPLv3+.tpl@
wrapped_wget_sha512

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_GPLv3+.tpl@
wrapped_wget_sha512

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_LGPLv3+.tpl@
wrapped_wget_sha512
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_counts.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_filters.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_maps.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_overwrite_if_not_equal.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_black_complement.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_isort_complement.libr.sh@
wrapped_wget_sha512
```

```
225
226     LFBFL_file_name="repository_data.libr.sh"
227     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
228     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
229     @sha512_repository_data.libr.sh@
230     wrapped_wget_sha512
231
232     LFBFL_file_name="shell_checks_complement.libr.sh"
233     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
234     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
235     @sha512_shell_checks_complement.libr.sh@
236     wrapped_wget_sha512
237
238     LFBFL_file_name="split_score.exec.php"
239     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
240     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
241     @sha512_split_score.exec.php@
242     wrapped_wget_sha512
243
244     LFBFL_file_name="split_score.libr.php"
245     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
246     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
247     @sha512_split_score.libr.php@
248     wrapped_wget_sha512
249
250     LFBFL_file_name="strings_functions.libr.sh"
251     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
252     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
253     @sha512_strings_functions.libr.sh@
254     wrapped_wget_sha512
255
256     LFBFL_file_name="too_long_code_lines.libr.sh"
257     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
258     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
259     @sha512_too_long_code_lines.libr.sh@
260     wrapped_wget_sha512
261
262     LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
263     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
264     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
265     @sha512_update_or_check_files_names_listing.exec.sh@
266     wrapped_wget_sha512
267     chmod +x "./${LFBFL_file_path}"
268
269     # shellcheck source=check_shell_scripts_beginnings.libr.sh
270     source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
```

```
271     # shellcheck source=check_shell_scripts_indentation.libr.sh
272     source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
273     # shellcheck source=check_URLs.libr.sh
274     source "./${LFBFL_subdir}/check_URLs.libr.sh"
275     # shellcheck source=common_options.libr.sh
276     source "./${LFBFL_subdir}/common_options.libr.sh"
277     # shellcheck source=comparisons.libr.sh
278     source "./${LFBFL_subdir}/comparisons.libr.sh"
279     # shellcheck source=generate_from_template.libr.sh
280     source "./${LFBFL_subdir}/generate_from_template.libr.sh"
281     # shellcheck source=get_common_text_glob_patterns.libr.sh
282     source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
283     # shellcheck source=grammar_and_spelling_check.libr.sh
284     source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
285     # shellcheck source=lines_counts.libr.sh
286     source "./${LFBFL_subdir}/lines_counts.libr.sh"
287     # shellcheck source=lines_filters.libr.sh
288     source "./${LFBFL_subdir}/lines_filters.libr.sh"
289     # shellcheck source=lines_maps.libr.sh
290     source "./${LFBFL_subdir}/lines_maps.libr.sh"
291     # shellcheck source=overwrite_if_not_equal.libr.sh
292     source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
293     # shellcheck source=python_black_complement.libr.sh
294     source "./${LFBFL_subdir}/python_black_complement.libr.sh"
295     # shellcheck source=python_isort_complement.libr.sh
296     source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
297     # shellcheck source=shell_checks_complement.libr.sh
298     source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
299     # shellcheck source=strings_functions.libr.sh
300     source "./${LFBFL_subdir}/strings_functions.libr.sh"
301     # shellcheck source=too_long_code_lines.libr.sh
302     source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"
303
304     enhanced_set_bash_option extglob
305     # shellcheck source=repository_data.libr.sh
306     source "./${LFBFL_subdir}/repository_data.libr.sh"
307     # shellcheck disable=SC2154,SC2309
308     [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
309       && enhanced_unset_bash_option extglob
310
311     declare -a LFBFL_return_traps_stack
312     local LFBFL_previous_return_trap
313     init_return_trap
314
315     enhanced_set_shell_option pipefail --trap-unset
316
```

```
  declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
  keep_options LFBFL_some_common_options2 --verbose

  printf "Building license headers\n"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Building README.md\n"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    --work-directory="${LFBFL_work_directory}"\
    --base-name=README\
    "${LFBFL_verbose}"

  printf "Building other MarkDown files\n"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf -- "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
      LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        --work-directory="${LFBFL_some_directory}"\
        --base-name="${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  printf "Building dependencies_notes\n"
  # https://github.com/php/php-src/issues/21538
  local LFBFL_annoying_warning
  LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
  LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
  LFBFL_annoying_warning=" by settings in"
  readonly LFBFL_annoying_warning
  php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
    -- "${LFBFL_work_directory}"\
```

```
363       2>&1\
364       | grep --invert-match "${LFBFL_annoying_warning}"
365
366     declare -i LFBFL_i_directory_changed
367     pushd_to_work_directory
368     LFBFL_i_directory_changed=$?
369     can_continue_after_enhanced_pushd || return 1
370
371     local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"
372
373     declare -i LFBFL_i_max_line_length
374     grep_variable "${LFBFL_data_file_path}" max_line_length\
375       --result-variable-prefix=LFBFL_i_
376
377     LFBFL_some_common_options+=(
378       "--max-line-length=${LFBFL_i_max_line_length}"
379     )
380     LFBFL_some_common_options2+=(
381       "--max-line-length=${LFBFL_i_max_line_length}"
382     )
383
384     declare -i LFBFL_i_upgrade_venvs=0
385     local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
386     readonly LFBFL_upgrade_venvs_ts_file
387     declare -i LFBFL_i_upgrade_venvs_ts
388     declare -i LFBFL_i_current_ts
389     local LFBFL_upgrade_venvs_answer
390
391     local LFBFL_upgrade_venvs_time_interval_in_seconds=""
392     get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
393       "${LFBFL_some_common_options2[@]}"
394
395     if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
396       LFBFL_i_upgrade_venvs_ts=$(
397         stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
398       )
399       LFBFL_i_current_ts=$(date +%s)
400       ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
401       ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
402       if [[ LFBFL_i_current_ts -gt 0 ]]; then
403         LFBFL_i_upgrade_venvs=1
404       fi
405     else
406       LFBFL_i_upgrade_venvs=1
407     fi
408     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
```

```
    local LFBFL_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Analyzing too long lines\n"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_some_common_options2[@]}"

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»..."
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
    "[^${LFBFL_usual_characters}]" .\
    | relevant_grep

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_path}"\
    "${LFBFL_some_common_options2[@]}"

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
        -- "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
```

```
455     shell_checks_complement "${LFBFL_some_common_options2[@]}"
456
457     printf "Analyzing shell scripts beginnings\n"
458     check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
459       | relevant_grep
460
461     printf "Analyzing shell scripts indentation\n"
462     check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
463       | relevant_grep
464
465     printf -- "---Python---\n"
466
467     maybe_deactivate(){
468       if command -v deactivate
469       then
470         deactivate
471       fi
472     }
473
474     source_without_return_trap(){
475       declare -i LFBFL_i_no_return_traps=1
476       # shellcheck disable=SC1090,SC1091
477       source -- "$1"
478     }
479
480     printf "Running isort\n"
481     local LFBFL_isort_venv=""
482     grep_variable "${LFBFL_data_file_path}" isort_venv\
483       --result-variable-prefix=LFBFL_\
484       --replace-line-returns-by=""
485     if [[ -n "${LFBFL_isort_venv}" ]]; then
486       maybe_deactivate
487       source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
488     fi
489     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
490       pip install --upgrade isort
491     fi
492     isort --profile=black .
493     if [[ -n "${LFBFL_isort_venv}" ]]; then
494       deactivate
495     fi
496     printf "Running python_isort_complement\n"
497     python_isort_complement "${LFBFL_some_common_options2[@]}"
498
499     printf "Running black\n"
500     # First, we update the configuration file with max_line_length.
```

```
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
  sed --regexp-extended --expression="${LFBFL_update_max_length}"\
    "${LFBFL_subdir3}/black.toml"\
    > "${LFBFL_subdir3}/black.toml.temp"
  overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
    "${LFBFL_subdir3}/black.toml.temp"

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_path}" black_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_path}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
```

```
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
        mypy -- "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_path}" bandit_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
    --recursive .
  # Saving new baseline in temp if necessary.
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --format=json\
    --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
    --recursive .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_path}" pylint_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
```

```
593   if [[ -n "${LFBFL_pylint_venv}" ]]; then
594     maybe_deactivate
595     source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
596   fi
597   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
598     pip install --upgrade pylint
599   fi
600   pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
601   if [[ -n "${LFBFL_pylint_venv}" ]]; then
602     deactivate
603   fi
604
605   printf "Running ruff\n"
606   LFBFL_ruff_venv=""
607   grep_variable "${LFBFL_data_file_path}" ruff_venv\
608     --result-variable-prefix=LFBFL_\
609     --replace-line-returns-by=""
610   if [[ -n "${LFBFL_ruff_venv}" ]]; then
611     maybe_deactivate
612     source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
613   fi
614   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
615     pip install --upgrade ruff
616   fi
617   ruff check --config="${LFBFL_subdir3}/ruff.toml"
618   if [[ -n "${LFBFL_ruff_venv}" ]]; then
619     deactivate
620   fi
621   printf -- "---Python end---\n"
622
623   printf -- "---PHP---\n"
624   printf "Running PHP native linter\n"
625   LFBFL_s_files_paths=$(
626     find . -type f -name "*.php"\
627     | relevant_find
628   )
629   if [[ -n "${LFBFL_s_files_paths}" ]]; then
630     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
631     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
632       php --syntax-check -f="${LFBFL_file_path}"
633     done
634   fi
635
636   printf "Running PHP_CodeSniffer\n"
637   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
638     composer global require squizlabs/php_codesniffer
```

```
  fi
  phpcs --report=code\
    --standard=build_and_checks_variables/phpcs_ruleset.xml

  printf "Running PHPMD\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
  local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

  local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
  LFBFL_phpmd_rulesets+="design,naming,unusedcode"
  readonly LFBFL_phpmd_rulesets

  phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  # Saving new baseline in temp if necessary.
  rm "${LFBFL_temp_phpmd_baseline}"
  phpmd --color --generate-baseline\
    --baseline-file="${LFBFL_temp_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  sed --in-place --expression='s/" file="/"\n    file="/g'\
    "${LFBFL_temp_phpmd_baseline}"
  sed --in-place --regexp-extended\
    --expression='s~(file=".*)/>~\1\n  />~g'\
    "${LFBFL_temp_phpmd_baseline}"
  diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"

  printf "Running phpDocumentor\n"
  # shellcheck disable=SC2312
  docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
    --result-variable-prefix=LFBFL_
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
```

```
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd -- "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
    "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_path}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_path}" vnu_jar_path\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
    ((
```

```
      LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
    ))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    # vnu has false positives for xhtml
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml\
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose -- "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch -- "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.10 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
```

```
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get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

# RETURN traps handling
# Add the following lines at the start of a function.
#   declare -a LFBFL_return_traps_stack
#   local LFBFL_previous_return_trap
#   init_return_trap

init_return_trap(){
  local LFBFL_previous_return_trap2
  LFBFL_previous_return_trap2=$(trap -p RETURN)
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
```

```
    get_where_was_i 2
    printf -- "%s previous_return_trap is: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap2}"
  fi
  if [[ -z "${LFBFL_previous_return_trap2}" ]]; then
    LFBFL_previous_return_trap2="trap - RETURN"
  fi
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  LFBFL_previous_return_trap="${LFBFL_function_depth}:"
  LFBFL_previous_return_trap+="${LFBFL_previous_return_trap2}"
  trap 'execute_return_traps' RETURN
}

execute_return_traps(){
  if [[ "${FUNCNAME[1]}" == "init_return_trap" ]]; then
    return
  fi
  # ----------------------------------------------------------------------
  # See https://lists.gnu.org/archive/html/help-bash/2026-03/msg00052.html
  if [[ "${FUNCNAME[1]}" == "source" ]]; then
    # Not Working
    return
  fi
  if [[ ${LFBFL_i_no_return_traps:-0} -eq 1 ]]; then
    return
  fi
  # ----------------------------------------------------------------------
  local LFBFL_some_return_trap
  local LFBFL_where_was_i
  get_where_was_i 2
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  while [[ -n "${LFBFL_return_traps_stack}" ]]; do
    LFBFL_some_return_trap="${LFBFL_return_traps_stack[-1]}"
    if [[ "${LFBFL_some_return_trap}" != ${LFBFL_function_depth}:* ]]; then
      # We only treat the top of the stack that should correspond to the
      # current function.
      break
    fi
    LFBFL_some_return_trap=${LFBFL_some_return_trap#*:}
    # shellcheck disable=SC2154,SC2309
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf -- "%s executing some return trap: %s.\n"\
        "${LFBFL_where_was_i}"\
        "${LFBFL_some_return_trap}"
    fi
```

```
    eval -- "${LFBFL_some_return_trap}"
    unset 'LFBFL_return_traps_stack[-1]'
  done
  if [[ "${LFBFL_previous_return_trap}" != ${LFBFL_function_depth}:* ]];
  then
    # We only set the previous return trap if such a trap was replaced
    # by an init_return_trap.
    return
  fi
  LFBFL_previous_return_trap=${LFBFL_previous_return_trap#*:}
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf -- "%s setting back to previous_return_trap: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap}"
  fi
  eval -- "${LFBFL_previous_return_trap}"
}


is_top_dirstack_directory(){
  # Altough "${DIRSTACK[0]}" may seem to be already a realpath,
  # possible use of symbolic links makes this function necessary.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath -- "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf -- "enhanced_pushd requested: %s\n" "$1"
  declare -gi i_enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
```

```
    i_enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  fi
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated1\n"
  is_top_dirstack_directory "$1" && {
    i_enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated2\n"

  pushd -- "$1" || {
    i_enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf --\
      "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3" "$1"
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed\n"
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf -- "enhanced_popd requested: %s\n" "$1"
  if [[ -z "$1" ]]; then
    return 0
  fi

  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
```

```
    get_where_was_i "$2"
    printf -- "%s popd failed.\n" "${LFBFL_where_was_i}"
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed\n"
  return 0
}

can_continue_after_enhanced_pushd(){
  [[ i_enhanced_pushd_result -eq 0
  || i_enhanced_pushd_result -eq 110
  || i_enhanced_pushd_result -eq 111
  ]]
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_some_flag(){
  # $1=variable_name
  # $2=some_flag_long_name
  # $3=read_only 0 or 1 to make the variable readonly
```

```
  # This command is to be called in another one with same arguments on top
  # of the three previous.
  # This test:
  # if [[ "$*" == *--verbose* ]]; then
  # is faster than below.
  # But it is not bug free in a calling function,
  # and even incompatible with the current function where
  # the flag_long_name is a mandatory argument.
  printf -v "$1" "%s" "0"
  declare -i LFBFL_i
  local LFBFL_argument
  for ((LFBFL_i = 4; LFBFL_i <= $#; ++LFBFL_i)); do
    LFBFL_argument="${!LFBFL_i}"
    if [[ "${LFBFL_argument}" == "$2" ]]; then
      printf -v "$1" "%s" "1"
      break
    fi
  done
  if [[ $3 -eq 1 ]]; then
    readonly "$1"
  fi
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  get_some_flag LFBFL_i_verbose --verbose 1 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf -- "%s %s\n" "${LFBFL_where_was_i}" "$*"
  fi
}


get_some_option(){
  # $1=variable_name
  # $2=some_option_long_name
  # $3=option_default_value
  # $4=read_only 0 or 1 to make the variable readonly
  # This command is to be called in another one with same arguments on top
  # of the four previous.
  local LFBFL_argument
  for LFBFL_argument in "$@"; do
```

```
    if [[ "${LFBFL_argument}" == "$2"=* ]]; then
      printf -v "$1" "%s" "${LFBFL_argument#"$2"=}"
      break
    fi
  done
  printf -v "$1" "%s" "${!1:-$3}"
  if [[ $4 -eq 1 ]]; then
    readonly "$1"
  fi
}

keep_options(){
  # $1=variable_name of array of options
  # Other arguments: the options to keep.
  declare -a LFBFL_array_of_filtered_options
  declare -n LFBFL_array_of_options=$1
  local LFBFL_some_argument
  local LFBFL_some_argument2
  for LFBFL_some_argument in "${LFBFL_array_of_options[@]}"; do
    for LFBFL_some_argument2 in "$@"; do
      if [[
        "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"
        || "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"=*
      ]]; then
        LFBFL_array_of_filtered_options+=("${LFBFL_some_argument}")
      fi
    done
  done
  LFBFL_array_of_options=("${LFBFL_array_of_filtered_options[@]}")
}


# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return 1
# Or:
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory --trap-popd
#   can_continue_after_enhanced_pushd || return 1

get_work_directory_option(){
```

```
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  get_some_option LFBFL_work_directory --work-directory "." 0 "$@"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
  # Options:
  #   --trap-popd
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
  if [[ i_enhanced_pushd_result -eq 0 ]]; then
    declare -i LFBFL_i_trap_popd
    get_some_flag LFBFL_i_trap_popd --trap-popd 1 "$@"
    if [[ LFBFL_i_trap_popd -eq 1 ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="popd_from_work_directory 4"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
    fi
  fi
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

popd_from_work_directory(){
  # $1=offset for where_was_i
  enhanced_popd "${LFBFL_work_directory}" "${1:-3}"
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


get_offset_option(){
  # This command is to be called in another one with same arguments on top
  # of the default value.
  # $1=option_default_value
  # Options:
  #   --offset=2
  get_some_option LFBFL_i_offset --offset "$1" 1 "$@"
}
```

```

get_max_line_length_option(){
  # This command is to be called in another one with same arguments.
  # Its most useful feature is that it factorizes the default maximum line
  # length of 70 characters.
  # Options:
  #   --max-line-length
  get_some_option LFBFL_i_max_line_length --max-line-length 70 1 "$@"
}


# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
#       && trap 'enhanced_set_shell_option pipefail' RETURN
#   if you need to unset it in some function.
#   or better if you use my RETURN traps handler:
#     enhanced_set_shell_option pipefail --trap-unset
#     enhanced_unset_shell_option pipefail --trap-set
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN
#   or better if you use my RETURN traps handler:
#     enhanced_set_bash_option globstar --trap-unset
#     enhanced_unset_bash_option globstar --trap-set

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
```

```
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf -- "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  declare -i LFBFL_i_trap_unset
  get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
  if [[ LFBFL_i_trap_unset -eq 1 ]]; then
    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
    local LFBFL_some_return_trap
    LFBFL_some_return_trap="${LFBFL_function_depth}:"
    LFBFL_some_return_trap+="enhanced_unset_shell_option '$1' --offset=3"
    readonly LFBFL_some_return_trap
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set +o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf --\
      "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
```

```
467   declare -i LFBFL_i_trap_set
468   get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
469   if [[ LFBFL_i_trap_set -eq 1 ]]; then
470     declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
471     local LFBFL_some_return_trap
472     LFBFL_some_return_trap="${LFBFL_function_depth}:"
473     LFBFL_some_return_trap+="enhanced_set_shell_option '$1' --offset=3"
474     readonly LFBFL_some_return_trap
475     LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
476   fi
477   # shellcheck disable=SC2086
478   return ${!LFBFL_result_name}
479 }
480
481 enhanced_set_bash_option(){
482   # $1=optname some bash option for shopt
483   # Options:
484   #   --offset=2
485   #   --trap-unset
486   declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
487   declare -gi "${LFBFL_result_name}"=0
488   # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
489   # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
490   if shopt -q "$1"; then
491     printf -v "${LFBFL_result_name}" "%d" "1"
492     # shellcheck disable=SC2086
493     return ${!LFBFL_result_name}
494   fi
495   shopt -s "$1"
496   declare -i LFBFL_i_offset=2
497   get_offset_option 2 "$@"
498   # shellcheck disable=SC2154,SC2309
499   if [[ LFBFL_i_verbose -eq 1 ]]; then
500     local LFBFL_where_was_i
501     # shellcheck disable=SC2248
502     get_where_was_i ${LFBFL_i_offset}
503     printf -- "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
504   fi
505   declare -i LFBFL_i_trap_unset
506   get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
507   if [[ LFBFL_i_trap_unset -eq 1 ]]; then
508     declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
509     local LFBFL_some_return_trap
510     LFBFL_some_return_trap="${LFBFL_function_depth}:"
511     LFBFL_some_return_trap+="enhanced_unset_bash_option '$1' --offset=3"
512     readonly LFBFL_some_return_trap
```

```
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf --\
      "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  declare -i LFBFL_i_trap_set
  get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
  if [[ LFBFL_i_trap_set -eq 1 ]]; then
    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
    local LFBFL_some_return_trap
    LFBFL_some_return_trap="${LFBFL_function_depth}:"
    LFBFL_some_return_trap+="enhanced_set_bash_option '$1' --offset=3"
    readonly LFBFL_some_return_trap
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

3.11 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
  # l'idée d'un principe de première différence ternaire.
  declare -i LFBFL_i_first=1
  local LFBFL_current
  local LFBFL_argument
  for LFBFL_argument in "$@"; do
    if [[ LFBFL_i_first -eq 1 ]]; then
      LFBFL_current="${LFBFL_argument}"
      LFBFL_i_first=0
      continue
    fi
```

```
    if [[ "${LFBFL_current}" != "${LFBFL_argument}" ]]; then
      return 0
    fi
  done
  return 1
}

some_distinct(){
  ! equal "$@"
}

all_distinct(){
  # Returns 1 if all the arguments strings are distinct.
  declare -ir LFBFL_ij_max=$#
  declare -i LFBFL_i
  declare -i LFBFL_j
  for ((LFBFL_i = 1; LFBFL_i < LFBFL_ij_max; ++LFBFL_i)); do
    for ((LFBFL_j = LFBFL_i + 1; LFBFL_j <= LFBFL_ij_max; ++LFBFL_j)); do
      [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
    done
  done
  return 1
}

all_distinct2(){
  # Returns 1 if all the arguments strings are distinct.
  local LFBFL_element
  local LFBFL_previous_element
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_sorted_arguments=$(
    printf -- "%s\n" "$@" \
    | sort
  )
  declare -a LFBFL_arr_sorted_arguments
  mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
  declare -i LFBFL_i
  declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
  LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
  for ((LFBFL_i = 1; LFBFL_i < LFBFL_i_max; ++LFBFL_i)); do
    LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
    if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
      return 0
    fi
    LFBFL_previous_element="${LFBFL_element}"
```

```
  done
  return 1
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf -- "%s\n" "$@" \
    | sort\
    | uniq --repeated
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
```

```
137 }
138
139 max(){
140   # Returns the maximum string among the arguments.
141   # Renvoie la chaîne la plus grande parmi celles proposées.
142   # $1=sort_command with the flags you want to be applied.
143   # $ max sort a b c
144   # c
145   if [[ $# -eq 0 ]]; then
146     return 1
147   fi
148   enhanced_set_shell_option pipefail\
149     && trap 'enhanced_unset_shell_option pipefail' RETURN
150   printf -- "%s\n" "${@:2}" \
151     | eval -- "$1 --reverse"\
152     | head --lines=1
153 }
154
155 min(){
156   # Returns the minimum string among the arguments.
157   # Renvoie la chaîne la plus petite parmi celles proposées.
158   # $1=sort_command with the flags you want to be applied.
159   # $ min sort a b c
160   # a
161   if [[ $# -eq 0 ]]; then
162     return 1
163   fi
164   enhanced_set_shell_option pipefail\
165     && trap 'enhanced_unset_shell_option pipefail' RETURN
166   printf -- "%s\n" "${@:2}" \
167     | eval -- "$1"\
168     | head --lines=1
169 }
170
171 is_substring(){
172   # $1=string
173   # $2=substring
174   declare -r LFBFL_var_1=$(
175     printf -- "%s" "$1"\
176     | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
177   )
178   declare -r LFBFL_var_2=$(
179     printf -- "%s" "$2"\
180     | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
181   )
182   local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
```

```
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php --run="${LFBFL_var_3}"
  return $?
}

is_subfile(){
  # $1=file_path
  # $2=sub_file_path
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php --run="${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

3.12 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  local LFBFL_subdir2="build_and_checks_variables"
  local LFBFL_data_file_path="${LFBFL_subdir2}/repository_data.txt"
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_path}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

```

```
68    local LFBFL_repository_web_url=""
69    grep_variable "${LFBFL_data_file_path}" repository_web_url\
70      --result-variable-prefix="LFBFL_"\
71      --replace-line-returns-by=""
72
73    local LFBFL_repository_git_url=""
74    grep_variable "${LFBFL_data_file_path}" repository_git_url\
75      --result-variable-prefix="LFBFL_"\
76      --replace-line-returns-by=""
77
78    local LFBFL_abstract=""
79    grep_variable "${LFBFL_data_file_path}" abstract\
80      --result-variable-prefix="LFBFL_"
81
82    local LFBFL_acknowledgments=""
83    grep_variable "${LFBFL_data_file_path}" acknowledgments\
84      --result-variable-prefix="LFBFL_"
85
86    local LFBFL_author_email=""
87    grep_variable "${LFBFL_data_file_path}" author_email\
88      --result-variable-prefix="LFBFL_"\
89      --replace-line-returns-by=""
90
91    local LFBFL_author_full_name=""
92    grep_variable "${LFBFL_data_file_path}" author_full_name\
93      --result-variable-prefix="LFBFL_"\
94      --replace-line-returns-by=" "
95
96    local LFBFL_author_website=""
97    grep_variable "${LFBFL_data_file_path}" author_website\
98      --result-variable-prefix="LFBFL_"\
99      --replace-line-returns-by=""
100
101   declare -i LFBFL_i_max_line_length
102   grep_variable "${LFBFL_data_file_path}" max_line_length\
103     --result-variable-prefix="LFBFL_i_"
104
105   declare -r LFBFL_current_date=$(date -I"date")
106
107   declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)
108
109   declare -ir LFBFL_i_number_of_commits=$(
110     git shortlog\
111     | space_starting_lines\
112     | wc --lines
113   )
```

```
114
115   local LFBFL_redirect="/dev/stdout"
116   if [[ LFBFL_i_verbose -eq 0 ]]; then
117     LFBFL_redirect="/dev/null"
118   fi
119   {
120     code_lines_count_all "${LFBFL_common_options[@]}"
121     code_lines_count_not_empty "${LFBFL_common_options[@]}"
122     code_lines_count_empty "${LFBFL_common_options[@]}"
123   } &> "${LFBFL_redirect}"
124
125   local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
126   LFBFL_number_of_lines+=" total lines,"
127   LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
128   LFBFL_number_of_lines+=" not empty lines,"
129   LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
130   LFBFL_number_of_lines+=" empty lines."
131   readonly LFBFL_number_of_lines
132
133   declare -r LFBFL_temp_path="${LFBFL_subdir2}/temp"
134
135   # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
136   # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
137   # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
138   tree -a --gitignore\
139     -I "node_modules/"\
140     -I "__pycache__/"\
141     -I ".mypy_cache/"\
142     -I ".ruff_cache/"\
143     -I ".git/"\
144     | replace_non_ascii_spaces\
145     > "${LFBFL_temp_path}/current_tree_light.txt"
146
147   tree -a -DFh --gitignore\
148     -I "node_modules/"\
149     -I "__pycache__/"\
150     -I ".mypy_cache/"\
151     -I ".ruff_cache/"\
152     -I ".git/"\
153     | replace_non_ascii_spaces\
154     > "${LFBFL_temp_path}/current_tree.txt.temp"
155
156   local LFBFL_s_files_paths
157   # Remove line returns that are here to keep lines short.
158   LFBFL_s_files_paths=$(
159     sed --regexp-extended --null-data\
```

```
      --expression='s/(.)\\\n/\1/Mg'\
      -- "${LFBFL_subdir2}/files_names_listing.txt"\
    | grep --invert-match '^//'
  )
  readonly LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi
  readonly LFBFL_arr_files_paths

  local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
  LFBFL_temp_files_listing+="files_listing.tex.sub.temp"
  readonly LFBFL_temp_files_listing
  : > "${LFBFL_temp_files_listing}"
  local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
  LFBFL_temp_files_listing2+="files_listing.html.sub.temp"
  readonly LFBFL_temp_files_listing2
  : > "${LFBFL_temp_files_listing2}"
  # HTML <li> elements, hence "lis".
  local LFBFL_temp_files_lis
  LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.sub.temp"
  readonly LFBFL_temp_files_lis
  : > "${LFBFL_temp_files_lis}"
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_line_length} /
  declare -r LFBFL_score_command="${get_split_score_result}"
  declare -i LFBFL_i_score_command_properties
  LFBFL_i_score_command_properties=${i_get_split_score_result2}
  readonly LFBFL_i_score_command_properties
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
  declare -r LFBFL_score_command2="${get_split_score_result}"
  declare -i LFBFL_i_score_command_properties2
  LFBFL_i_score_command_properties2=${i_get_split_score_result2}
  readonly LFBFL_i_score_command_properties2
  declare -r LFBFL_suffix='%'
  declare -i LFBFL_i=0
  local LFBFL_cleaned_path1
  local LFBFL_cleaned_path2
  local LFBFL_new_lines
  local LFBFL_new_lines2
  local LFBFL_new_lines3
  local LFBFL_file_path
  local LFBFL_s_format
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
```

```
    printf -- "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
    # printf -- "Listing file for tex : %s\n" "${LFBFL_file_path}"
    # LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
    LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
    LFBFL_cleaned_path2=$(
      printf -- "%s" "${LFBFL_file_path}"\
      | sed --expression='s/\//:/g;s/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        ${LFBFL_i_score_command_properties2}
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_path}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    LFBFL_s_format="\\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
    LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
    LFBFL_s_format+="{\n%s\n}\n\n\n"
    # shellcheck disable=SC2059
```

```
    printf -- "${LFBFL_s_format}"\
      "${LFBFL_new_lines}"\
      "${LFBFL_new_lines2}"\
      "${LFBFL_new_lines3}"\
      >> "${LFBFL_temp_files_listing}"

    # printf -- "Listing file for HTML : %s\n" "${LFBFL_file_path}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_path}"
    if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_i_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      LFBFL_s_format="<h3 id=\"subsection3.%s\">3.%s\n%s\n</h3>\n"
      LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
      # shellcheck disable=SC2059
      printf -- "${LFBFL_s_format}"\
        "${LFBFL_i}"\
        "${LFBFL_i}"\
        "${LFBFL_new_lines}"
      sed --expression='s/&/\&amp;/g;s/</\&lt;/g'\
        < "${LFBFL_file_path}"
      printf "</pre>\n\n\n"
    } >> "${LFBFL_temp_files_listing2}"
    LFBFL_s_format="      <li><a href=\"#subsection3.%s\">\n%s\n"
    LFBFL_s_format+="      </a></li>\n"
    # shellcheck disable=SC2059
    printf -- "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
      >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.tex.sub"\
    "${LFBFL_temp_files_listing}"
  # Example of generated content of files_listing.tex.sub:
  # ----------------------------------------------------------------------
  # \subsection{
  #   build\_and\_checks\_dependencies/licenses\_templates/%
  # build\_licenses\_templates.exec.sh
  # }
  # \label{
```

```
#   build_and_checks_dependencies:licenses_templates:%
# build_licenses_templatesexecsh
# }
#
# \VerbatimInput[numbers=left,xleftmargin=-5mm]{
# build_and_checks_dependencies/licenses_templates/%
# build_licenses_templates.exec.sh
# }
# ----------------------------------------------------------------------
sed --in-place --expression="s|\\n</pre>|</pre>|g"\
  -- "${LFBFL_temp_files_listing2}"
overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.html.sub"\
  "${LFBFL_temp_files_listing2}"
# Example of generated content of files_listing.html.sub:
# ----------------------------------------------------------------------
# <h3 id="subsection3.15">3.15
# build_and_checks_dependencies/licenses_templates/<!--
# -->build_licenses_templates.exec.sh
# </h3>
# ...
# <h3 id="subsection3.33">3.33
# build_and_checks_variables/.gitignore
# </h3>
# <pre class="numbered_lines">
# temp/
# upgrade_venvs_ts
# </pre>
# ----------------------------------------------------------------------
overwrite_if_not_equal "${LFBFL_temp_path}/files_lis.html.sub"\
  "${LFBFL_temp_files_lis}"
# Example of generated content of files_lis.html.sub:
# ----------------------------------------------------------------------
#       <li><a href="#subsection3.15">
# build_and_checks_dependencies/licenses_templates/<!--
# -->build_licenses_templates.exec.sh
#       </a></li>
# ----------------------------------------------------------------------

# We verify if some lines are beyond max_line_length characters
# in current_tree_light.txt and current_tree.txt.
declare -ar LFBFL_trees=(
  "current_tree_light.txt"
  "current_tree.txt.temp"
)
local LFBFL_tree_path
local LFBFL_line
```

```
344   local LFBFL_prefix
345   declare -i LFBFL_i_prefix_last_position
346   local LFBFL_char
347   local LFBFL_line_start
348   declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
349   local LFBFL_file_name
350   local LFBFL_s_lines
351   declare -a LFBFL_arr_lines
352   declare -i LFBFL_i_keep_length
353   for LFBFL_tree in "${LFBFL_trees[@]}"; do
354     LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
355     # shellcheck disable=SC2248
356     LFBFL_s_lines=$(
357       grep -- '.\{'${LFBFL_i_overlength}'\}' "${LFBFL_tree_path}"
358     )
359     if [[ -z "${LFBFL_s_lines}" ]]; then
360       continue
361     fi
362     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
363     for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
364       # printf -- "LFBFL_line: %s\n" "${LFBFL_line}"
365       LFBFL_prefix=$(
366         printf -- "%s" "${LFBFL_line}"\
367         | sed --regexp-extended --expression='s/(.*)——[^—]+$/\1/g'
368       )
369       # printf -- "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
370       LFBFL_i_prefix_last_position=$((${#LFBFL_prefix}-1))
371       LFBFL_char="${LFBFL_prefix:${LFBFL_i_prefix_last_position}:1}"
372       # printf "LFBFL_char: %s\n" "${LFBFL_char}"
373       LFBFL_prefix=$(
374         printf -- "%s" "${LFBFL_prefix}"\
375         | sed --regexp-extended --expression='s/[^ ]+$//g'
376       )
377       if [[ "${LFBFL_char}" == "└" ]]; then
378         LFBFL_prefix+="  "
379       else
380         LFBFL_prefix+="│ "
381       fi
382       # printf -- "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
383       LFBFL_file_name=$(
384         printf -- "%s" "${LFBFL_line}"\
385         | sed --regexp-extended --expression='s|.* ([^ ]+)$|\1|g'
386       )
387       # printf -- "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
388       LFBFL_i_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
389       LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
```

```
      LFBFL_line_start=$(
        printf -- "%s" "${LFBFL_line_start}"\
        | sed --expression='s/ *$//g'
      )
      # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_i_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      printf -- "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
        > "${LFBFL_tree_path}.some_line.temp"
      insert_file_at_token "${LFBFL_tree_path}"\
        "${LFBFL_line}"\
        "${LFBFL_tree_path}.some_line.temp"\
        ""\
        --quiet
    done
    rm --force -- "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_temp_path}/current_tree.txt.temp" 1 1

  local LFBFL_tex_path_start
  LFBFL_tex_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
  readonly LFBFL_tex_path_start
  local LFBFL_html_path_start
  LFBFL_html_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.html"
  readonly LFBFL_html_path_start

  if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]]; then
    # If there is a template, we init the process from it.
    cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]]; then
    # Otherwise if there is a tex file, we init the process from it.
    cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
```

```
436       printf -- "Neither .tex.tpl, nor .tex in ./%s/\n" "${LFBFL_subdir2}"
437     fi
438
439     # Same logic with repository HTML file.
440     if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]]; then
441       cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
442         "${LFBFL_html_path_start}.1"
443     elif [[ -f "${LFBFL_repository_name}.html" ]]; then
444       cp -- "${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
445     else
446       printf -- "Neither .html.tpl in ./%s/, nor .html in ./\n"\
447         "${LFBFL_subdir2}"
448     fi
449
450     # shellcheck disable=SC2001
451     printf -- "%s\n" "${LFBFL_abstract}"\
452       | sed --expression='s/\\n/\n/g'\
453       > "${LFBFL_temp_path}/abstract_temp"
454
455     # shellcheck disable=SC2001
456     printf -- "%s\n" "${LFBFL_acknowledgments}"\
457       | sed --expression='s/\\n/\n/g'\
458       > "${LFBFL_temp_path}/acknowledgments_temp"
459
460     # But the filling still occurs, in case the dev want to refill
461     # part of the tex/html file that he modified with @token@
462     # using some computed results.
463     # Tex filling:
464     if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
465       sed --in-place\
466         --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
467         --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
468         --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
469         --expression="s|@author_email@|${LFBFL_author_email}|g"\
470         --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
471         --expression="s|@author_website@|${LFBFL_author_website}|g"\
472         --expression="s|@current_date@|${LFBFL_current_date}|g"\
473         --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
474         --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
475         --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
476         -- "${LFBFL_tex_path_start}.1"
477
478       insert_file_at_token "${LFBFL_tex_path_start}.1"\
479         @abstract@\
480         "${LFBFL_temp_path}/abstract_temp"\
481         "${LFBFL_tex_path_start}.2"
```

```
482
483       insert_file_at_token "${LFBFL_tex_path_start}.2"\
484         @acknowledgments@\
485         "${LFBFL_temp_path}/acknowledgments_temp"\
486         "${LFBFL_tex_path_start}.3"
487
488       insert_file_at_token "${LFBFL_tex_path_start}.3"\
489         @current_tree_light@\
490         "${LFBFL_temp_path}/current_tree_light.txt"\
491         "${LFBFL_tex_path_start}.4"
492
493       insert_file_at_token "${LFBFL_tex_path_start}.4"\
494         @current_tree@\
495         "${LFBFL_temp_path}/current_tree.txt"\
496         "${LFBFL_tex_path_start}.5"
497
498       insert_file_at_token "${LFBFL_tex_path_start}.5"\
499         @dependencies_notes@\
500         "${LFBFL_temp_path}/dependencies_notes.tex.sub"\
501         "${LFBFL_tex_path_start}.6"
502
503       insert_file_at_token "${LFBFL_tex_path_start}.6"\
504         @files_listing_VerbatimInput@\
505         "${LFBFL_temp_path}/files_listing.tex.sub"\
506         "${LFBFL_tex_path_start}.7"
507
508       overwrite_if_not_equal\
509         "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
510         "${LFBFL_tex_path_start}.7" 1
511     fi
512
513     # HTML filling:
514     declare -i LFBFL_i_HTML_updated=0
515     if [[ -f "${LFBFL_html_path_start}.1" ]]; then
516       sed --in-place\
517         --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
518         --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
519         --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
520         --expression="s|@author_email@|${LFBFL_author_email}|g"\
521         --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
522         --expression="s|@author_website@|${LFBFL_author_website}|g"\
523         --expression="s|@current_date@|${LFBFL_current_date}|g"\
524         --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
525         --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
526         --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
527         -- "${LFBFL_html_path_start}.1"
```

```
    insert_file_at_token "${LFBFL_html_path_start}.1"\
      @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_html_path_start}.2"

    insert_file_at_token "${LFBFL_html_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_html_path_start}.3"

    insert_file_at_token "${LFBFL_html_path_start}.3"\
      @files_lis@\
      "${LFBFL_temp_path}/files_lis.html.sub"\
      "${LFBFL_html_path_start}.4"

    insert_file_at_token "${LFBFL_html_path_start}.4"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_html_path_start}.5"

    insert_file_at_token "${LFBFL_html_path_start}.5"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_html_path_start}.6"

    insert_file_at_token "${LFBFL_html_path_start}.6"\
      @dependencies_notes@\
      "${LFBFL_temp_path}/dependencies_notes.html.sub"\
      "${LFBFL_html_path_start}.7"

    insert_file_at_token "${LFBFL_html_path_start}.7"\
      @files_listing_HTMLPreInput@\
      "${LFBFL_temp_path}/files_listing.html.sub"\
      "${LFBFL_html_path_start}.8"

    overwrite_if_not_equal "${LFBFL_repository_name}.html"\
      "${LFBFL_html_path_start}.8" 1
    LFBFL_i_HTML_updated=$?
  fi

  # The HTML contains everything including listings with source code.
  # If it doesn't get updated,
  # then it is useless to recompute the PDF.
  if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
    return
```

```
  fi

  local LFBFL_latex_command=""
  grep_variable "${LFBFL_data_file_path}" latex_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  case ${LFBFL_latex_command} in
    pdflatex) : ;;
    lualatex) : ;;
    *) printf "Unsupported latex command" ; return 1 ;;
  esac

  local LFBFL_command
  LFBFL_command="${LFBFL_latex_command}"
  LFBFL_command+=" '${LFBFL_subdir2}/${LFBFL_repository_name}.tex'"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_command+=" > /dev/null"
  fi
  readonly LFBFL_command

  for ((i=0; i<3; i++)); do
    eval -- "${LFBFL_command}"
  done

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv -- "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

3.13 build_and_checks_dependencies/generate_from_template.libr.sh

```
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```
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    printf -- "%s\n" "$3"
    cat -- "$1"
    printf -- "%s\n" "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=line_comment_prefix
```

```
53    # $4=optional_post_processing
54    # $5=file_prefix
55    local LFBFL_temp
56    LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
57    readonly LFBFL_temp
58    if [[ -n "$5" ]]; then
59      printf -- "%s\n" "$5"\
60        > "${LFBFL_temp}"
61    fi
62    sed --expression="s/^/$3/g" -- "$1"\
63      >> "${LFBFL_temp}"
64    if [[ -n "$4" ]]; then
65      eval -- "$4"
66    fi
67    overwrite_if_not_equal "$2" "${LFBFL_temp}"
68  }
69
70  split_file_in_two(){
71    # $1=file_path
72    # $2=token assume that token is the only thing on his line,
73    #    except for a possible indentation.
74    # $3=path_of_part1_of_file
75    # $4=path_of_part2_of_file
76    # We cannot add --line-regexp with --fixed-strings because of the
77    # possible indentation in file.
78    declare -ir LFBFL_i_line_number=$(
79      grep_fixed_string_with_anchor "$1" "$2"\
80        --enforce-line-ends-with-fixed-string --fixed-strings --line-number\
81      | head --lines=1\
82      | cut --delimiter=':' --fields=1
83    )
84    declare -ir LFBFL_i_line_count=$(ll_wc -l -n "$1")
85    declare -ir LFBFL_i_lines_after=$((
86      LFBFL_i_line_count - LFBFL_i_line_number
87    ))
88    head --lines="$((LFBFL_i_line_number - 1))" -- "$1"\
89      > "$3"
90    tail --lines="${LFBFL_i_lines_after}" -- "$1"\
91      > "$4"
92  }
93
94  insert_file_at_token(){
95    # $1=file_path
96    # $2=token assume that token is the only thing on his line,
97    #    except for a possible indentation.
98    # $3=path_of_file_to_insert
```

```
  # $4=path_of_result_file
  # Options:
  #   $5=--quiet to suppress diff output in overwrite_if_not_equal
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
    "${LFBFL_end_file_name}"
  cat -- "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm -- "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}" "" "" "$5"
  fi
}
```

3.14 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
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#
```

```
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.sub'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.xml'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

```

```
  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.sub'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.xml'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

3.15 build_and_checks_dependencies/grammar_and_spelling_check.libr.sh

```
#!/usr/bin/env bash
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2026-03-22T15:30+01:00: This file was renamed from
# "grammar_and_spell_check.libr.sh"
# to
# "grammar_and_spelling_check.libr.sh".
# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spelling_check(){
  # $1=configuration_path build_and_checks_variables/repository_data.txt
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_grammar_and_spelling_checker_command=""
  grep_variable "$1" grammar_and_spelling_checker_command\
    --result-variable-prefix=LFBFL_\
```

```
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_and_spelling_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: No command in configuration.\n"
    fi
    return
  fi
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf -- "grammar_and_spelling_check: Command found: %s.\n"\
      "${LFBFL_command}"
  fi
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf -- "grammar_and_spelling_check: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "grammar_and_spelling_check: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval -- "${LFBFL_eval_string}"
    done
  done
}
```

3.16 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
```

```
52
53    enhanced_set_shell_option pipefail --trap-unset
54
55    local LFBFL_s_format
56
57    # Source
58    local LFBFL_license_source_subdir
59    LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
60    readonly LFBFL_license_source_subdir
61    local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
62    LFBFL_source_license_prefix+="license_file_header_"
63    readonly LFBFL_source_license_prefix
64    # Target
65    local LFBFL_license_target_subdir
66    LFBFL_license_target_subdir="${LFBFL_work_directory}/"
67    LFBFL_license_target_subdir="build_and_checks_variables/temp/"
68    readonly LFBFL_license_target_subdir
69    local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
70    LFBFL_target_license_prefix+="license_file_header_"
71    readonly LFBFL_target_license_prefix
72
73    declare -ar LFBFL_licenses=(
74      "AGPLv3+"
75      "GPLv3+"
76      "LGPLv3+"
77    )
78
79    # ------------------------------------------------------------------
80    # Various language definitions for text file formats
81    declare -ar LFBFL_block_comment_languages=(
82      "c"
83      "py"
84    )
85    declare -ar LFBFL_block_comment_enters=(
86      '/*'
87      '"""'
88    )
89    declare -ar LFBFL_block_comment_exits=(
90      '*/'
91      '"""'
92    )
93    equal "${#LFBFL_block_comment_languages[@]}"\
94          "${#LFBFL_block_comment_enters[@]}"\
95          "${#LFBFL_block_comment_exits[@]}"
96    declare -i LFBFL_i_equal_result=$?
97    if [[ LFBFL_i_equal_result -eq 0 ]]; then
```

```
98      printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
99      return 1
100   fi
101
102   declare -ar LFBFL_line_comment_languages=(
103     "sh"
104     "sql"
105     "tex"
106   )
107   declare -ar LFBFL_line_comment_prefixes=(
108     '# '
109     '-- '
110     '% '
111   )
112   equal "${#LFBFL_line_comment_languages[@]}"\
113         "${#LFBFL_line_comment_prefixes[@]}"
114   LFBFL_i_equal_result=$?
115   if [[ LFBFL_i_equal_result -eq 0 ]]; then
116     printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
117     return 1
118   fi
119   # ------------------------------------------------------------------
120
121   # ------------------------------------------------------------------
122   # We generate the license header templates adapted to various text
123   # files formats.
124   # Kamoulox do endfor, Jacques Beauregard XD
125   # (C'est toujours mieux que l'inverse :) !)
126   # for do endfor -> for do done (Fort Doux Donne).
127   local LFBFL_some_license
128   local LFBFL_license_prefix
129   local LFBFL_license_prefix2
130   local LFBFL_extension
131   local LFBFL_license_file_path
132   local LFBFL_enter_string
133   local LFBFL_exit_string
134   local LFBFL_temp2
135   local LFBFL_prefix_string
136   local LFBFL_intermediate_file_path
137   local LFBFL_file_prefix
138   declare -i LFBFL_i_generate_from_template_result
139   declare -i LFBFL_i
140   for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
141     LFBFL_license_prefix="${LFBFL_source_license_prefix}"
142     LFBFL_license_prefix+="${LFBFL_some_license}"
143     LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
```

```
144     LFBFL_license_prefix2+="${LFBFL_some_license}"
145     for ((
146       LFBFL_i = 0;
147       LFBFL_i < ${#LFBFL_block_comment_languages[@]};
148       ++LFBFL_i
149     )); do
150       LFBFL_extension=${LFBFL_block_comment_languages[LFBFL_i]}
151       LFBFL_license_file_path="${LFBFL_license_prefix2}"
152       LFBFL_license_file_path+=".${LFBFL_extension}"
153       LFBFL_enter_string=${LFBFL_block_comment_enters[LFBFL_i]}
154       LFBFL_exit_string=${LFBFL_block_comment_exits[LFBFL_i]}
155       generate_from_template_with_block_comments\
156         "${LFBFL_license_prefix}.tpl"\
157         "${LFBFL_license_file_path}.tpl"\
158         "${LFBFL_enter_string}" "${LFBFL_exit_string}"
159       LFBFL_i_generate_from_template_result=$?
160       if [[ LFBFL_i_verbose -eq 1 ]]; then
161         if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
162           printf -- "License template %s.tpl updated.\n"\
163             "${LFBFL_license_file_path}"
164         elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
165           printf -- "License template %s.tpl created.\n"\
166             "${LFBFL_license_file_path}"
167         fi
168       fi
169     done
170     LFBFL_temp2=".generate_from_template_with_line_comments.temp"
171     for ((
172       LFBFL_i = 0;
173       LFBFL_i < ${#LFBFL_line_comment_languages[@]};
174       ++LFBFL_i
175     )); do
176       LFBFL_extension=${LFBFL_line_comment_languages[LFBFL_i]}
177       LFBFL_license_file_path="${LFBFL_license_prefix2}"
178       LFBFL_license_file_path+=".${LFBFL_extension}"
179       LFBFL_prefix_string=${LFBFL_line_comment_prefixes[LFBFL_i]}
180       LFBFL_intermediate_file_path="${LFBFL_license_file_path}.tpl"
181       LFBFL_intermediate_file_path+="${LFBFL_temp2}"
182       LFBFL_file_prefix=""
183       if [[ "${LFBFL_extension}" == "sh" ]]; then
184         LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
185       fi
186       # The sed command is here to remove trailing spaces of previously
187       # empty lines after comment prefix is added to the line.
188       generate_from_template_with_line_comments\
189         "${LFBFL_license_prefix}.tpl"\
```

```
        "${LFBFL_license_file_path}.tpl"\
        "${LFBFL_prefix_string}"\
        "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_path}'"\
        "${LFBFL_file_prefix}"
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf -- "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_path}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
          printf -- "License template %s.tpl created.\n"\
            "${LFBFL_license_file_path}"
        fi
      fi
    done
  done
  # ------------------------------------------------------------------

  # Now that base license templates generated languages license
  # templates in work directory, we can go there.
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_path="build_and_checks_variables/"
  LFBFL_data_file_path+="repository_data.txt"
  readonly LFBFL_data_file_path
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_path}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  local LFBFL_license=""
  grep_variable "${LFBFL_data_file_path}" license\
    --result-variable-prefix="LFBFL_"
  local LFBFL_license2=""
  grep_variable "${LFBFL_data_file_path}" license2\
    --result-variable-prefix="LFBFL_"
  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_path}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "

  LFBFL_license_prefix="build_and_checks_variables/temp/"
  LFBFL_license_prefix+="license_file_header_"
  readonly LFBFL_license_prefix
  LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
  readonly LFBFL_license_prefix2
```

```
236   declare -r\
237     LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
238
239   # First year according to current state of git repository.
240   declare -ir LFBFL_i_first_year=$(
241     git log\
242     | grep 'Date:'\
243     | tail --lines=1\
244     | cut --delimiter=' ' --fields=8
245   )
246   # Last year according to current state of git repository.
247   declare -ir LFBFL_i_last_year=$(
248     git log\
249     | grep 'Date:'\
250     | head --lines=1\
251     | cut --delimiter=' ' --fields=8
252   )
253   local LFBFL_copyright_string
254   LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
255   LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
256   readonly LFBFL_copyright_string
257
258   prepare_filled_license_file_block(){
259     # $1=license_file_path
260     sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
261         --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
262         -- "$1.tpl"\
263       > "$1.temp"
264     overwrite_if_not_equal "$1" "$1.temp"
265     declare -ir LFBFL_i_result=$?
266     if [[ LFBFL_i_verbose -eq 1 ]]; then
267       if [[ LFBFL_i_result -eq 1 ]]; then
268         printf -- "License template %s updated.\n" "$1"
269       elif [[ LFBFL_i_result -eq 2 ]]; then
270         printf -- "License template %s created.\n" "$1"
271       fi
272     fi
273     head --lines=-1 -- "$1"\
274       | tail --lines=+2\
275       > "$1.temp"
276     # Strange world: tail --lines=-1 was working
277     # to exclude first line.
278     # I tested it months ago, and saw with my eyes the start and end
279     # block comments removed with:
280     # head --lines=-1 "$1" | tail --lines=-1.
281     # Now, man page says to use --lines=+2 instead.
```

```
    # Welcome in the world where AI change your environment and your
    # documentation on the fly...
  }

  declare -Ar LFBFL_all_block_comment_languages=(
    ["c"]="c"
    ["php"]="c"
    ["py"]="py"
    ["js"]="c"
  )

  local LFBFL_key
  local LFBFL_dest
  local LFBFL_license_file_path2
  declare -i LFBFL_i_not_subfile
  declare -i LFBFL_i_not_subfile2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_block "${LFBFL_license_file_path}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_path2="${LFBFL_license_prefix3}"
      LFBFL_license_file_path2+=".${LFBFL_dest}"
      prepare_filled_license_file_block "${LFBFL_license_file_path2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
        -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        LFBFL_s_format="build_licenses_templates: "
        LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
        # shellcheck disable=SC2059
        printf -- "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
      fi
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}"\
```

```
        "${LFBFL_license_file_path}.temp"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}"\
          "${LFBFL_license_file_path2}.temp"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf -- "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
    rm -- "${LFBFL_license_file_path}.temp"
    if [[ -n "${LFBFL_license2}" ]]; then
      rm -- "${LFBFL_license_file_path2}.temp"
    fi
  done

  prepare_filled_license_file_line(){
    # $1=LFBFL_license_file_path
    sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
        --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
        -- "$1.tpl"\
      > "$1.temp"
    overwrite_if_not_equal "$1" "$1.temp"
    declare -ir LFBFL_i_result=$?
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ LFBFL_i_result -eq 1 ]]; then
        printf -- "License template %s updated.\n" "$1"
      elif [[ LFBFL_i_result -eq 2 ]]; then
        printf -- "License template %s created.\n" "$1"
      fi
    fi
  }

  declare -Ar LFBFL_all_line_comment_languages=(
    ["sh"]="sh"
    ["sql"]="sql"
    ["tex"]="tex"
  )

  for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
```

```
    prepare_filled_license_file_line "${LFBFL_license_file_path}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_path2="${LFBFL_license_prefix3}"
      LFBFL_license_file_path2+=".${LFBFL_dest}"
      prepare_filled_license_file_line "${LFBFL_license_file_path2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
        -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        LFBFL_s_format="build_licenses_templates: "
        LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
        # shellcheck disable=SC2059
        printf -- "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
      fi
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path}"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path2}"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf -- "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
  done
}

build_licenses_templates "$@"
```

3.17 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```
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```

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```

3.18 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

3.19 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

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```

3.20 build_and_checks_dependencies/lines_counts.libr.sh

```
#!/usr/bin/env bash
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# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
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```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf -- "all_code_lines: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep --invert-match --perl-regexp --with-filename\
        -- 'a(?!a)a'
  done
}

all_self_code_lines(){
  # Options:
```

```
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  # below and another lines, see SC2145 false positive
  all_code_lines "${LFBFL_common_options[@]}" \
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  all_self_code_lines "${LFBFL_common_options[@]}" \
```

```
112     | empty_lines_after_file_name
113 }
114
115 # shellcheck disable=SC2120
116 all_self_not_empty_code_lines(){
117   # Options:
118   #   --verbose
119   #   --work-directory=""
120   declare -i LFBFL_i_verbose=0
121   get_verbose_option "$@"
122   local LFBFL_work_directory=""
123   get_work_directory_option "$@"
124   declare -a LFBFL_return_traps_stack
125   local LFBFL_previous_return_trap
126   init_return_trap
127   pushd_to_work_directory --trap-popd
128   can_continue_after_enhanced_pushd || return 1
129
130   enhanced_set_shell_option pipefail --trap-unset
131
132   declare -a LFBFL_common_options=("$@")
133   keep_options LFBFL_common_options --verbose
134
135   all_self_code_lines "${LFBFL_common_options[@]}" \
136     | not_empty_lines_after_file_name
137 }
138
139 code_lines_count_all(){
140   # Options:
141   #   --verbose
142   #   --work-directory=""
143   declare -i LFBFL_i_verbose=0
144   get_verbose_option "$@"
145   local LFBFL_work_directory=""
146   get_work_directory_option "$@"
147   declare -a LFBFL_return_traps_stack
148   local LFBFL_previous_return_trap
149   init_return_trap
150   pushd_to_work_directory --trap-popd
151   can_continue_after_enhanced_pushd || return 1
152
153   enhanced_set_shell_option pipefail --trap-unset
154
155   declare -gi i_code_lines_count_all_result
156   i_code_lines_count_all_result=$(
157     all_self_code_lines\
```

```
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_all_result}
}

code_lines_count_empty(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_empty_result=$(
    all_self_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_empty_result}
}

code_lines_count_not_empty(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
```

```
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_not_empty_result}
}
```

3.21 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

```

```
# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_i_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # printf "%s\n" "${ll_wc_var_arg}"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # printf "%s\n" "${ll_wc_var_i_number_only}"
```

```
  if [[ ll_wc_var_i_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat -- "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # printf "normal\n"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed --expression='s|domain|lists.gnu.org|' \
#       --expression='s|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  # Same arguments and options as grep.
```

```
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
  # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
  # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # printf "%s\n" "${LFBFL_i_lines_before}"
  # printf "%s\n" "${LFBFL_i_lines_after}"
  # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
  # Almost as fast (on large files) and correct in all cases :).
  # Thanks Paul Eggert for the suggestion :).
  # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
  if cmp --silent -- "${LFBFL_temp}" "${!#}"; then
    rm -- "${LFBFL_temp}"
    return
  fi
  mv -- "${LFBFL_temp}" "${!#}"
}

grep_fixed_string_with_anchor(){
  # $1=file_path
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below,
  #   you don't need to supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
  declare -i LFBFL_i_enforce_line_starts=0
  declare -i LFBFL_i_enforce_line_ends=0
  declare -i LFBFL_i_quiet=0
  local LFBFL_argument
  for LFBFL_argument in "$@"; do
    ((++LFBFL_i))
    if [[ LFBFL_i -lt 3 ]]; then
      continue
    fi
    if [[
      "${LFBFL_argument}" == "--enforce-line-starts-with-fixed-string"
    ]]; then
      LFBFL_i_enforce_line_starts=1
      continue
```

```
165       fi
166       if [[
167         "${LFBFL_argument}" == "--enforce-line-ends-with-fixed-string"
168       ]]; then
169         LFBFL_i_enforce_line_ends=1
170         continue
171       fi
172       if [[
173         "${LFBFL_argument}" == "-q"
174         || "${LFBFL_argument}" == "--quiet"
175       ]]; then
176         LFBFL_i_quiet=1
177         LFBFL_grep_options+=("${LFBFL_argument}")
178         continue
179       fi
180       LFBFL_grep_options2+=("${LFBFL_argument}")
181     done
182
183     if [[
184       LFBFL_i_enforce_line_starts -eq 0
185       && LFBFL_i_enforce_line_ends -eq 0
186     ]]; then
187       grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
188       return
189     fi
190     declare -r LFBFL_s_lines=$(
191       grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
192         "$2" "$1"
193     )
194     declare -i LFBFL_i_result=1
195     if [[ -z "${LFBFL_s_lines}" ]]; then
196       # shellcheck disable=SC2248
197       return ${LFBFL_i_result}
198     fi
199     declare -a LFBFL_arr_lines
200     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
201     readonly LFBFL_arr_lines
202
203     declare -i LFBFL_i_fixed_string_length=${#2}
204     local LFBFL_some_line
205     local LFBFL_line_part
206     for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
207       if [[ "${LFBFL_some_line}" == "$2" ]]; then
208         LFBFL_i_result=0
209         if [[ LFBFL_i_quiet -eq 1 ]]; then
210           break
```

```
211       else
212         printf -- "%s\n" "${LFBFL_some_line}"
213       fi
214       continue
215     fi
216     if [[
217       LFBFL_i_enforce_line_starts -eq 1
218       && LFBFL_i_enforce_line_ends -eq 1
219     ]]; then
220       continue
221     fi
222     if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
223       LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
224     fi
225     if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
226       LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
227     fi
228     if [[ "${LFBFL_line_part}" == "$2" ]]; then
229       LFBFL_i_result=0
230       if [[ LFBFL_i_quiet -eq 1 ]]; then
231         break
232       else
233         printf -- "%s\n" "${LFBFL_some_line}"
234       fi
235     fi
236   done
237   LFBFL_i_result=$((LFBFL_i_result | $?))
238   # shellcheck disable=SC2248
239   return ${LFBFL_i_result}
240 }
241
242 grep_variable(){
243   # $1=file_path
244   # $2=variable_name
245   # Options :
246   #   --replace-line-returns-by=""
247   #   --result-variable-prefix="LFBFL_" for example
248   declare -r LFBFL_regexp="(?<=^$2=).*$"
249   # printf "%s\n" "${LFBFL_regexp}"
250   local LFBFL_variable_value
251   local LFBFL_s_replace
252   get_some_option LFBFL_s_replace --replace-line-returns-by $'\n' 1 "$@"
253   if [[ "${LFBFL_s_replace}" != $'\n' ]]; then
254     # sed -Ez -e...
255     LFBFL_variable_value=$(
256       grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"\
```

```
      | sed --null-data --regexp-extended\
        --expression='s/\n/'"${LFBFL_s_replace}"'/Mg'
    )
    # We remove the trailing replace string,
    # since the intent is only to replace the intermediate \n.
    if [[ -n "${LFBFL_s_replace}" ]]; then
    LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
    fi
  else
    LFBFL_variable_value=$(
      grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"
    )
  fi
  local LFBFL_prefix=""
  get_some_option LFBFL_prefix --result-variable-prefix '' 1 "$@"
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # printf "%s : %s\n " "${LFBFL_variable_name}" "${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # printf "%s\n" "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep --invert-match '^$'
}

space_starting_lines(){
  grep '^ '
}

not_space_starting_lines(){
  grep --invert-match '^ '
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep --extended-regexp '^( |$)'
}

```

```
not_spampty_lines(){
  grep '^[^ ]'
  # grep --extended-regexp --invert-match '^( |$)'
}

all_space_lines(){
  grep '^[ ]\+$'
}

not_all_space_lines(){
  grep --invert-match '^[ ]\+$'
}

# All space or empty lines = aspampty
aspampty_lines(){
  grep '^[ ]*$'
}

not_aspampty_lines(){
  grep '[^ ]'
  # grep --invert-match '^[ ]*$'
}

empty_lines_after_file_name(){
  grep '^[^:]\+:$'
}

not_empty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:$'
}

space_starting_lines_after_file_name(){
  grep '^[^:]\+: '
}

not_space_starting_lines_after_file_name(){
  grep --invert-match '^[^:]\+: '
}

spampty_lines_after_file_name(){
  grep --extended-regexp '^[^:]+:( |$)'
}

not_spampty_lines_after_file_name(){
  # The two variants results differ on lines without filename before.
  # Since the intent of the function is only on lines "blabla.sh:whatever",
```

```
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep --extended-regexp --invert-match '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}

not_all_space_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]\+$'
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# -E = --show-ends :). Nice command :).

not_JS_dependencies_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)node_modules/"\
    --regexp="(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?node_modules/"\
    --regexp="^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
```

```
395   not_JS_dependencies_grep
396 }
397
398 not_php_cache_find(){
399   grep --extended-regexp --invert-match\
400     --regexp="(^|/)\.phpdoc/cache/"
401 }
402
403 not_php_cache_grep(){
404   grep --extended-regexp --invert-match\
405     --regexp="^([^:]+/)?\.phpdoc/cache/"
406 }
407
408 not_python_cache_find(){
409   grep --extended-regexp --invert-match\
410     --regexp="(^|/)__pycache__/"\
411     --regexp="(^|/)\.mypy_cache/"\
412     --regexp="(^|/)\.ruff_cache/"
413 }
414
415 not_python_cache_grep(){
416   grep --extended-regexp --invert-match\
417     --regexp="^([^:]+/)?__pycache__/"\
418     --regexp="^([^:]+/)?\.mypy_cache/"\
419     --regexp="^([^:]+/)?\.ruff_cache/"
420 }
421
422 not_cache_find(){
423   not_php_cache_find\
424     | not_python_cache_find
425 }
426
427 not_cache_grep(){
428   not_php_cache_grep\
429     | not_python_cache_grep
430 }
431
432 not_git_find(){
433   grep --extended-regexp --invert-match "(^|/)\.git/"
434 }
435
436 not_git_grep(){
437   grep --extended-regexp --invert-match "^([^:]+/)?\.git/"
438 }
439
440 not_archive_find(){
```

```
  grep --extended-regexp --invert-match "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)COPYING$"\
    --regexp="(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?COPYING:"\
    --regexp="^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
```

```
487   grep --extended-regexp --invert-match\
488     --regexp="(^|/)${LFBFL_repository_name}\.html$"
489 }
490
491 not_main_html_grep(){
492   local LFBFL_repository_name=""
493   grep_variable build_and_checks_variables/repository_data.txt\
494     repository_name\
495     --result-variable-prefix="LFBFL_"\
496     --replace-line-returns-by=""
497   grep --extended-regexp --invert-match\
498     --regexp="^([^:]+/)?${LFBFL_repository_name}\.html:"
499 }
500
501 not_temp_file_find(){
502   grep --extended-regexp --invert-match\
503     --regexp="(^|/)build_and_checks_variables/temp/"
504 }
505
506 not_temp_file_grep(){
507   grep --extended-regexp --invert-match\
508     --regexp="^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
509 }
510
511 not_unchecked_builds_find(){
512   grep --extended-regexp --invert-match\
513     --regexp="(^|/)\.phpdoc/build/"
514 }
515
516 not_unchecked_builds_grep(){
517   grep --extended-regexp --invert-match\
518     --regexp="^([^:]+/)?\.phpdoc/build/"
519 }
520
521 relevant_find(){
522   enhanced_set_shell_option pipefail\
523     && trap 'enhanced_unset_shell_option pipefail' RETURN
524   not_dependencies_find\
525     | not_cache_find\
526     | not_git_find\
527     | not_archive_find\
528     | not_temp_file_find\
529     | not_unchecked_builds_find
530 }
531
532 relevant_grep(){
```

```
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep\
    | not_unchecked_builds_grep
}
```

3.22 build_and_checks_dependencies/lines_maps.libr.sh

```
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# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
```

```
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed\
    --expression='s/\xc2\xa0/ /g'\
    --expression='s/\xe2\x80\x80/ /g'\
    --expression='s/\xe2\x80\x81/ /g'\
    --expression='s/\xe2\x80\x82/ /g'\
    --expression='s/\xe2\x80\x83/ /g'\
    --expression='s/\xe2\x80\x84/ /g'\
    --expression='s/\xe2\x80\x85/ /g'\
    --expression='s/\xe2\x80\x86/ /g'\
    --expression='s/\xe2\x80\x87/ /g'\
    --expression='s/\xe2\x80\x88/ /g'\
    --expression='s/\xe2\x80\x89/ /g'\
    --expression='s/\xe2\x80\x8a/ /g'\
    --expression='s/\xe2\x80\x8b/ /g'\
    --expression='s/\xe2\x80\x8c/ /g'\
    --expression='s/\xe2\x80\x8d/ /g'\
    --expression='s/\xe2\x80\x8e/ /g'\
    --expression='s/\xe2\x80\x8f/ /g'\
    --expression='s/\xe2\x80\xa8/ /g'\
    --expression='s/\xe2\x80\xa9/ /g'\
    --expression='s/\xe2\x80\xaa/ /g'\
    --expression='s/\xe2\x80\xab/ /g'\
    --expression='s/\xe2\x80\xac/ /g'\
    --expression='s/\xe2\x80\xad/ /g'\
    --expression='s/\xe2\x80\xae/ /g'\
    --expression='s/\xe2\x80\xaf/ /g'\
    --expression='s/\xe2\x81\x9f/ /g'\
    --expression='s/\xe2\x81\xa0/ /g'\
    --expression='s/\xe2\x81\xa1/ /g'\
    --expression='s/\xe2\x81\xa2/ /g'\
    --expression='s/\xe2\x81\xa3/ /g'\
    --expression='s/\xe2\x81\xa4/ /g'\
    --expression='s/\xe2\x81\xa5/ /g'\
    --expression='s/\xe2\x81\xa6/ /g'\
    --expression='s/\xe2\x81\xa7/ /g'\
    --expression='s/\xe2\x81\xa8/ /g'\
    --expression='s/\xe2\x81\xa9/ /g'\
    --expression='s/\xe2\x81\xaa/ /g'\
    --expression='s/\xe2\x81\xab/ /g'\
    --expression='s/\xe2\x81\xac/ /g'\
    --expression='s/\xe2\x81\xad/ /g'\
    --expression='s/\xe2\x81\xae/ /g'\
    --expression='s/\xe2\x81\xaf/ /g'
```

```
}
```

3.23 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
  # $1=target_file_path
  # $2=temp_file_path
  # $3=keep_temp
  # $4=tree_mode : when the two files are the output of tree command,
  # we ignore if all the differences are only on directories timestamps.
  # $5=--quiet : we even don't want the message of diff saying that
  #              files are distinct
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  if [[ ! -f "$1" ]]; then
    if [[ -n "$3" ]]; then
      cp --preserve=mode,ownership,timestamps -- "$2" "$1"
    else
      mv -- "$2" "$1"
```

```
43      fi
44      return 2
45    fi
46    declare -i LFBFL_i_is_equal
47    if [[ -n "$4" ]]; then
48      # Not setting pipefail since the result would still be incorrect.
49      # shellcheck disable=SC2312
50      diff -- "$1" "$2"\
51        | grep --extended-regexp "^(>|<)"\
52        | grep --invert-match "/$"
53      LFBFL_i_is_equal=${PIPESTATUS[2]}
54      if [[ LFBFL_i_is_equal -eq 1 ]]; then
55        grep --only-matching --regexp="[^ ]*/$" -- "$1"\
56          > "overwrite_if_not_equal.file1.temp"
57        grep --only-matching --regexp="[^ ]*/$" -- "$2"\
58          > "overwrite_if_not_equal.file2.temp"
59        if [[ -n "$5" ]]; then
60          diff overwrite_if_not_equal.file1.temp\
61            overwrite_if_not_equal.file2.temp\
62            > /dev/null
63        else
64          diff overwrite_if_not_equal.file1.temp\
65            overwrite_if_not_equal.file2.temp
66        fi
67        LFBFL_i_is_equal=1-$?
68        rm overwrite_if_not_equal.file1.temp\
69          overwrite_if_not_equal.file2.temp
70      fi
71    else
72      if [[ -n "$5" ]]; then
73        diff --brief -- "$1" "$2" > /dev/null
74      else
75        diff --brief -- "$1" "$2"
76      fi
77      LFBFL_i_is_equal=1-$?
78    fi
79    if [[ LFBFL_i_is_equal -eq 1 ]]; then
80      if [[ -z "$3" ]]; then
81        rm -- "$2"
82      fi
83      return 0
84    fi
85    if [[ -n "$3" ]]; then
86      cp --preserve=mode,ownership,timestamps -- "$2" "$1"
87    else
88      mv -- "$2" "$1"
```

```
  fi
  return 1
}
```

3.24 build_and_checks_dependencies/python_black_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
```

```
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking empty lines after Python function docstrings\n"
  pcre2grep --multiline --\
    $'def ([^"]|"(?!"))*"""([^"]|"(?!"))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_no_empty_line_after_python_function_docstring\
    "${LFBFL_common_options[@]}"
}
```

3.25 build_and_checks_dependencies/python_isort_complement.libr.sh

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # This function should be useless in a few months:
  # https://github.com/PyCQA/isort/issues/2250
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  [[ LFBFL_i_verbose -eq 1 ]]\
```

```
52      && printf "Checking import of _collections_abc is at the right place\n"
53
54    enhanced_set_shell_option pipefail --trap-unset
55
56    declare -r LFBFL_temp=".check_collections_abc_place.temp"
57    local LFBFL_file_path
58    declare -r LFBFL_s_files_paths=$(
59      find . -type f -name "*.py" -printf '%P\n'\
60      | relevant_find
61    )
62    if [[ -z "${LFBFL_s_files_paths}" ]]; then
63      [[ LFBFL_i_verbose -eq 1 ]]\
64        && printf "check_collections_abc_place: No file found.\n"
65      return
66    fi
67    declare -a LFBFL_arr_files_paths
68    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
69    readonly LFBFL_arr_files_paths
70    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
71      sed --regexp-extended --null-data\
72        --expression='s/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
73        -- "${LFBFL_file_path}"\
74        > "${LFBFL_file_path}${LFBFL_temp}"
75      overwrite_if_not_equal "${LFBFL_file_path}"\
76        "${LFBFL_file_path}${LFBFL_temp}"
77    done
78  }
79
80  python_isort_complement(){
81    # Options:
82    #   --verbose
83    #   --work-directory=""
84    declare -i LFBFL_i_verbose=0
85    get_verbose_option "$@"
86    local LFBFL_work_directory=""
87    get_work_directory_option "$@"
88    declare -a LFBFL_return_traps_stack
89    local LFBFL_previous_return_trap
90    init_return_trap
91    pushd_to_work_directory --trap-popd
92    can_continue_after_enhanced_pushd || return 1
93
94    declare -a LFBFL_common_options=("$@")
95    keep_options LFBFL_common_options --verbose
96
97    check_collections_abc_place "${LFBFL_common_options[@]}"
```

```
}
```

3.26 build_and_checks_dependencies/repository_data.libr.sh

```
#!/usr/bin/env bash
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#
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_path
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
```

```

  enhanced_set_bash_option extglob --trap-unset

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf -- "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf -- "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}
```

3.27 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
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```

```
#
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  local LFBFL_s="Checking that shell scripts do not"
  LFBFL_s+=" contain wrong array size expansion.\n"
  # shellcheck disable=SC2059
  [[ LFBFL_i_verbose -eq 1 ]] && printf "${LFBFL_s}"

  pcre2grep --multiline -- $'\$\{#(|.*_)arr_(?!.*\[@\])' **/*.sh
}

check_no_misplaced_then(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1
```

```
68
69    declare -i LFBFL_i_max_line_length
70    get_max_line_length_option "$@"
71    declare -i LFBFL_i_max_previous_line_length
72    LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))
73
74    enhanced_set_bash_option globstar --trap-unset
75
76    [[ LFBFL_i_verbose -eq 1 ]]\
77      && printf\
78          "Checking that shell scripts do not contain misplaced 'then'.\n"
79
80    local LFBFL_regexp
81    LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
82    LFBFL_regexp+="(?<=\]\];\s{0,${LFBFL_i_max_previous_line_length}}"
83    LFBFL_regexp+="(#[^\n]{0,${LFBFL_i_max_previous_line_length}})?)"
84    LFBFL_regexp+="(?<!^\s{0,${LFBFL_i_max_previous_line_length}}\]\];)"
85    LFBFL_regexp+="\n\s*then"
86    # pcre2grep -M --
87    # '^[^\n]{1,70}(?<=\]\];\s{0,70}(#[^\n]{0,70})?)'
88    # '(?<!^\s{0,70}\]\];)\n\s*then'
89    pcre2grep --multiline -- "${LFBFL_regexp}" **/*.sh
90  }
91
92  check_no_negation_before_bash_test(){
93    # Options:
94    #   --verbose
95    #   --work-directory=""
96    declare -i LFBFL_i_verbose=0
97    get_verbose_option "$@"
98    local LFBFL_work_directory=""
99    get_work_directory_option "$@"
100   declare -a LFBFL_return_traps_stack
101   local LFBFL_previous_return_trap
102   init_return_trap
103   pushd_to_work_directory --trap-popd
104   can_continue_after_enhanced_pushd || return 1
105
106   enhanced_set_bash_option globstar --trap-unset
107
108   [[ LFBFL_i_verbose -eq 1 ]]\
109     && printf "Checking that shell scripts do not contain ! before [[.\n"
110
111   pcre2grep --multiline -- $'!\s*\[\[' **/*.sh
112 }
113
```

```
shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose --max-line-length

  check_no_size_of_array_first_element "${LFBFL_common_options[@]}"
  check_no_misplaced_then "${LFBFL_common_options[@]}"
  check_no_negation_before_bash_test "${LFBFL_common_options[@]}"
}
```

3.28 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php

/**
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```
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".

@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once './build_and_checks_dependencies/split_score.libr.php';

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// phpcs:disable Generic.WhiteSpace.ArbitraryParenthesesSpacing
// The first cast "(int) $func..." is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1],
    (int) $argv[2],
    explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
// phpcs:enable Generic.WhiteSpace.ArbitraryParenthesesSpacing
```

```
?>
```

3.29 build_and_checks_dependencies/split_score.libr.php

```
<?php

/**
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".

@category Library
@package DevOrSysAdminScripts
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*/

declare(strict_types=1);
declare(encoding='UTF-8');

```

```
43
44
45 /**
46 Given the 3 following arguments, this function returns a closure that can
47 be used to compute the split score given the current string
48 (hopefully matching a delimiter), the current cut/split position,
49 and if that split position is after or before the current string.
50
51 @param bool          $b_larger_after
52                      Score is larger when splitting after,
53                      previous name of this parameter was $b_after_before.
54 @param int           $i_max_length
55                      The maximum length allowed (goal length).
56 @param array<string> $a_delimiter_strings_domain
57                      An array of "delimiters".
58
59 @return \Closure {
60   The "split_score" closure to use repeatedly afterward.
61
62   @param string $s_delimiter_string The string tested to be a delimiter.
63   @param int    $i_cut_position     The offset of the cut in the string.
64   @param bool   $b_is_cut_after     Is the offset taken after the string?
65
66   @return int The split score.
67 }
68 */
69 function generate_split_score(
70   bool $b_larger_after,
71   int $i_max_length,
72   array $a_delimiter_strings_domain,
73 ) : Closure {
74   return function (
75     string $s_delimiter_string,
76     int $i_cut_position,
77     bool $b_is_cut_after,
78   ) use (
79     $b_larger_after,
80     $i_max_length,
81     $a_delimiter_strings_domain,
82   ) : int {
83     if(
84       !in_array(
85         $s_delimiter_string,
86         $a_delimiter_strings_domain,
87         true,
88       )
```

```
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_larger_after){
      // When $b_larger_after,
      // always way better to cut after '/' than before it.
      // When !$b_larger_after (aka larger_before),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
}//end generate_split_score()
?>
```

3.30 build_and_checks_dependencies/strings_functions.libr.sh

```
#!/usr/bin/env bash
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#
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# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
```

```
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

split_line_at(){
  # $1=line
  # $2=position : number of characters in split_line_at_result_beginning
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # printf -- "%s\n" "$split_line_at_result_beginning"
  # printf -- "%s\n" "$split_line_at_result_end"
}

#---------------------------------------------------------------------------
#---------------------------------------------------------------------------
# Properties for the score for a possible split/cut in the string between
# two characters.
#---------------------------------------------------------------------------
# Some preliminary remarks:
# a) If the character before the split position is a forbidden previous
#   character, the score is -1.
#   (Usually, either we have no or one forbidden previous character that is
#   a backslash.)
# b) If the previous character is not forbidden,
#   and either the character before the position is among the delimiters,
#   or the character after the position is among the delimiters,
#   then the score is computed according to the delimiter found,
#   the position, and the fact that the position is after or before the
#   delimiter.
# c) If two consecutive delimiters are found in the string,
#   the maximum score between the one after the first delimiter
#   and before the second delimiter is taken for the score between the
#   two delimiters.
# d) Otherwise, by default,
#   the score for cutting/splitting at some position is 0.
# e) A substring is given by the goal length given for the resulting
#   strings/lines after split.
```

```
# f) When some scores are equal,
#   we split at the rightmost position with maximum score in the substring.
# g) Remark f) implies that below we do not need to consider strict
#    inequalities for the score.
# h) The score function SF(position) which will dictate our choice for
#   split/cut position is in reality
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   )
#   but at the cost of expliciting its internal and adding verbosity,
#   we can see that the score really depends on the following arguments
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   )
#   If it is needed,
#   then matched_delimiter_before or matched_delimiter_after is NULL.
#   Somehow previous_character can be seen as position.previous_character,
#   and next_character can be seen as position.next_character.
#   We can see that the previous or next character without
#   knowing that it matches a delimiter is irrelevant.
#   The fact that there is 1 or more delimiters may be interesting.
#   The fact that there is 0 or more forbidden previous characters
#   may be interesting, but we haven't integrated it in the split_score
#   functions.
#   SF1a when the previous character is not forbidden
#   needs a score function SF2a with the following parameters
#   SF2a(position, current_character, delimiter, is_after):
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   ) = max(
#     For all delimiter in delimiters,
#     SF2a(position, previous_character, delimiter, TRUE),
#     SF2a(position, next_character, delimiter, FALSE),
#   )
#   =  max(
#     For all delimiter in delimiters,
#     For all couple C in {
#       (position.previous_character, TRUE),
#       (position.next_character, FALSE)
#     }
#     SF2a(position, C.1, delimiter, C.2),
#   )
#   Similarly,
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
```

```
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2b(position, matched_delimiter_before, TRUE),
#     SF2b(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2b has the following parameters
#   SF2b(cut_position, matched_delimiter, is_after).
#   We have also:
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2c(position-1, matched_delimiter_before, TRUE),
#     SF2c(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2c has the following parameters
#   SF2c(character_position, matched_delimiter, is_after).
#   current_character is only used to check if it is equal to the
#   delimiter, otherwise the score is zero;
#   but when expressing properties it makes sense to compare after
#   and before the same character position.
#   position, delimiter, and is_after may all three impact the score
#   once the current character matches the delimiter.
#   Position is an integer from 1 to goal length.
#   Cutting at position 0 would be stupid,
#   and moreover there would be no previous character.
#   Since we ask for a split, the current length is above goal length,
#   and there is a next character at the last considered position.
#   So, in search of properties that may have an impact on the algorithm,
#   we discard the characters to only keep matched delimiters.
#   Let us use p1 and p2 for position1 and position2,
#   mdb1 and mdb2 for matched_delimiter_before1, etc.,
#   mda1 and mda2 for matched_delimiter_after1, etc.,
#   pcif for previous_character_is_forbidden, etc.,
#   md1 and md2 for matched_delimiter1, etc.,
#   ia1 and ia2 for is_after1, etc.
#
#   What we consider as properties are of 3 forms.
#   The first form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all mdb1 in delimiters union {NULL},
#     for all mdb2 in delimiters union {NULL},
#     for all mda1 in delimiters union {NULL},
#     for all mda2 in delimiters union {NULL},
```

```
#     RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
#     implies RelationS(
#       SF1b(p1, mdb1, mda1, false),
#       SF1b(p2, mdb2, mda2, false),
#     )
#   There are many choices for RelationP, RelationM and RelationS.
#   We can apply the without loss of generality argument only once when
#   breaking symmetry between equivalent properties.
#   Breaking symmetry on another relation that RelationS first
#   is a challenge that we do not tackle.
#   Indeed RelationS is the only relation where all variables appear.
#   And breaking symmetry on RelationP would not allow to break symmetry
#   on RelationM freely since some variables are correlated by the
#   suffixes 1 or 2.
#   But some choices for some relation may allow to break symmetry on
#   another relation.
#   Breaking symmetry on the result, there are only 2 choices for
#   RelationS:
#   RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
#   RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
#   But then there are 5+1 choices for RelationP:
#   RP1 : p1 = p2,
#   RP2 : p1 < p2,
#   RP3 : p1 <= p2,
#   RP4 : p1 > p2,
#   RP5 : p1 >= p2,
#   plus the choice that p1 and p2 are not necessarily related
#   (empty RP or RPe).
#   We can add RP6 : p1 = p2 + 1 and RP7 : p2 = p1 + 1, since it implies
#   some equalities between matched delimiters.
#   Note that  p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
#   logical relations compared to p1 < p2 or p2 < p1;
#   it only yields subslices of other properties and open a slightly silly
#   door on infinity.
#   If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
#   and maybe RP6.
#   Note that RSe or empty RS would have no interest at all.
#   There are many choices for RelationM:
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
#   interesting at all since both scores are 0 or -1.
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL
#   implies that second score is always larger, that’s not an additional
#   property, that is a theorem.
#   Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
#   must (have the possibility to) be non-NULL.
#   If we start listing the remaining relations, we have:
```

```
#   1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#   RMA1A2
#   2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 = mda2
#   RMA1A2=
#   3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 != mda2 -> This case has no meaning:
#     If we only know they are different, we cannot say anything on
#     properties related to delimiters that are different unless the score
#     doesn't depend on the delimiter, and that case is already handled by
#     1). The same remark applies to some other choices.
#     The only thing that could be said would be if we had an order on
#     delimiters scores that are uniform for the position, or the score
#     before some delimiter would always be =, >=, <=, etc. the score
#     before or after another one. So it would only make sense with RPe,
#     and mda1 <= mda2 for example, but that would trivially imply RS<=.
#     It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
#     ignores NULL value. You do not deduce or specify anything new here.
#   4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
#     AND mda2 is NULL
#   RMB1B2
#   and so on. So if we use the short notation RM followed by A1, resp. A2,
#   resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
#   is not null, we have the following possibilities:
#   RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
#   RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
#   with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
#   and _4 (XX = YY = ZZ)
#   for equalities on the 1st, 2nd or 3rd couple or between the three
#   variables by transitivity,
#   RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
#   RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
#   for one edge;
#   RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
#   RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
#   RMALL_ALL square with diagonals.
#   So we see that we listed at least 39 possibilities only for RM.
#   We did not list yet the possibilities where the variables not present
#   in some equality can be NULL or not NULL, etc.
#   We see 9 of them : RMe (40), and the following 8:
#   RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
#   RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
#   RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
```

```
#   RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
#   RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
#   RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
#   RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
#   RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
#   We don't do complete analysis at the moment.
#   If we wanted to filter out some possibilities,
#   we would keep only the RM possibilities, among the 38,
#   that are symmetric between 1 and 2:
#   RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
#   RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
#   RMALL_ALL. Still 10 possibilities.
#   We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
#   combinations (life is too short ;) :) );
#   if we keep 48 RM possibilities, there is 624 combinations,
#   but we did not analyze totally which RM relations are useless.
#
#   The second form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2b(p1, md1, ia1)
#       SF2b(p2, md2, ia2)
#     )
#   The third form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2c(p1, md1, ia1)
#       SF2c(p2, md2, ia2)
#     )
#   This time the combinatorics is much more pleasant :).
#   As above, we have RS= and RS<=, RPe, RP1 to RP7;
#   For RelationMM, assume that the possible delimiters correspond to
#   integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer
#   in [0,i], either 0 for NULL or the delimiter index.
#   Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
#   information we can use is k = l or k != l.
#   If we allow that md1 != 0 and md2 = 0,
#   then both RS= and RS<= are false.
```

```
#   Thus RMMe is not possible.
#   If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
#   whilst RS= is false.
#   Thus properties with RS<= can trivially be extended with
#   RMM-"allow-md1-0", not useful to explicit this for the moment.
#   If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
#   nothing more.
#   Thus we can focus on md1 and md2 are not NULL,
#   and are equal RMM= or not RMM!=. It seems that a property with RMM!=
#   could be extended to a property RMM=or!=/RMMNotNULL.
#   and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
#   3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
#   or FALSE. Wrong ! Not pleasant ! XD
#   So for RIA, we keep either: empty RIAe (RIA8),
#   ia1 = ia2 RIA= (RIA0 or RIA4),
#   ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
#   ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
#   ia1 != ia2 = TRUE RIA!=T2 (RIA1),
#   ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
#   ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
#   even less pleasant XD : 11 possibilities.
#   In total, we have now 1 (RS=) times 5 times 2 times 11
#   + 1 (RS<=) times 8 times 2 times 11 = 286
#   combinations (life is even shorter ;) :) ).
#   We only listed below properties of the third form.
# i) By the previous remark, there is a good chance that we missed some
#   possible property.
#   We hope we did found the properties that had a simple meaning,
#   in particular related to the algorithm where either you
#   loop on the string forward and keep last position with maximum score,
#   or you loop backward and abort as soon as possible as soon you know
#   that you found some position at the end that must have maximum score.
# j) These properties are "boolean" flags (integers that are powers of 2).
#   We first put the exact integers,
#   then wondered about switching everything to exponentiation 2**i,
#   then we took the solution that minimizes the character length between
#   [1-9][0-9]* and 2**i, then reverted to initial before committing.
#   Most efficient interpreted code is what we did first,
#   if there is no opcode cache for constants.
#   Most safe code is when the language hides the details and handles
#   families of flags,
#   but beware if some flag goes into the database.
#   Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
#   or $((2<<0))...
#   What I would love to see in many languages is a flag type, as a subtype
#   of integers, flag32, flag64 for C for example.
```

```
#   I think there is no such thing because sooner or later you play with
#   pseudo-flags that are combination of flags, hence integers.
#   But just for the declaration to let the type checker check that the
#   integer is a power of 2: flag32 MY_FLAG = 4;
#   Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
#   Another fact is that practice makes perfect, so working with the powers
#   of 2 written out in base 10 gets you accustomed to them.
#   Safe code should be favored since (pre-)compilation optimizes it,
#   but for this project, I prefer to keep the integers in decimal and
#   check that it doubles when reading my listing, because I'm familiar
#   with that and the powers of 2 doesn't go beyond what I can check fast
#   enough in my brain.
# k) Some of the properties below are union of other properties,
#   and are here to simplify not only code but understanding.
#   split_score_properties_logical_closure() function below checks that
#   all flags that are consequences of already activated flags are set.
# l) We use SSP_ prefix for SPLIT_SCORE_PROPERTY_.
# Some of these remarks are repeated below in the corresponding functions,
# and some remarks are added just after the constants.
#-----------------------------------------------------------------------
# Any score after a character that matches a delimiter
# is larger than or equal to
# any score before a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T2 RS<=
declare -gir SSP_ALWAYS_LARGER_AFTER=1

# Any score before a character that matches a delimiter
# is larger than or equal to
# any score after a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T1 RS<=
declare -gir SSP_ALWAYS_LARGER_BEFORE=2

# The score doesn't depend on the delimiter that is found.
# RPP1 RMMNotNull RIA= RS=
declare -gir SSP_DELIMITER_UNIFORM=4

# Whenever a character that matches a delimiter is found,
# the score after is equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is equal to
# the score before the character.
# RPP1 RMM= RIA!= RS=
declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
```

```
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=

# Whenever a character that matches a delimiter is found,
# the score after is larger than or equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is larger than or equal to
# the score before the character.
# RPP1 RMM= RIA!=T2 RS<=
declare -gir SSP_LARGER_AFTER=16
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=

# Whenever a character that matches a delimiter is found,
# the score before is larger than or equal to the score after.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score before the character
# is larger than or equal to
# the score after the character.
# RPP1 RMM= RIA!=T1 RS<=
declare -gir SSP_LARGER_BEFORE=32
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=

# The score after, resp. before, a character that matches a delimiter
# doesn't decrease when the position in the string increase.
# (RPP2 or RPP3) RMM= RIA= RS<=
declare -gir SSP_POSITION_NOT_DECREASING=64

# The score after a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=T RS<=
declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128

# The score before a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=F RS<=
declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256

# The score after, resp. before, a character that matches a delimiter
# doesn't depend on the position in the string.
# (RPP2 or RPP3) RMM= RIA= RS=
declare -gir SSP_POSITION_UNIFORM=512

```

```
# The score after a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=T RS=
declare -gir SSP_POSITION_UNIFORM_AFTER=1024

# The score before a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=F RS=
declare -gir SSP_POSITION_UNIFORM_BEFORE=2048

# For any delimiter,
# any score after a character that matches this delimiter
# is equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!= RS=
# <=> RPPe RMM= RIAe RS=
declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096

# For any delimiter,
# any score after a character that matches this delimiter
# is larger than or equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!=T2 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192

# For any delimiter,
# any score before a character that matches this delimiter
# is larger than or equal to
# any score after a character that matches this delimiter.
# RPPe RMM= RIA!=T1 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!= RS=
declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is larger than or equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T2 RS<=
```

```
declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score before a character at this position that matches a delimiter
# is larger than or equal to
# the score after a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T1 RS<=
declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072

# For all possible split/cut positions, the score is equal as long as
# at least one of the previous and next characters is a delimiter.
# Or equivalently:
# Any score after a character that matches a delimiter
# is equal to
# any score before a character that matches a delimiter.
# is_after doesn't change anything.
# So we cut on the last position adjacent to a delimiter without
# forbidden previous character (edge case, on last position).
# RPPe RMMNotNull RIAe RS=
declare -gir SSP_UNIFORM=262144
# declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1

#-----------------------------------------------------------------------
# m) Note that a property telling that the score is always the same
#   even if the current character isn't among delimiters is not needed,
#   since, in that case, a null split score command will split at the goal
#   length.
#-----------------------------------------------------------------------
#-----------------------------------------------------------------------

split_score_properties_logical_closure(){
  # $1=split_score_properties
  # Since this code is not performance critical,
  # we followed lexicographic order on constants,
  # instead of an order to minimize the operations.
  # It is also safer to avoid forgetting some relation.
  declare -i LFBFL_i_result=$(($1))
  declare -i LFBFL_i_result_old=0
  declare -i LFBFL_i_all_flags=$((
    SSP_ALWAYS_LARGER_AFTER
    | SSP_ALWAYS_LARGER_BEFORE
    | SSP_DELIMITER_UNIFORM
    | SSP_EQUAL_AFTER_OR_BEFORE
    | SSP_LARGER_AFTER
    | SSP_LARGER_BEFORE
```

```
532       | SSP_POSITION_NOT_DECREASING
533       | SSP_POSITION_NOT_DECREASING_AFTER
534       | SSP_POSITION_NOT_DECREASING_BEFORE
535       | SSP_POSITION_UNIFORM
536       | SSP_POSITION_UNIFORM_AFTER
537       | SSP_POSITION_UNIFORM_BEFORE
538       | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
539       | SSP_SAME_DELIMITER_LARGER_AFTER
540       | SSP_SAME_DELIMITER_LARGER_BEFORE
541       | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
542       | SSP_SAME_POSITION_LARGER_AFTER
543       | SSP_SAME_POSITION_LARGER_BEFORE
544       | SSP_UNIFORM
545     ))
546     until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
547       LFBFL_i_result_old=$((LFBFL_i_result))
548       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
549         LFBFL_i_result=$((
550           LFBFL_i_result
551           | SSP_LARGER_AFTER
552           | SSP_SAME_DELIMITER_LARGER_AFTER
553           | SSP_SAME_POSITION_LARGER_AFTER
554         ))
555       fi
556       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
557         LFBFL_i_result=$((
558           LFBFL_i_result
559           | SSP_LARGER_BEFORE
560           | SSP_SAME_DELIMITER_LARGER_BEFORE
561           | SSP_SAME_POSITION_LARGER_BEFORE
562         ))
563       fi
564       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
565         LFBFL_i_result=$((
566           LFBFL_i_result
567           | SSP_POSITION_NOT_DECREASING_AFTER
568           | SSP_POSITION_NOT_DECREASING_BEFORE
569         ))
570       fi
571       if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
572         LFBFL_i_result=$((
573           LFBFL_i_result
574           | SSP_POSITION_UNIFORM_AFTER
575           | SSP_POSITION_UNIFORM_BEFORE
576           | SSP_POSITION_NOT_DECREASING
577           | SSP_POSITION_NOT_DECREASING_AFTER
```

```
578         | SSP_POSITION_NOT_DECREASING_BEFORE
579       ))
580     fi
581     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
582       LFBFL_i_result=$((
583         LFBFL_i_result
584         | SSP_POSITION_NOT_DECREASING_AFTER
585       ))
586     fi
587     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
588       LFBFL_i_result=$((
589         LFBFL_i_result
590         | SSP_POSITION_NOT_DECREASING_BEFORE
591       ))
592     fi
593     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
594       LFBFL_i_result=$((
595         LFBFL_i_result
596         | SSP_EQUAL_AFTER_OR_BEFORE
597         | SSP_LARGER_AFTER
598         | SSP_LARGER_BEFORE
599         | SSP_SAME_DELIMITER_LARGER_AFTER
600         | SSP_SAME_DELIMITER_LARGER_BEFORE
601       ))
602     fi
603     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
604       LFBFL_i_result=$((
605         LFBFL_i_result
606         | SSP_LARGER_AFTER
607       ))
608     fi
609     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
610       LFBFL_i_result=$((
611         LFBFL_i_result
612         | SSP_LARGER_BEFORE
613       ))
614     fi
615     if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
616       LFBFL_i_result=$((
617         LFBFL_i_result
618         | SSP_DELIMITER_UNIFORM
619         | SSP_EQUAL_AFTER_OR_BEFORE
620         | SSP_LARGER_AFTER
621         | SSP_LARGER_BEFORE
622         | SSP_SAME_POSITION_LARGER_AFTER
623         | SSP_SAME_POSITION_LARGER_BEFORE
```

```
624         ))
625       fi
626       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
627         LFBFL_i_result=$((
628           LFBFL_i_result
629           | SSP_LARGER_AFTER
630         ))
631       fi
632       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
633         LFBFL_i_result=$((
634           LFBFL_i_result
635           | SSP_LARGER_BEFORE
636         ))
637       fi
638       if ((LFBFL_i_result & SSP_UNIFORM)); then
639         LFBFL_i_result=$((LFBFL_i_all_flags))
640         break
641       fi
642       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
643       && ((
644         LFBFL_i_result & (
645           SSP_ALWAYS_LARGER_BEFORE
646           | SSP_EQUAL_AFTER_OR_BEFORE
647           | SSP_LARGER_BEFORE
648           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
649           | SSP_SAME_DELIMITER_LARGER_BEFORE
650           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
651           | SSP_SAME_POSITION_LARGER_BEFORE
652         )
653       )); then
654         LFBFL_i_result=$((LFBFL_i_all_flags))
655         break
656       fi
657       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
658       && ((
659         LFBFL_i_result & (
660           SSP_EQUAL_AFTER_OR_BEFORE
661           | SSP_LARGER_AFTER
662           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
663           | SSP_SAME_DELIMITER_LARGER_AFTER
664           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
665           | SSP_SAME_POSITION_LARGER_AFTER
666         )
667       )); then
668         LFBFL_i_result=$((LFBFL_i_all_flags))
669         break
```

```
670     fi
671     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
672     && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
673       LFBFL_i_result=$((
674         LFBFL_i_result
675         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
676       ))
677     fi
678     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
679     && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
680       LFBFL_i_result=$((
681         LFBFL_i_result
682         | SSP_SAME_POSITION_LARGER_AFTER
683       ))
684     fi
685     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
686     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
687       LFBFL_i_result=$((
688         LFBFL_i_result
689         | SSP_SAME_POSITION_LARGER_BEFORE
690       ))
691     fi
692     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
693     && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
694       LFBFL_i_result=$((LFBFL_i_all_flags))
695       break
696     fi
697     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
698     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
699       LFBFL_i_result=$((
700         LFBFL_i_result
701         | SSP_ALWAYS_LARGER_AFTER
702       ))
703     fi
704     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
705     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
706       LFBFL_i_result=$((
707         LFBFL_i_result
708         | SSP_ALWAYS_LARGER_BEFORE
709       ))
710     fi
711     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
712     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
713       LFBFL_i_result=$((
714         LFBFL_i_result
715         | SSP_EQUAL_AFTER_OR_BEFORE
```

```
716         ))
717       fi
718       if ((LFBFL_i_result & SSP_LARGER_AFTER))\
719       && ((
720         LFBFL_i_result & (
721           SSP_POSITION_UNIFORM
722           | SSP_POSITION_UNIFORM_AFTER
723         )
724       )); then
725         LFBFL_i_result=$((
726           LFBFL_i_result
727           | SSP_SAME_DELIMITER_LARGER_AFTER
728         ))
729       fi
730       if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
731       && ((
732         LFBFL_i_result & (
733           SSP_POSITION_UNIFORM
734           | SSP_POSITION_UNIFORM_BEFORE
735         )
736       )); then
737         LFBFL_i_result=$((
738           LFBFL_i_result
739           | SSP_SAME_DELIMITER_LARGER_BEFORE
740         ))
741       fi
742       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
743       && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
744         LFBFL_i_result=$((
745           LFBFL_i_result
746           | SSP_POSITION_NOT_DECREASING
747         ))
748       fi
749       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
750       && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
751         LFBFL_i_result=$((
752           LFBFL_i_result
753           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
754         ))
755       fi
756       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
757       && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
758         LFBFL_i_result=$((LFBFL_i_all_flags))
759         break
760       fi
761       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
```

```
762     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
763       LFBFL_i_result=$((
764         LFBFL_i_result
765         | SSP_ALWAYS_LARGER_AFTER
766       ))
767     fi
768     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
769     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
770       LFBFL_i_result=$((
771         LFBFL_i_result
772         | SSP_ALWAYS_LARGER_BEFORE
773       ))
774     fi
775     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
776     && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
777       LFBFL_i_result=$((
778         LFBFL_i_result
779         | SSP_POSITION_UNIFORM
780       ))
781     fi
782     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
783     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
784       LFBFL_i_result=$((
785         LFBFL_i_result
786         | SSP_ALWAYS_LARGER_AFTER
787       ))
788     fi
789     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
790     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
791       LFBFL_i_result=$((
792         LFBFL_i_result
793         | SSP_ALWAYS_LARGER_BEFORE
794       ))
795     fi
796     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
797     && ((
798       LFBFL_i_result & (
799         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
800         | SSP_SAME_POSITION_LARGER_AFTER
801         | SSP_SAME_POSITION_LARGER_BEFORE
802       )
803     )); then
804       LFBFL_i_result=$((LFBFL_i_all_flags))
805       break
806     fi
807     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
```

```
808       && ((
809         LFBFL_i_result & (
810           SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
811           | SSP_SAME_POSITION_LARGER_BEFORE
812         )
813       )); then
814         LFBFL_i_result=$((LFBFL_i_all_flags))
815         break
816       fi
817       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
818       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
819         LFBFL_i_result=$((
820           LFBFL_i_result
821           | SSP_ALWAYS_LARGER_AFTER
822         ))
823       fi
824       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
825       && ((
826         LFBFL_i_result & (
827           SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
828           |  SSP_SAME_POSITION_LARGER_AFTER
829         )
830       )); then
831         LFBFL_i_result=$((LFBFL_i_all_flags))
832         break
833       fi
834       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
835       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
836         LFBFL_i_result=$((
837           LFBFL_i_result
838           | SSP_ALWAYS_LARGER_BEFORE
839         ))
840       fi
841       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER))\
842       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
843         LFBFL_i_result=$((
844           LFBFL_i_result
845           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
846         ))
847       fi
848     done
849     declare -gi i_split_score_properties_logical_closure_result=$((
850       LFBFL_i_result
851     ))
852   }
853
```

```
split_score_simple(){
  # $1=larger_after (The name was "after_before" previously,
  #                  the meaning was not modified.
  #                  Remark not repeated below.)
  # $2=delimiters_strings_domain concatenated characters/delimiters
  # $3=delimiter_string a single character
  # $4=is_cut_after
  declare -gi i_split_score_result
  if [[ "$2" != *$3* ]]; then
    i_split_score_result=0
    return
  fi
  if [[ "$4" == "$1" ]]; then
    i_split_score_result=2
    return
  fi
  i_split_score_result=1
}

get_split_score_simple(){
  # $1=larger_after
  # $2=max_length
  # $3=delimiters_strings_domain
  declare -g get_split_score_result="split_score_simple $1 '$3'"
  declare -gi i_get_split_score_result2
  # i_get_split_score_result2=$((11-$1*4))
  # if [[ $1 -eq 1 ]]; then
  #   i_get_split_score_result2=7
  # else
  #   i_get_split_score_result2=11
  # fi
  i_get_split_score_result2=$((
    SSP_DELIMITER_UNIFORM
    + SSP_POSITION_UNIFORM
    + SSP_LARGER_BEFORE
    + (
      $1 * (
        SSP_LARGER_AFTER
        - SSP_LARGER_BEFORE
      )
    )
  ))
  # shellcheck disable=SC2248
  split_score_properties_logical_closure ${i_get_split_score_result2}
  i_get_split_score_result2=$((
    i_split_score_properties_logical_closure_result
```

```
900   ))
901 }
902
903 get_split_score_exec(){
904   declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
905   SPLIT_SCORE_EXEC+="/split_score.exec.php"
906 }
907
908 split_score(){
909   # $1=larger_after
910   # $2=max_length
911   # $3=delimiters_strings_domain concatenated characters/delimiters
912   # $4=delimiter_string a single character
913   # $5=cut_position
914   # $6=is_cut_after
915   get_split_score_exec
916   declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
917   declare -gi i_split_score_result
918   i_split_score_result=$(eval -- "${LFBFL_command}")
919 }
920
921 # This function is currently not called,
922 # hence we need this shellcheck disable.
923 # shellcheck disable=SC2034
924 get_split_score(){
925   # $1=larger_after
926   # $1=max_length
927   # $2=delimiters_strings_domain
928   declare -g get_split_score_result="split_score $1 $2 '$3'"
929   declare -gi i_get_split_score_result2
930   # i_get_split_score_result2=$((9-$1*4))
931   i_get_split_score_result2=$((
932     SSP_DELIMITER_UNIFORM
933     + SSP_LARGER_BEFORE
934     + (
935       $1 * (
936         SSP_LARGER_AFTER
937         - SSP_LARGER_BEFORE
938       )
939     )
940   ))
941   # shellcheck disable=SC2248
942   split_score_properties_logical_closure ${i_get_split_score_result2}
943   i_get_split_score_result2=$((
944     i_split_score_properties_logical_closure_result
945   ))
```

```
}

split_line_at_most(){
  # $1=line
  # $2=max_length of beginning result string
  # $3=split_score_command
  #   negative score means do not consider as a possible split
  # $4=split_score_command_properties :
  #   These properties are flags.
  #   If someday some property is not a flag,
  #   then create split_score_command_properties2.
  #   Old values to get a simple idea without reading everything at the
  #   beginning of file:
  #   - null or 0 no property;
  #   - 1 split score doesn't depend on which delimiter matches;
  #   - 2 split score doesn't depend on the current position;
  #   - 4 any split score after is always larger
  #     than any split score before;
  #   - 8 any split score before is always larger
  #     than any split score after;
  # $5=forbidden_previous_character
  # $6=split_score_command_delimiter_strings_domain
  #   optimisation if small domain
  #   - this argument can be "null", then we will use $3 only instead,
  #     in that case, only mono-character strings are considered,
  #   - this argument can be an array of characters,
  #     optimize accordingly,
  #   - this argument can be an array of strings,
  #     optimize accordingly,
  #   - this argument can be an array of regexps,
  #     optimize accordingly.
  #     Not sure an array of regexps makes more sense than a single
  #     one. Maybe they could be both cases with corresponding
  #     optimizations. Same for a single character or substring.
  #
  # Note that we're always looking for the last position to split
  # among the positions with maximum score;
  # if some position gets a score from 2 delimiters,
  # its score is always the maximum of the two corresponding scores.
  declare -g split_line_at_most_result_start
  declare -g split_line_at_most_result_end
  declare -A LFBFL_positions
  # printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
  # For my use case in Bash scripts, I will need only an array of
  # characters. See get_split_score().
  if [[ -n "$6" ]]; then
```

```
992       printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
993     fi
994     if [[ $2 -ge ${#1} ]]; then
995       split_line_at_most_result_start=$1
996       split_line_at_most_result_end=""
997       return
998     fi
999     # We know that ${#1} > $2, hence ${#1} - 1 >= $2
1000    # Edge cases:
1001    # There is always at least one character that is overlength,
1002    # but we need to look only at the score before that character that is
1003    # overlength, the score after that character is invalid,
1004    # whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
1005    # That first character that is overlength has index $2.
1006    # The cut position after that character has index $2 + 1.
1007    # Looping from character position going from 0 to $2, we need to drop
1008    # score before character 0 and drop score after character $2.
1009    # We do this with forbidden score -1.
1010    declare -ir LFBFL_i_max=$(($2))
1011    declare -i LFBFL_i
1012    declare -i LFBFL_j
1013    local LFBFL_current_char
1014
1015    # At the beginning, each position has score 0.
1016    LFBFL_positions["0"]="-1"
1017    for ((LFBFL_i = 1; LFBFL_i <= LFBFL_i_max; ++LFBFL_i)) do
1018      LFBFL_positions["${LFBFL_i}"]="0"
1019    done
1020    LFBFL_positions["${LFBFL_i}"]="-1"
1021
1022    # If we have a forbidden previous character,
1023    # we mark the forbidden positions with -1.
1024    if [[ -n "$5" ]]; then
1025      for ((LFBFL_i = 0; LFBFL_i < LFBFL_i_max;)) do
1026        LFBFL_current_char="${1:${LFBFL_i}:1}"
1027        ((++LFBFL_i))
1028        if [[ "${LFBFL_current_char}" == "$5" ]]; then
1029          LFBFL_positions["${LFBFL_i}"]="-1"
1030        fi
1031      done
1032    fi
1033
1034    declare -r LFBFL_sort_command="sort --numeric-sort"
1035
1036    if (($4 & SSP_DELIMITER_UNIFORM))\
1037    && (($4 & SSP_LARGER_AFTER))\
```

```
1038    && (($4 & SSP_POSITION_NOT_DECREASING_AFTER)); then
1039      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1040        LFBFL_current_char="${1:${LFBFL_i}:1}"
1041        eval -- "$3 '${LFBFL_current_char}' 1"
1042        # needs j=i+1 printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1043        if [[ i_split_score_result -ge 1 ]]; then
1044          LFBFL_j=$((LFBFL_i + 1))
1045          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1046          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1047            LFBFL_positions["${LFBFL_j}"]=$(
1048              max "${LFBFL_sort_command}"\
1049                "${LFBFL_positions["${LFBFL_j}"]}"\
1050                "${i_split_score_result}"
1051            )
1052            break
1053          fi
1054        fi
1055        eval -- "$3 '${LFBFL_current_char}' 0"
1056        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1057        if [[ i_split_score_result -ge 1 ]]; then
1058          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1059          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1060            LFBFL_positions["${LFBFL_i}"]=$(
1061              max "${LFBFL_sort_command}"\
1062                "${LFBFL_positions["${LFBFL_i}"]}"\
1063                "${i_split_score_result}"
1064            )
1065          fi
1066        fi
1067      done
1068    elif (($4 & SSP_DELIMITER_UNIFORM))\
1069    && (($4 & SSP_LARGER_BEFORE))\
1070    && (($4 & SSP_POSITION_NOT_DECREASING_BEFORE)); then
1071      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1072        LFBFL_current_char="${1:${LFBFL_i}:1}"
1073        # We still need to break in second if, after computing score after,
1074        # because SSP_LARGER_BEFORE implies large inequality instead of
1075        # strict inequality.
1076        # And since the algorithm is to take the largest position with
1077        # maximum score, there is a possibility that we'll break on a found
1078        # score before but for splitting on a score after.
1079        eval -- "$3 '${LFBFL_current_char}' 1"
1080        # needs j=i+1 printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1081        if [[ i_split_score_result -ge 1 ]]; then
1082          LFBFL_j=$((LFBFL_i + 1))
1083          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
```

```
1084          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1085            LFBFL_positions["${LFBFL_j}"]=$(
1086              max "${LFBFL_sort_command}"\
1087                "${LFBFL_positions["${LFBFL_j}"]}"\
1088                "${i_split_score_result}"
1089            )
1090          fi
1091        fi
1092        eval -- "$3 '${LFBFL_current_char}' 0"
1093        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1094        if [[ i_split_score_result -ge 1 ]]; then
1095          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1096          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1097            LFBFL_positions["${LFBFL_i}"]=$(
1098              max "${LFBFL_sort_command}"\
1099                "${LFBFL_positions["${LFBFL_i}"]}"\
1100                "${i_split_score_result}"
1101            )
1102            break
1103          fi
1104        fi
1105      done
1106    else
1107      for ((LFBFL_i = 0; LFBFL_i <= LFBFL_i_max;)) do
1108        LFBFL_current_char="${1:${LFBFL_i}:1}"
1109        eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
1110        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1111        if [[ i_split_score_result -ge 1 ]]; then
1112          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1113          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1114            LFBFL_positions["${LFBFL_i}"]=$(
1115              max "${LFBFL_sort_command}"\
1116                "${LFBFL_positions["${LFBFL_i}"]}"\
1117                "${i_split_score_result}"
1118            )
1119          fi
1120        fi
1121        ((++LFBFL_i))
1122        eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 1"
1123        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1124        if [[ i_split_score_result -ge 1 ]]; then
1125          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1126          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1127            LFBFL_positions["${LFBFL_i}"]=$(
1128              max "${LFBFL_sort_command}"\
1129                "${LFBFL_positions["${LFBFL_i}"]}"\
```

```
1130                "${i_split_score_result}"
1131            )
1132          fi
1133        fi
1134      done
1135    fi
1136    declare -ir LFBFL_i_max_score=$(
1137      max 'sort --numeric-sort' "${LFBFL_positions[@]}"
1138    )
1139    local LFBFL_value
1140    declare -i LFBFL_i_best_position
1141    for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1142      if [[ LFBFL_i -eq 0 ]]; then
1143        printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
1144      fi
1145      local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
1146      # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
1147      if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
1148        LFBFL_i_best_position=$((LFBFL_i))
1149        break
1150      fi
1151    done
1152    # shellcheck disable=SC2248
1153    split_line_at "$1" ${LFBFL_i_best_position}
1154    split_line_at_most_result_start="${split_line_at_result_beginning}"
1155    split_line_at_most_result_end="${split_line_at_result_end}"
1156    # printf "%s\n" "$split_line_at_most_result_start"
1157    # printf "%s\n" "$split_line_at_most_result_end"
1158  }
1159
1160  split_last_line(){
1161    # $1=new_lines
1162    # $2=prefix
1163    # $3=max_length
1164    # $4=suffix
1165    # $5=split_score_command
1166    # $6=split_score_command_properties
1167    # $7=forbidden_previous_character ('\' usually)
1168    # Options:
1169    #   --verbose
1170    # In general, a prefix/suffix can be a required line prefix/suffix
1171    # for all final lines like a comment prefix,
1172    # or it can be a continuation line prefix/suffix applied only
1173    # when splitting a line in 2 lines,
1174    # or it can be a prefix/suffix for the string made of all the lines.
1175    # Clearly, there is no gain to couple handling a prefix/suffix for
```

```
1176    # the string made of all the lines in this function.
1177    # Since we add a prefix/suffix when splitting,
1178    # it is easy to see that we do not need to distinguish between
1179    # required line prefix/suffix and continuation line prefix/suffix.
1180    # The $2 and $4 arguments are given as concatenation of the
1181    # required and continuation line prefixes/suffixes as needed by the
1182    # user.
1183    # Thus lines arguments should be given already prefixed
1184    # and suffixed.
1185    # This is logic, since it would be cumbersome to handle prefixing
1186    # or not already splitted lines in this function
1187    # (see repeated_split_last_line).
1188    # The given max_length is the goal.
1189    # This function adapts the effective max position for the split by
1190    # taking into account the length of the suffix.
1191    declare -i LFBFL_i_verbose=0
1192    get_verbose_option "$@"
1193    declare -a LFBFL_return_traps_stack
1194    local LFBFL_previous_return_trap
1195    init_return_trap
1196
1197    enhanced_set_shell_option pipefail --trap-unset
1198
1199    declare -g split_last_line_result="$1"
1200    declare -ir LFBFL_i_overlength=$(($3 + 1))
1201    declare -ir LFBFL_i_length2=$(($3 - ${#4}))
1202    # shellcheck disable=SC2248
1203    get_overlength_regexp ${LFBFL_i_overlength}
1204
1205    # Testing if some line is too long.
1206    printf -- "%s" "$1"\
1207      | grep --quiet "${get_overlength_regexp_result}"
1208    if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
1209      return
1210    fi
1211
1212    declare -r LFBFL_start=$(
1213      printf -- "%s" "$1"\
1214      | head --lines=-1
1215    )
1216    # printf "start: %s\n" "${LFBFL_start}"
1217    declare -r LFBFL_last_line=$(
1218      printf -- "%s" "$1"\
1219      | tail --lines=1
1220    )
1221    # printf "last_line: %s\n" "${LFBFL_last_line}"
```

```
  split_last_line_result=""
  if [[ -n "${LFBFL_start}" ]]; then
    split_last_line_result="${LFBFL_start}"
    # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
    # printf "%s %s\n" "${#split_last_line_result}"
    #   "${split_last_line_result}"
    split_last_line_result+=$'\n'
    # something="${split_last_line_result}"
    # printf "%s %s\n" "${#something}" "${something}"
  fi
  if [[ -n "$5" ]]; then
    # shellcheck disable=SC2248
    split_line_at_most "${LFBFL_last_line}" ${LFBFL_i_length2}\
      "$5" "$6" "$7"
    split_last_line_result+="${split_line_at_most_result_start}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${split_line_at_most_result_end}"
  else
    split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
  fi
  # printf "result: %s\n" "${split_last_line_result}"
}

repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
  local LFBFL_result="$1x"
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_i_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
```

```
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

3.31 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

get_overlength_regexp(){
  declare -g get_overlength_regexp_result=".\{$1\}"
  # declare -g get_overlength_regexp_result="^.\{$1\}"
  # declare -g get_overlength_regexp_result=".\{$1\}\$"
```

```
  # First one seems to be faster by a few percents when testing like this:
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '^.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}$' COPYING; done
}

too_long_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_max_line_length
  get_max_line_length_option "$@"

  declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
  # shellcheck disable=SC2248
  get_overlength_regexp ${LFBFL_i_overlength}

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  local LFBFL_long_line
  local LFBFL_file_path
  local LFBFL_line
  local LFBFL_extension
  local LFBFL_base_name
  local LFBFL_s_long_lines
  declare -a LFBFL_arr_long_lines
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf -- "too_long_code_lines: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    # shellcheck disable=SC2248
    LFBFL_s_long_lines=$(
```

```
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_license_find\
      | xargs grep --line-number --with-filename\
        --regexp="${get_overlength_regexp_result}" --
    )
    if [[ -z "${LFBFL_s_long_lines}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "too_long_code_lines: No file found with long lines.\n"
      continue
    fi
    mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
    for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
      LFBFL_file_path=${LFBFL_long_line%%:*} # Drop long ':*' suffix
      # Remove file_path and line_number
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      LFBFL_line=${LFBFL_line#*:} # Drop short '*:' prefix
      # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_path##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_path%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.sub"
        then
          continue
        fi
        printf -- "%s\n" "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if [[ ! -f "${LFBFL_base_name}.md.tpl" ]]; then
          printf -- "%s\n" "${LFBFL_long_line}"
        fi
      else
        printf -- "%s\n" "${LFBFL_long_line}"
      fi
    done
  done
}
```

3.32 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

source "overwrite_if_not_equal.libr.sh"

update_common_build_and_checks(){
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf -- "%s %s\n" "$0" "$*"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_i_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  cp -- "./${LFBFL_common_file_name}.tpl"\
    "./${LFBFL_common_file_name}.temp"

  declare -ar LFBFL_files_names=(
    "build_dependencies_notes.exec.php"
    "build_dependencies_notes.libr.php"
```

```
    "build_md_from_printable_md.exec.sh"
    "check_shell_scripts_beginnings.libr.sh"
    "check_shell_scripts_indentation.libr.sh"
    "check_URLs.libr.sh"
    "common_options.libr.sh"
    "comparisons.libr.sh"
    "create_PDF.exec.sh"
    "generate_from_template.libr.sh"
    "get_common_text_glob_patterns.libr.sh"
    "grammar_and_spelling_check.libr.sh"
    "licenses_templates/build_licenses_templates.exec.sh"
    "licenses_templates/license_file_header_AGPLv3+.tpl"
    "licenses_templates/license_file_header_GPLv3+.tpl"
    "licenses_templates/license_file_header_LGPLv3+.tpl"
    "lines_counts.libr.sh"
    "lines_filters.libr.sh"
    "lines_maps.libr.sh"
    "overwrite_if_not_equal.libr.sh"
    "python_black_complement.libr.sh"
    "python_isort_complement.libr.sh"
    "repository_data.libr.sh"
    "shell_checks_complement.libr.sh"
    "split_score.exec.php"
    "split_score.libr.php"
    "strings_functions.libr.sh"
    "too_long_code_lines.libr.sh"
    "update_or_check_files_names_listing.exec.sh"
  )

  local LFBFL_file_name
  local LFBFL_file_sha512
  local LFBFL_file_sha512_all
  local LFBFL_base_file_name
  for LFBFL_file_name in "${LFBFL_files_names[@]}"; do
    LFBFL_file_sha512=$(
      sha512sum "./${LFBFL_file_name}"\
      | cut --delimiter=' ' --fields=1
    )
    LFBFL_file_sha512_all="LFBFL_correct_sha512="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename -- "${LFBFL_file_name}")
    sed --in-place\
```

```
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut --delimiter=' ' --fields=1
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed --in-place\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

3.33 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
#!/usr/bin/env bash
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# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_write
  get_some_flag LFBFL_i_write --write 1 "$@"
  declare -i LFBFL_i_append
  get_some_flag LFBFL_i_append --append 1 "$@"
```

```
60
61    enhanced_set_shell_option pipefail --trap-unset
62
63    LFBFL_subdir2="build_and_checks_variables"
64    declare -r LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
65    declare -r LFBFL_data_file_path="./${LFBFL_subdir2}/repository_data.txt"
66
67    declare -i LFBFL_i_max_line_length
68    grep_variable "${LFBFL_data_file_path}" max_line_length\
69      --result-variable-prefix="LFBFL_i_"
70
71    if [[ LFBFL_i_write -eq 1 ]]; then
72      : > "${LFBFL_listing}"
73    fi
74    # Remove line returns that are here to keep lines short.
75    sed --regexp-extended --null-data --expression='s/(.)\\\n/\1/Mg'\
76      -- "${LFBFL_listing}"\
77      > "${LFBFL_listing}.temp"
78    # shopt -s dotglob was needed at some point but I don't see why now.
79    # shellcheck disable=SC2248
80    get_split_score_simple 1 ${LFBFL_i_max_line_length} /
81    declare -r LFBFL_split_score_command="${get_split_score_result}"
82    declare -i LFBFL_i_split_score_command_properties
83    LFBFL_i_split_score_command_properties=${i_get_split_score_result2}
84    readonly LFBFL_i_split_score_command_properties
85    local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
86    readonly LFBFL_suffix
87    # There is always a $'\n' final suffix from >>.
88    # But when origin string ends with the suffix we must add something.
89    declare -r LFBFL_final_suffix=$'\n\\'
90
91    local LFBFL_s_files_paths
92    LFBFL_s_files_paths=$(
93      find . -type f -printf '%P\n'\
94      | relevant_find\
95      | sort
96    )
97    # printf "%s\n" "${LFBFL_s_files_paths}"
98    declare -a LFBFL_arr_files_paths
99    if [[ -n "${LFBFL_s_files_paths}" ]]; then
100     if [[ LFBFL_i_verbose -eq 1 ]]; then
101       printf\
102         "update_or_check_files_names_listing: Some files to process.\n"
103     fi
104     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
105   fi
```

```
  # printf "%s\n" "${LFBFL_arr_files_paths[@]}"

  local LFBFL_file_path
  local LFBFL_base_file_name
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    # if [[ LFBFL_i_verbose -eq 1 ]]; then
    #   printf --\
    #     "update_or_check_files_names_listing: Maybe ignored file: %s\n"\
    #     "${LFBFL_file_path}"
    # fi
    git check-ignore --quiet -- "${LFBFL_file_path}" && continue
    LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
    [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
      || continue
    if [[ "${LFBFL_base_file_name}" == *.md ]]; then
      if [[ -f "${LFBFL_file_path}.tpl" ]]; then
        continue
      fi
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf --\
        "update_or_check_files_names_listing: Non-ignored file: %s\n"\
        "${LFBFL_file_path}"
    fi
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_file_path}"\
      ""\
      ${LFBFL_i_max_line_length}\
      "${LFBFL_suffix}"\
      "${LFBFL_split_score_command}"\
      ${LFBFL_i_split_score_command_properties}\
      ""\
      "${LFBFL_final_suffix}"
    if [[ LFBFL_i_write -eq 1 ]]; then
      # shellcheck disable=SC2001
      printf -- "%s\n" "${repeated_split_last_line_result}"\
        >> "${LFBFL_listing}"
      continue
    fi
    if grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
        "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
    then
      :
    else
      printf -- "The file %s is not listed in %s.\n"\
        "${LFBFL_file_path}"\
```

```
        "${LFBFL_listing}"
      if [[ LFBFL_i_append -eq 1 ]]; then
        # shellcheck disable=SC2001
        printf -- "%s\n" "${repeated_split_last_line_result}"\
          >> "${LFBFL_listing}"
      fi
    fi
  done

  LFBFL_s_files_paths=$(cat -- "${LFBFL_listing}.temp")
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi

  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ "${LFBFL_file_path}" != '//'* ]] || continue
    if [[ ! -f "${LFBFL_file_path}" ]]; then
      printf -- "The non-file %s is listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
    fi
  done

  rm -- "${LFBFL_listing}.temp"

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    cat -- "${LFBFL_listing}"
  fi
}

update_or_check_files_names_listing "$@"
```

3.34 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts
```

3.35 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

3.36 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

3.37 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

3.38 build_and_checks_variables/dependencies_data.json

```
[
  { "dependency_name": "Bash", "URLs": [
      "https://www.gnu.org/software/bash/",
      "https://tiswww.case.edu/php/chet/bash/bashtop.html",
      "https://savannah.gnu.org/projects/bash/"
  ] },
  { "dependency_name": "GNU coreutils", "URLs": [
      "https://www.gnu.org/software/coreutils/",
      "https://savannah.gnu.org/projects/coreutils/"
  ] },
  { "dependency_name": "uutils coreutils", "URLs": [
      "https://uutils.github.io/coreutils/",
      "https://github.com/uutils/coreutils"
  ] },
  { "dependency_name": "diff (GNU diffutils)", "URLs": [
      "https://www.gnu.org/software/diffutils/",
      "https://savannah.gnu.org/projects/diffutils/"
  ] },
  { "dependency_name": "find, xargs (GNU findutils)", "URLs": [
      "https://www.gnu.org/software/findutils/",
      "https://savannah.gnu.org/projects/findutils/"
  ] },
  { "dependency_name": "grep (GNU grep)", "URLs": [
      "https://www.gnu.org/software/grep/",
      "https://savannah.gnu.org/projects/grep/"
  ] },
  { "dependency_name": "sed (GNU sed)", "URLs": [
      "https://www.gnu.org/software/sed/",
      "https://savannah.gnu.org/projects/sed/"
  ] },
  { "dependency_name": "ShellCheck", "URLs": [
      "https://www.shellcheck.net/",
      "https://github.com/koalaman/shellcheck"
  ] },
  { "dependency_name": "GNU Wget", "URLs": [
      "https://www.gnu.org/software/wget/",
      "https://savannah.gnu.org/projects/wget/"
  ] },
  { "dependency_name": "more of util-linux", "URLs": [
```

```
40          "https://www.kernel.org/pub/linux/utils/util-linux/",
41          "https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/",
42          {
43            "URL": "https://github.com/util-linux/util-linux",
44            "comment": "official?"
45          }
46      ] },
47      { "dependency_name": "pandoc", "URLs": [
48          "https://pandoc.org/",
49          "https://github.com/jgm/pandoc"
50      ] },
51      { "dependency_name": "pdftex/pdflatex", "URLs": [
52          "https://www.tug.org/applications/pdftex/",
53          "https://tug.org/texlive/svn/"
54      ] },
55      { "dependency_name": "tree", "URLs": [
56          "https://oldmanprogrammer.net/source.php?dir=projects/tree",
57          "https://gitlab.com/OldManProgrammer/unix-tree",
58          "https://github.com/Old-Man-Programmer/tree"
59      ] },
60      { "dependency_name": "git", "URLs": [
61          "https://git-scm.com/about",
62          "https://github.com/git/git"
63      ] },
64      { "dependency_name": "pylint", "URLs": [
65          "https://www.pylint.org/",
66          "https://pylint.readthedocs.io/en/stable/",
67          "https://github.com/pylint-dev/pylint",
68          "https://pypi.org/project/pylint/"
69      ] },
70      { "dependency_name": "notify-send", "URLs": [
71          "https://gitlab.gnome.org/GNOME/libnotify/",
72          "https://gnome.pages.gitlab.gnome.org/libnotify/"
73      ] },
74      { "dependency_name": "black", "URLs": [
75          "https://github.com/psf/black",
76          "https://black.readthedocs.io/en/stable/",
77          "https://pypi.org/project/black/"
78      ] },
79      { "dependency_name": "isort", "URLs": [
80          "https://pycqa.github.io/isort/",
81          "https://github.com/pycqa/isort/",
82          "https://pypi.org/project/isort/"
83      ] },
84      { "dependency_name": "mypy", "URLs": [
85          "https://mypy-lang.org/",
```

```
86          "https://mypy.readthedocs.io/en/stable/",
87          "https://github.com/python/mypy",
88          "https://pypi.org/project/mypy/"
89      ] },
90      { "dependency_name": "phpdoc (phpDocumentor)", "URLs": [
91          "https://phpdoc.org/",
92          "https://docs.phpdoc.org/",
93          "https://github.com/phpdocumentor/phpdocumentor"
94      ] },
95      { "dependency_name": "pip", "URLs": [
96          "https://pip.pypa.io/en/stable/",
97          "https://github.com/pypa/pip",
98          {
99            "URL": "https://pypi.org/project/pip/",
100           "comment": "(egg and chicken ;) )"
101         }
102     ] },
103     { "dependency_name": "ruff", "URLs": [
104         "https://docs.astral.sh/ruff/",
105         "https://github.com/astral-sh/ruff",
106         "https://pypi.org/project/ruff/"
107     ] },
108     { "dependency_name": "bandit", "URLs": [
109         "https://bandit.readthedocs.io/en/latest/",
110         "https://github.com/PyCQA/bandit",
111         "https://pypi.org/project/bandit/"
112     ] },
113     { "dependency_name": "phpcs (PHP_CodeSniffer)", "URLs": [
114         "https://github.com/PHPCSStandards/PHP_CodeSniffer",
115         "https://packagist.org/packages/squizlabs/php_codesniffer"
116     ] },
117     { "dependency_name": "phpmd", "URLs": [
118         "https://phpmd.org/",
119         "https://github.com/phpmd/phpmd",
120         "https://packagist.org/packages/phpmd/phpmd"
121     ] },
122     { "dependency_name": "php", "URLs": [
123         "https://www.php.net/",
124         "https://github.com/php/php-src"
125     ] },
126     { "dependency_name": "tex", "URLs": [
127         "https://ctan.org/tex-archive/systems/knuth/dist/tex"
128     ] },
129     { "dependency_name": "pcregrep, pcre2grep", "URLs": [
130         "https://www.pcre.org/",
131         "https://pcre2project.github.io/pcre2/",
```

```
      "https://github.com/PCRE2Project/pcre1",
      "https://github.com/PCRE2Project/pcre2",
      "https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
  ] },
  { "dependency_name": "python3", "URLs": [
      "https://www.python.org/",
      "https://docs.python.org/3/"
  ] }
]
```

3.39 build_and_checks_variables/dependencies_notes.txt

```
Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
```

```
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
```

```
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

phpdoc (phpDocumentor) can be found at the following URLs:
https://phpdoc.org/
https://docs.phpdoc.org/
https://github.com/phpdocumentor/phpdocumentor

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer
https://packagist.org/packages/squizlabs/php_codesniffer

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
```

```
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

3.40 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
```

```
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have an accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
ol.level2{
  counter-reset: level2;
}
ol.level1 li:before{
```

```
  content: counter(level1) ". ";
  counter-increment: level1;
}
ol.level2 li:before{
  content: counter(level1) "." counter(level2) " ";
  counter-increment: level2;
}
</style>
<title>@repository_name@</title>
</head>
<body >
<table class="title">
  <tr>
    <td style="padding:1ex">
      <h1 class="titlemain">@repository_name@</h1>
      <h2 class="titlerest">
        @author_full_name@
        <br>
        <a href="@author_website@">
          <span style="font-family:monospace">
          @author_website@
          </span>
        </a>
        <br>
        <a href="mailto:@author_email@">
          <span style="font-family:monospace">
          @author_email@
          </span>
        </a>
      </h2>
    </td>
  </tr>
</table>

<blockquote class="abstract">
  <span style="font-weight:bold">Abstract:</span>
  @abstract@
</blockquote>
<p>Current version: @current_date@</p>
<p>Current number of commits: @number_of_commits@</p>
<p>Current git SHA1: @current_git_SHA1@</p>
<p>
Code lines: @number_of_lines@
</p>


```

```
<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files Tree</a></li>
  <li><a href="#section2">Notes on licenses and dependencies</a></li>
  <li><a href="#section3">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Notes on licenses and dependencies</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
```

```
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, phpdoc, pip, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpcs, phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

@dependencies_notes@


<h2 id="section3">3 Listing of files</h2>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>
```

```

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

3.41 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
```

```
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

```

```
\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files tree}
\label{section:tree}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Notes on licenses and dependencies}
\label{section:notes_on}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
```

```
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, phpdoc, pip, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpcs, phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

@dependencies_notes@

\section{Listing of files}
\label{section:listing}

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

3.42 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
```

```
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_dependencies_notes.exec.php
build_and_checks_dependencies/build_dependencies_notes.libr.php
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spelling_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
```

```
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/dependencies_data.json
build_and_checks_variables/dependencies_notes.txt
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpcs_ruleset.xml
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/repository_data.txt.tpl
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
```

```
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234 5
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
56
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
56\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
5\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890 123\
\
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/spampty_data.txt
miscellaneous/string_escaping.html
miscellaneous/string_escaping.xhtml
// miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
string_escaping.libr.php
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

3.43 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
```

```
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, phpdoc, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
```

3.44 build_and_checks_variables/phpcs_ruleset.xml

```
<?xml version="1.0"?>
<ruleset name="LFBFLCustomStandard" namespace="LFBFL\CS\CustomStandard">
  <description>A custom coding standard</description>
  <config name="tab_width" value="2"/>

  <file>../</file>

  <arg name="extensions" value="php"/>
  <arg name="report" value="summary"/>
  <arg name="colors"/>
  <arg value="sp"/>

  <!--
```

```
I added all the standards, then excluded all the rules that gave me
false positives or that seemed unappropriate when looking at the
documentation.
Some are bugs, some are conflicting with my own coding style.
You may delete an exclude and see what pops up to maybe understand
my choices.
-->

<rule ref="Generic">
  <!-- You have to allow long or short or both syntaxes. -->
  <exclude name="Generic.Arrays.DisallowShortArraySyntax"/>
  <exclude name="Generic.Commenting.DocComment.ParamGroup"/>
  <exclude name="Generic.Commenting.DocComment.TagsNotGrouped"/>
  <exclude name="Generic.Commenting.DocComment.TagValueIndent"/>
  <!-- Back to my habits at Teliae with always a closing tag. -->
  <!-- <exclude name="Generic.PHP.ClosingPHPTag.NotFound"/> -->
  <!--
  Not standard in Unix/Linux world to keep files not ending with \n.
  This is always an option of the code editors.
  Since PHP doesn't output the final \n after ?>, it is ok.
  -->
  <exclude name="Generic.Files.EndFileNoNewline.Found"/>
  <exclude name="Generic.Files.ExecutableFile.Executable"/>
  <exclude name="Generic.Formatting.MultipleStatementAlignment"/>
  <exclude
   name="Generic.Functions.OpeningFunctionBraceBsdAllman.BraceOnSameLine"
  />
  <exclude
    name="Generic.NamingConventions.CamelCapsFunctionName.NotCamelCaps"
  />
  <!-- Applies only to true, false and null. -->
  <exclude name="Generic.PHP.LowerCaseConstant.Found"/>
  <exclude name="Generic.PHP.UpperCaseConstant.Found"/>
  <!-- const_bidule_HTML_truc would be better. Everything is prefix. -->
  <exclude name="Generic.WhiteSpace.DisallowSpaceIndent.SpacesUsed"/>
</rule>
<rule ref="PEAR">
  <exclude name="PEAR.Commenting.FileComment.MissingVersion"/>
  <exclude name="PEAR.Commenting.FileComment.MissingLinkTag"/>
  <!--
  Look at the comment before generate_split_score.
  PEAR and Squiz require to align the parameter names parts in the same
  comment, and also the parameter comments parts in the same comment,
  but with a long parameter name like $a_delimiter_strings_domain
  and a line length limit at 75 characters, parameter comments parts
  have only 24 characters left per line, which is ridiculous.
```

```
    Our choice to put the parameter comments parts starting on another line
    with same "indent" as parameter names parts is way better in those
    cases. Alignment has always been a trap with short lines.
    Since phpDocumentor handles the parameter comments parts correctly,
    it is enough for me.
    Worse is the handling of "nested doc block comment" for the Closure
    returned by generate_split_score().
    Our solution makes at least the source code readable.
    Maybe one day the dev tools will handle it correctly.
    -->
    <exclude name="PEAR.Commenting.FunctionComment.DuplicateReturn"/>
    <exclude name="PEAR.Commenting.FunctionComment.ExtraParamComment"/>
    <exclude name="PEAR.Commenting.FunctionComment.MissingParamComment"/>
    <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamName"/>
    <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamType"/>
    <exclude name="PEAR.ControlStructures.ControlSignature.Found"/>
    <exclude name="PEAR.ControlStructures.MultiLineCondition.Alignment"/>
    <exclude
      name="PEAR.ControlStructures.MultiLineCondition.SpaceBeforeOpenBrace"
    />
    <exclude
     name="PEAR.ControlStructures.MultiLineCondition.SpacingAfterOpenBrace"
    />
    <exclude
      name="PEAR.ControlStructures.MultiLineCondition.StartWithBoolean"
    />
    <!--
    Don’t laugh when looking at the false positives of the two exclude
    below ;) XD.
    They don’t check for multiline function declarations apparently.
    -->
    <exclude
      name="PEAR.Functions.FunctionDeclaration.NewlineBeforeOpenBrace"
    />
    <exclude
      name="PEAR.Functions.FunctionDeclaration.BraceOnSameLine"
    />
    <exclude
      name="PEAR.NamingConventions.ValidFunctionName.FunctionNoCapital"
    />
    <exclude
      name="PEAR.NamingConventions.ValidFunctionName.FunctionNameInvalid"
    />
  </rule>
  <rule ref="PSR1">
  </rule>
```

```
106   <rule ref="PSR2">
107     <!-- Back to my habits at Teliae with always a closing tag. -->
108     <exclude name="PSR2.Files.ClosingTag.NotAllowed"/>
109     <exclude
110 name="PSR2.ControlStructures.ControlStructureSpacing.SpacingAfterOpenBrace"
111     />
112   </rule>
113   <rule ref="PSR12">
114     <exclude
115  name="PSR12.ControlStructures.ControlStructureSpacing.FirstExpressionLine"
116     />
117     <exclude name="PSR12.Files.FileHeader.SpacingAfterDeclareBlock"/>
118     <!--
119     3 false positives, see:
120     https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1402
121     -->
122     <exclude
123       name="PSR12.Files.DeclareStatement.SpaceFoundAfterDirectiveValue"
124     />
125     <exclude
126       name="PSR12.Files.DeclareStatement.SpaceFoundBeforeDirectiveValue"
127     />
128     <exclude
129       name="PSR12.Files.DeclareStatement.SpaceFoundBeforeSemicolon"
130     />
131     <!-- and even a fourth and fifth false positives -->
132     <exclude
133       name="PSR12.Files.DeclareStatement.CloseBracketNotAligned"
134     />
135     <exclude
136       name="PSR12.Files.DeclareStatement.ExtraSpaceFoundAfterBracket"
137     />
138     <exclude name="PSR12.Classes.ClosingBrace.StatementAfter"/>
139     <exclude
140       name="PSR12.Functions.ReturnTypeDeclaration.SpaceBeforeColon"
141     />
142     <exclude name="PSR2.Namespaces.NamespaceDeclaration.BlankLineAfter"/>
143     <exclude name="PSR2.Namespaces.UseDeclaration.SpaceAfterLastUse"/>
144     <!-- I agree except for string concatenation. -->
145     <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceBefore"/>
146     <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceAfter"/>
147   </rule>
148   <rule ref="Squiz">
149     <exclude name="Squiz.Arrays.ArrayDeclaration.SingleLineNotAllowed"/>
150     <!--
151     It doesn't know the //*/ trick...
```

```
    https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1403
    -->
    <exclude name="Squiz.Commenting.BlockComment.CloserSameLine"/>
    <exclude name="Squiz.Commenting.BlockComment.FirstLineIndent"/>
    <exclude name="Squiz.Commenting.BlockComment.LineIndent"/>
    <exclude name="Squiz.Commenting.BlockComment.NoCapital"/>
    <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineAfter"/>
    <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineBefore"/>
    <!-- Problem for doc block comment before namespace -->
    <exclude name="Squiz.Commenting.BlockComment.WrongStart"/>
    <exclude name="Squiz.Commenting.DocCommentAlignment.SpaceBeforeStar"/>
    <exclude name="Squiz.Commenting.FileComment.CopyrightTagOrder"/>
    <exclude name="Squiz.Commenting.FileComment.IncorrectAuthor"/>
    <exclude name="Squiz.Commenting.FileComment.IncorrectCopyright"/>
    <exclude name="Squiz.Commenting.FileComment.MissingSubpackageTag"/>
    <exclude name="Squiz.Commenting.FileComment.PackageTagOrder"/>
    <exclude name="Squiz.Commenting.FileComment.SpacingAfterOpen"/>
    <exclude name="Squiz.Commenting.FileComment.SubpackageTagOrder"/>
    <exclude name="Squiz.Commenting.FunctionComment.DuplicateReturn"/>
    <exclude name="Squiz.Commenting.FunctionComment.ExtraParamComment"/>
    <!-- asks for boolean instead of bool which is incorrect -->
    <exclude
      name="Squiz.Commenting.FunctionComment.IncorrectParamVarName"
    />
    <!-- asks for boolean instead of bool which is incorrect -->
    <exclude name="Squiz.Commenting.FunctionComment.InvalidReturn"/>
    <exclude name="Squiz.Commenting.FunctionComment.MissingParamComment"/>
    <exclude name="Squiz.Commenting.FunctionComment.ParamCommentFullStop"/>
    <exclude
      name="Squiz.Commenting.FunctionComment.SpacingAfterParamName"
    />
    <exclude
      name="Squiz.Commenting.FunctionComment.SpacingAfterParamType"
    />
    <!-- Problem for doc block comment before namespace -->
    <exclude name="Squiz.Commenting.InlineComment.DocBlock"/>
    <exclude name="Squiz.Commenting.InlineComment.NotCapital"/>
    <!-- It doesn't know the //*/ trick... -->
    <exclude name="Squiz.Commenting.InlineComment.NoSpaceBefore"/>
    <exclude name="Squiz.Commenting.InlineComment.InvalidEndChar"/>
    <exclude name="Squiz.Commenting.LongConditionClosingComment.Invalid"/>
    <exclude name="Squiz.Commenting.PostStatementComment.Found"/>
    <exclude
      name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseBrace"
    />
    <exclude
```

```
198   name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseParenthesis"
199      />
200      <exclude
201        name="Squiz.ControlStructures.ControlSignature.SpaceAfterKeyword"
202      />
203      <exclude name="Squiz.Formatting.OperatorBracket.MissingBrackets"/>
204      <exclude name="Squiz.Functions.GlobalFunction.Found"/>
205      <exclude
206        name="Squiz.Functions.MultiLineFunctionDeclaration.BraceOnSameLine"
207      />
208      <exclude
209   name="Squiz.Functions.MultiLineFunctionDeclaration.NewlineBeforeOpenBrace"
210      />
211      <exclude
212        name="Squiz.NamingConventions.ValidFunctionName.NotCamelCaps"
213      />
214      <exclude
215        name="Squiz.NamingConventions.ValidVariableName.NotCamelCaps"
216      />
217      <exclude name="Squiz.Operators.ComparisonOperatorUsage.NotAllowed"/>
218      <exclude name="Squiz.Operators.ComparisonOperatorUsage.ImplicitTrue"/>
219      <exclude name="Squiz.PHP.CommentedOutCode"/>
220      <exclude name="Squiz.Strings.EchoedStrings.HasBracket"/>
221      <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceBefore"/>
222      <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceAfter"/>
223      <exclude name="Squiz.WhiteSpace.OperatorSpacing.SpacingBefore"/>
224      <exclude
225        name="Squiz.WhiteSpace.FunctionClosingBraceSpace.SpacingBeforeClose"
226      />
227    </rule>
228    <rule ref="Zend">
229      <!-- Back to my habits at Teliae with always a closing tag. -->
230      <exclude name="Zend.Files.ClosingTag.NotAllowed"/>
231      <exclude
232        name="Zend.NamingConventions.ValidVariableName.NotCamelCaps"
233      />
234    </rule>
235
236    <rule ref="Generic.Arrays.ArrayIndent">
237      <properties>
238        <property name="indent" value="2" />
239      </properties>
240    </rule>
241    <rule ref="Generic.Files.LineLength">
242      <properties>
243        <property name="lineLimit" value="75"/>
```

```
244         <property name="absoluteLineLimit" value="75"/>
245       </properties>
246     </rule>
247     <rule ref="Generic.Formatting.SpaceAfterNot">
248       <properties>
249         <property name="spacing" value="0" />
250       </properties>
251     </rule>
252     <rule ref="Generic.Functions.OpeningFunctionBraceKernighanRitchie">
253       <properties>
254         <property name="checkFunctions" value="true"/>
255         <property name="checkClosures" value="true"/>
256       </properties>
257     </rule>
258     <rule ref="Generic.Metrics.CyclomaticComplexity">
259       <properties>
260         <property name="complexity" value="20" />
261         <property name="absoluteComplexity" value="40" />
262       </properties>
263     </rule>
264     <rule ref="Generic.Strings.UnnecessaryStringConcat">
265       <properties>
266         <property name="allowMultiline" value="true" />
267       </properties>
268     </rule>
269     <rule ref="Generic.WhiteSpace.ArbitraryParenthesesSpacing">
270       <properties>
271         <property name="ignoreNewlines" value="true" />
272       </properties>
273     </rule>
274     <rule ref="Generic.WhiteSpace.ScopeIndent">
275       <properties>
276         <property name="ignoreIndentationTokens" type="array">
277           <element value="T_COMMENT"/>
278           <element value="T_DOC_COMMENT_OPEN_TAG"/>
279         </property>
280         <property name="indent" value="2"/>
281       </properties>
282     </rule>
283
284     <rule ref="PEAR.Functions.FunctionCallSignature">
285       <properties>
286         <property name="indent" value="2"/>
287       </properties>
288     </rule>
289     <rule ref="PEAR.Functions.FunctionDeclaration">
```

```
290       <properties>
291         <property name="indent" value="2"/>
292       </properties>
293     </rule>
294     <rule ref="PEAR.WhiteSpace.ScopeIndent">
295       <properties>
296         <property name="ignoreIndentationTokens" type="array">
297           <element value="T_COMMENT"/>
298           <element value="T_DOC_COMMENT_OPEN_TAG"/>
299         </property>
300         <property name="indent" value="2"/>
301       </properties>
302     </rule>
303
304     <rule ref="PSR2.Methods.FunctionCallSignature">
305       <properties>
306         <property name="indent" value="2"/>
307       </properties>
308     </rule>
309
310     <rule ref="PSR12.ControlStructures.ControlStructureSpacing">
311       <properties>
312         <property name="indent" value="2"/>
313       </properties>
314     </rule>
315
316     <rule ref="Squiz.Functions.MultiLineFunctionDeclaration">
317       <properties>
318         <property name="indent" value="2"/>
319       </properties>
320     </rule>
321     <rule ref="Squiz.Strings.ConcatenationSpacing">
322       <properties>
323         <property name="ignoreNewlines" value="true" />
324       </properties>
325     </rule>
326     <rule ref="Squiz.WhiteSpace.FunctionSpacing">
327       <properties>
328         <property name="spacing" value="3" />
329         <property name="spacingBeforeFirst" value="0" />
330         <property name="spacingAfterLast" value="0" />
331       </properties>
332     </rule>
333   </ruleset>
```

3.45 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\CleanCode\ElseExpression"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="string_escaping.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="string_escaping.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="unicode.libr.php"
  />
```

```
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\CleanCode\MissingImport"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="unicode.libr.php"
  />
</phpmd-baseline>
```

3.46 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

3.47 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

latex_command=pdflatex

grammar_and_spelling_checker_command=


Why one venv for each Python dependency ?
```

```
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
        }
      },
      "D": {
        "0.1.0": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "4.4.4" : {}
          }
        }
      }
    }
  },
  "B": {
    "2.2.2": {
```

```
        "C": {
          // Dumb, you did not found it by yourself? Failed exam.
          "4.4.4" : {}
        }
      }
    }
}
Wonderful someone rediscovered nested structures XD
Seriously so many engineers or computer scientists are dumb to cry.

bandit_venv=../venv_bandit
black_venv=../venv_black
isort_venv=../venv_isort
mypy_venv=../venv_mypy
pylint_venv=../venv_pylint
ruff_venv=../venv_ruff


upgrade venvs can be auto or ask, any other value is interpreted
as ask.

upgrade_venvs=ask


time interval can be 0 to always upgrade or ask to upgrade,
or a number at most bash max int
https://superuser.com/questions/1030122/
what-is-the-maximum-value-of-a-numeric-bash-shell-variable
it should be enough to simulate never "upgrade or ask for upgrade"
unless you claim to live many billions years ;)
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).
```

```

upgrade_vnu_jar_time_interval_in_seconds=604800

use_vnu_for_xhtml=false
```

3.48 build_and_checks_variables/repository_data.txt.tpl

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.
The goal of this template is to list all possible variables.

repository_name=
repository_web_url=
repository_git_url=

abstract=
abstract=

license=
license2=

author_email=
author_full_name=
author_last_name=
author_name=
author_website=

acknowledgments=
acknowledgments=

max_line_length=

latex_command=

grammar_and_spelling_checker_command=

bandit_venv=
black_venv=
isort_venv=
mypy_venv=
pylint_venv=
ruff_venv=

upgrade_venvs=
```

```

upgrade_venvs_time_interval_in_seconds=

vnu_jar_path=

upgrade_vnu_jar_time_interval_in_seconds=

use_vnu_for_xhtml=

npm_lint_directories=
npm_lint_directories=
```

3.49 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
```

```
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

3.50 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"
```

3.51 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
           False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
           False positive when c contains \n.
- SC2016 : Expressions don't expand in single quotes,
           use double quotes for that.
- SC2034 : Unused variable
           False positive for globals that are declared in functions
           that are not currently called.
           Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
```

```
            Use printf '..%s..' "$foo".
            False positive when using a variable to build the format string.
- SC2086 : Double quote to prevent globbing and word splitting.
            False positive when returning integer variable
            via an indirection like return ${!foo}.
            See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
            False positive when passing arguments would yield
            a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
            False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
            False positive when assigning a variable declared with -i.
            https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
            False positive when using "$@" line return and pipes.
            https://github.com/koalaman/shellcheck/issues/3192
            Example:
              # shellcheck disable=SC2145
              all_self_code_lines "$@"\
                | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
            Very useful but false positive when I play with
            variable variable names.
- SC2162X: read without -r will mangle backslashes.
            False positive when read is used without variable:
            https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
            False positive on returning integer values.
            https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
              Use = to compare as string (or expand explicitly with $var).
            Very useful but false positive when I play with
            variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
            value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
            https://github.com/koalaman/shellcheck/issues/3419

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2016$" -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$"\
  -e "=SC2120$" -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$"\
  -e "=SC2309$" -e "=SC2312$" -e "=SC2154,SC2309$"
```

3.52 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
```

```
45      LFBFL_verbose="--verbose"
46    fi
47    readonly LFBFL_verbose
48
49    local LFBFL_check_download=""
50    if [[ "$*" == *--check-download=keep* ]]; then
51      LFBFL_check_download="--check-download=keep"
52    elif [[ "$*" == *--check-download=y* ]]; then
53      LFBFL_check_download="--check-download=y"
54    fi
55    readonly LFBFL_check_download
56
57    source ./wget_sha512.libr.sh
58
59    LFBFL_subdir="build_and_checks_dependencies"
60    mkdir --parents "${LFBFL_subdir}/licenses_templates"
61    local LFBFL_variables_directory
62    LFBFL_variables_directory="${LFBFL_work_directory}/"
63    LFBFL_variables_directory+="build_and_checks_variables"
64    readonly LFBFL_variables_directory
65    mkdir --parents -- "${LFBFL_variables_directory}/temp"
66    if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
67      printf "temp/\nupgrade_venvs_ts\n"\
68        > "${LFBFL_variables_directory}/.gitignore"
69    fi
70
71    local LFBFL_dependencies_URL
72    LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
73    LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
74    readonly LFBFL_dependencies_URL
75
76    declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
77    local LFBFL_script_download_URL
78    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
79    LFBFL_script_download_URL+="${LFBFL_common_file_name}"
80    readonly LFBFL_script_download_URL
81    declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
82    local LFBFL_correct_sha512
83    LFBFL_correct_sha512='337bfc1a23adb5b7f46784a1f494ca41317d87a44f129'
84    LFBFL_correct_sha512+='c04edc0e4efda77fadbba69b231a940d8f57202548e0'
85    LFBFL_correct_sha512+='419a80711410725556ccd80ea9d1d3be3051654'
86    wget_sha512 "${LFBFL_file_path}"\
87      "${LFBFL_script_download_URL}"\
88      "${LFBFL_correct_sha512}"\
89      "${LFBFL_verbose}"\
90      "${LFBFL_check_download}"
```

```
  chmod +x "${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.53 COPYING

```
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```

```
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```

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```
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       1) Use a suitable shared library mechanism for linking with the
       Library.  A suitable mechanism is one that (a) uses at run time
```

```
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        system, and (b) will operate properly with a modified version
        of the Library that is interface-compatible with the Linked
        Version.

    e) Provide Installation Information, but only if you would otherwise
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```

```
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permanent authorization for you to choose that version for the
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```

3.55 git_helpers/git_checks.libr.sh

```
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# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
```

```
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
  declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
  local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
  LFBFL_send_body_2+=" dès la création du modèle"
  readonly LFBFL_send_body_2
  declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
  declare -r LFBFL_send_body_3=\
"Les nouveaux preloadings sont interdits -> #Prefetch()"
  declare -i LFBFL_i_error=0
  git diff --cached --name-only\
    > /tmp/DOSAS_django_git_check1.temp
  if grep models /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
    LFBFL_i_error=1
  fi
  if grep serializer /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check1.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_no_abusive_trailing_comma(){
  # Useful for Python code, although it regularly yields
  # false positives in Python code
  # (but it is easy to distinguish by hand).
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION"
  local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
  LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
```

```
  readonly LFBFL_send_body_1
  declare -i LFBFL_i_error=0
  git diff --cached\
    > /tmp/DOSAS_django_git_check2.temp
  if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check2.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_s_files_paths=$(
    git diff --cached --name-only\
    | grep '\.py'
  )
  printf -- "%s\n" "${LFBFL_s_files_paths}"
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  declare -i LFBFL_i_error=0
  # Not very useful since LFBFL_file should be local to the loop by
  # default.
  local LFBFL_file_path
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Black will check formatting on file %s\n" "${LFBFL_file_path}"
    if ! black --check --diff -- "${LFBFL_file_path}";
    then
      LFBFL_i_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

3.56 git_helpers/post-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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#
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# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "post-commit pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=post_commit_renaming_comment.libr.sh
source "./git_helpers/post_commit_renaming_comment.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./build_and_checks_dependencies/lines_filters.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "post-commit no popd\n"
  # shellcheck disable=SC2248
```

```
  exit ${LFBFL_i_popd_result3}
}

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

3.57 git_helpers/post_commit_renaming_comment.libr.sh

```
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# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
```

```
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath -- "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname -- "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_i_pushd_result2
pushd -- "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result2=$?
  printf --\
    "post_commit_renaming_comment.libr.sh pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result2}
}
# shellcheck source=common_options.libr.sh
source "build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_i_popd_result2
popd || {
  LFBFL_i_popd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# For this reason, this function will only handle when all files
# impacted in the preceding commit are only renamed.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
commit_a_file_renamed_comment(){
  # Options:
  #   --verbose
  #   --log-directory-change
  #   --max-comment-line-length
  declare -i LFBFL_i_verbose=0
```

```
  get_verbose_option "$@"
  declare -i LFBFL_i_log_directory_change
  get_some_flag LFBFL_i_log_directory_change --log-directory-change 1 "$@"
  declare -i LFBFL_i_max_comment_line_length
  get_some_option LFBFL_i_max_comment_line_length\
    --max-comment-line-length 70 1 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
    LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
    # shellcheck disable=SC2059,SC2248
    printf -- "${LFBFL_s_format}"\
      "${LFBFL_i_log_directory_change}"\
      ${LFBFL_i_max_comment_line_length}
  fi
  local LFBFL_s_format
  #-------------------------------------------------------------------

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_renaming_lines=$(
    git log --max-count=1 --patch\
    | grep --after-context=1 '^diff --git'
  )
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf -- "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
  fi

  declare -a LFBFL_renaming_lines_array
  mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
  readonly LFBFL_renaming_lines_array

  declare -r LFBFL_timestamp=$(date --iso-8601="minute")
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
  fi
  local LFBFL_split_score_command1
  declare -i LFBFL_i_split_score_command_properties1
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} _
  LFBFL_split_score_command1="${get_split_score_result}"
  LFBFL_i_split_score_command_properties1=${i_get_split_score_result2}
  readonly LFBFL_split_score_command1
  readonly LFBFL_i_split_score_command_properties1
  local LFBFL_split_score_command2
  declare -i LFBFL_i_split_score_command_properties2
```

```
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} /
  LFBFL_split_score_command2="${get_split_score_result}"
  LFBFL_i_split_score_command_properties2=${i_get_split_score_result2}
  readonly LFBFL_split_score_command2
  readonly LFBFL_i_split_score_command_properties2
  declare -i LFBFL_i_renaming_happened=0
  declare -i LFBFL_i
  declare -i LFBFL_j
  # declare -r LFBFL_special_file_name="files_names_listing.txt"
  declare -r LFBFL_sfn="files_names_listing.txt"
  local LFBFL_diff_line
  local LFBFL_similarity_line
  local LFBFL_old_file_path
  local LFBFL_old_file_name
  local LFBFL_old_file_directory
  local LFBFL_new_file_path
  local LFBFL_new_file_name
  local LFBFL_new_file_directory
  local LFBFL_useful_file_name
  local LFBFL_extension
  local LFBFL_comment_prefix
  local LFBFL_comment_prefix2
  local LFBFL_old_file_name2
  local LFBFL_old_file_directory2
  local LFBFL_new_file_name2
  local LFBFL_new_file_directory2
  local LFBFL_new_comment
  declare -i LFBFL_i_copyright_line_number
  declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
  declare -i LFBFL_i_line_count
  declare -i LFBFL_i_lines_after
  local LFBFL_temp_file_path
  for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
    LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
    LFBFL_i=$((LFBFL_i + 1))
    LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
    LFBFL_i=$((LFBFL_i + 1))
    # We need to increment 3 times since grep adds "--" lines between.
    LFBFL_i=$((LFBFL_i + 1))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf -- "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
        "${LFBFL_diff_line}"\
        "${LFBFL_similarity_line}"
    fi
    if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
```

```
      continue
    fi

    # Extract old file name from diff line. --------------------------
    LFBFL_old_file_path=$(
      printf -- "%s" "${LFBFL_diff_line}"\
      | cut --delimiter=' ' --fields=3
    )
    # It starts by "a/".
    LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
    LFBFL_old_file_name=$(basename -- "${LFBFL_old_file_path}")
    LFBFL_old_file_directory=$(dirname -- "${LFBFL_old_file_path}")
    # Extract new file name from diff line. --------------------------
    LFBFL_new_file_path=$(
      printf -- "%s" "${LFBFL_diff_line}"\
      | cut --delimiter=' ' --fields=4
    )
    # It starts by "b/".
    LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
    LFBFL_new_file_name=$(basename -- "${LFBFL_new_file_path}")
    LFBFL_new_file_directory=$(dirname -- "${LFBFL_new_file_path}")
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      LFBFL_s_format="LFBFL_old_file_path: %s\n"
      LFBFL_s_format+="LFBFL_old_file_name: %s\n"
      LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
      LFBFL_s_format+="LFBFL_new_file_path: %s\n"
      LFBFL_s_format+="LFBFL_new_file_name: %s\n"
      LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
      # shellcheck disable=SC2059
      printf -- "${LFBFL_s_format}"\
        "${LFBFL_old_file_path}"\
        "${LFBFL_old_file_name}"\
        "${LFBFL_old_file_directory}"\
        "${LFBFL_new_file_path}"\
        "${LFBFL_new_file_name}"\
        "${LFBFL_new_file_directory}"
    fi
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. ------------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf -- "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
        "${LFBFL_useful_file_name}"\
        "${LFBFL_extension}"
    fi
```

```
    for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf -- "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
          "${LFBFL_useful_file_name}"\
          "${LFBFL_extension}"
      fi
    done
    LFBFL_comment_prefix=""
    if [[ "${LFBFL_extension}" == "sh" ]]; then
      LFBFL_comment_prefix="# "
    fi
    if [[ "${LFBFL_extension}" == "sql" ]]; then
      LFBFL_comment_prefix="-- "
    fi
    if [[ "${LFBFL_extension}" == "tex" ]]; then
      LFBFL_comment_prefix="% "
    fi
    if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
      LFBFL_comment_prefix="// "
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf -- "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
    fi
    #-----------------------------------------------------------------
    # Split if necessary old_file_name, new_file_name, etc. to have
    # comment lines below max_comment_line_length.
    LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
    # -----
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
```

```
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. -----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_i_log_directory_change -eq 1
      && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
    ]];
```

```
    then
      if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
        LFBFL_new_comment+=$'\n'
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf -- "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. -------------------------------------
    LFBFL_i_copyright_line_number=$(
      grep --line-number -- '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut --delimiter=':' --fields=1
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_copyright_line_number: %s\n"\
        "${LFBFL_i_copyright_line_number}"
    fi
    if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
      notify-send "${LFBFL_send_summary_2}"\
        "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "No copyright line. Skipping file.\n"
      fi
      continue
    fi
    # Get total number of lines. -------------------------------------
    LFBFL_i_line_count=$(
      wc --lines\
      < "${LFBFL_new_file_path}"
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
    fi
    # Compute number of lines after. ---------------------------------
    LFBFL_i_lines_after=$((
      LFBFL_i_line_count - LFBFL_i_copyright_line_number
```

```
    ))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
    fi
    # Update file. ---------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_i_copyright_line_number}"\
        -- "${LFBFL_new_file_path}"
      printf -- "%s\n" "${LFBFL_new_comment}"
      tail --lines="${LFBFL_i_lines_after}" -- "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv -- "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. ---------------------------------------------
    git add -- "${LFBFL_new_file_path}"
    LFBFL_i_renaming_happened=1
  done
  readonly LFBFL_i_renaming_happened
  if [[ LFBFL_i_renaming_happened -eq 1 ]]; then
    # Commit ---------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    printf -- "Commit message:\n%s\n" "${LFBFL_message}"
    git commit --no-verify --message="${LFBFL_message}"
  fi
}
```

3.58 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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```

```
#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "pre-commit pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=git_checks.libr.sh
source "./git_helpers/git_checks.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "pre-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

declare -gi LFBFL_i_git_hook_error=0
if ! check_files; then
  LFBFL_i_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_i_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_i_git_hook_error=1
fi
readonly LFBFL_i_git_hook_error
```

```
# shellcheck disable=SC2248
exit ${LFBFL_i_git_hook_error}
```

3.59 IMAP_transfert.py

```
"""
This file is part of DevOrSysAdminScripts library.

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Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent, try badminton.
    """

```

```
class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don't need "pass" XD.
    """


def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
```

```
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print("Creating new folder in to account.")
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
```

```
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
        ),
    )
    parser.add_argument(
        "from_login",
        type=str,
        help="Email account login of user one.",
```

```
180     )
181     parser.add_argument(
182         "from_password",
183         type=str,
184         help="Email account password of user one.",
185     )
186     parser.add_argument(
187         "to_login",
188         type=str,
189         help="Email account login of user two.",
190     )
191     parser.add_argument(
192         "to_password",
193         type=str,
194         help="Email account password of user two.",
195     )
196     parser.add_argument(
197         "--to_new_folder",
198         type=str,
199         default="",
200         help="Transfert emails into a new folder of this name.",
201     )
202     parser.add_argument(
203         "--from_host",
204         type=str,
205         default="imap.gmail.com",
206         help="Host of first email account",
207     )
208     parser.add_argument(
209         "--from_port",
210         type=int,
211         default=993,
212         help="Port on host of first email account",
213     )
214     parser.add_argument(
215         "--to_host",
216         type=str,
217         default="imap.gmail.com",
218         help="Host of second email account",
219     )
220     parser.add_argument(
221         "--to_port",
222         type=int,
223         default=993,
224         help="Port on host of second email account",
225     )
```

```
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
            "-----------------------------------------------------\n"
            "IMAP transfert:\n"
            "-----------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)
```

```


if __name__ == "__main__":
    main()
```

3.60 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
https://pubs.opengroup.org/onlinepubs/9699919799/
When looking `man sed` GNU sed says to use POSIX -E instead of
-r or --regexp-extended for portability.
But the link above says that still now only -e, -f, -n are POSIX options
for sed.
For this reason we replaced all short option names by long option names
when possible, even for sed.
```

3.61 miscellaneous/funny.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```
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unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
    fi
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

3.62 miscellaneous/spampty_data.txt

```



a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

3.63 miscellaneous/string_escaping.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Truc</title>
</head>
<body>
<header id="title-block-header">
<h1 class="title">Truc</h1>
</header>
&lt;abc> &amp; &lt;de>
<p title="<&>’
 abcd‘\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>’&quot;
abcd‘\ machin"
This is some text
</pre>

<script>
let a = ’abcd & < > \\ \n  \"\’’;
alert(a);
let b = "abcd & < > \\ \n  \"\’";
alert(b);
</script>

</body>
</html>
```

3.64 miscellaneous/string_escaping.xhtml

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
          "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="fr" lang="fr">
<head>
<meta content="text/html; charset=UTF-8" http-equiv="Content-Type" />
  <title>Truc</title>
</head>
<body>
<h1 class="title">Truc</h1>
&lt;abc> &amp; &lt;de>
```

```

<p title="&lt;&amp;>'&#10;
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script type="text/javascript">
let a = 'abcd &amp; &lt; > \\ \n  \"\'';
alert(a);
let b = "abcd &amp; &lt; > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.65 README.md.tpl

```
# DevOrSysAdminScripts

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[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:
```

```

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- But the files "pre-commit" and "post-commit" were not renamed,
  since it is not possible to give them other names to use them as
  pre-commit/post-commit hooks in `git`.
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.
- We used `_?i_` or `_?j_`, etc. in integer variables names in Bash
  scripts.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts

[Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
```

```
3dd6ef85e8db4460991d1ebc404b68dd

[Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

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languages/top/llyaudet/DevOrSysAdminScripts

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languages/code-size/llyaudet/DevOrSysAdminScripts

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security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

3.66 string_escaping.libr.php

```php
<?php

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*/

declare(strict_types=1);
declare(encoding='UTF-8');



/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for HTML (surprised?)...

@package DevOrSysAdminScripts
```

```
@subpackage string_escaping\HTML
*/
namespace string_escaping\HTML {
  /**
  When you want that the string appears as is in HTML text block.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_text()



  /**
  When you want that the string appears as is in an HTML tag attribute
  that starts and ends with double quotes: my_attribute="something".
  Example: a title attribute.
  /!\ in double quotes /!\ my_attribute='something' is not supported.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_attribute(string $s_string) : string {
    return str_replace(
      ['"'],
      ['&quot;'],
      $s_string,
    );
  }//end escape_attribute()



```

```
  /**
  When you want that the string appears verbatim inside HTML <pre> tags.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_pre(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_pre()
}//end namespace string_escaping\HTML




/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for XHTML (still surprised?)...

@package DevOrSysAdminScripts
@subpackage string_escaping\XHTML
*/
namespace string_escaping\XHTML {
  /**
  When you want that the string appears as is in XHTML text block.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
```

```
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_text()



  /**
  When you want that the string appears as is in an XHTML tag attribute
  that starts and ends with double quotes: my_attribute="something".
  Example: a title attribute.
  /!\ in double quotes /!\ my_attribute='something' is not supported.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_attribute(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
        '"',
        "\n",
      ],
      [
        '&amp;',
        '&lt;',
        '&quot;',
        '&#10;',
      ],
      $s_string,
    );
  }//end escape_attribute()



  /**
  When you want that the string appears verbatim inside XHTML <pre> tags.

  @param string $s_string The string you want to escape.

```

```
  @return string The escaped string.
  */
  function escape_pre(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_pre()
}//end namespace string_escaping\XHTML




/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for JS (you're hopeless?)...

@package DevOrSysAdminScripts
@subpackage string_escaping\JS
*/
namespace string_escaping\JS {

  use function string_escaping\XHTML\escape_text as XHTML_escape_text;



  /**
  When you want that the string can be put in a JS string definition.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape(string $s_string) : string {
    return str_replace(
      [
        '\\',
        "\n",
```

```
        "'",
        '"',
      ],
      [
        '\\\\',
        '\n',
        '\\\'',
        '\\"',
      ],
      $s_string,
    );
  }//end escape()



  /**
  When you want that the string can be put in a JS string definition
  to be included inside an XHTML file <script> tag.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_for_string_definition_inside_XHTML(string $s_string)
  : string {
    return XHTML_escape_text(escape($s_string));
  }//end escape_for_string_definition_inside_XHTML()
}//end namespace string_escaping\JS




/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for TeX.

@package DevOrSysAdminScripts
@subpackage string_escaping\TeX
*/
namespace string_escaping\TeX {
  /**
  When you want that the string can be written to a TeX file inside a
  part where TeX is in text mode.

  @param string $s_string The string you want to escape.
```

```

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    // https://tex.stackexchange.com/a/34586/270015
    return str_replace(
      [
        '\\',
        '&',
        '%',
        '$',
        '#',
        '_',
        '{',
        '}',
        '~',
        '^',
      ],
      [
        '\\textbackslash',
        '\\&',
        '\\%',
        '\\$',
        '\\#',
        '\\_',
        '\\{',
        '\\}',
        '\\textasciitilde',
        '\\textasciicircum',
      ],
      $s_string,
    );
  }//end escape_text()
}//end namespace string_escaping\TeX

/*
$ php
<?php
require_once('string_escaping.libr.php');
var_dump(string_escaping\HTML\escape_text('<&"\'truc\\'));
var_dump(string_escaping\HTML\escape_attribute("<&\"\n'truc\\"));
var_dump(string_escaping\HTML\escape_pre("<&\"\n'truc\\"));
var_dump(string_escaping\XHTML\escape_text('<&"\'truc\\'));
var_dump(string_escaping\XHTML\escape_attribute("<&\"\n'truc\\"));
var_dump(string_escaping\XHTML\escape_pre("<&\"\n'truc\\"));
var_dump(string_escaping\JS\escape("abcd & < > \\ \n  \"'"));
```

```
var_dump(string_escaping\JS\escape_for_string_definition_inside_XHTML(
  "abcd & < > \\ \n  \"'"
));
*/
?>
```

3.67 unicode.libr.php

```
<?php

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This file was renamed from "unicode.php" to "unicode.libr.php".

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*/

declare(strict_types=1);
declare(encoding='UTF-8');


```

```

/**
Converts an int to the byte-string corresponding to the UTF-8 encoding
of the Unicode code point of same integer value.
decimal_code_point is a slight abuse of language to say that you write the
code point in decimal notation in PHP code and you get the UTF-8.
If you provide the integer of the code point without using decimal
notation, it will also work.

@param int $i_code_point_in_decimal_notation The code point.

@throws Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
*/
function decimal_code_point_to_UTF8(
  int $i_code_point_in_decimal_notation
) : string {
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
  if($i_code_point_in_decimal_notation < 128){
    return chr($i_code_point_in_decimal_notation);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 2048){  // 2**11
    $i_first_byte_base_value = 192;
    $i_last_byte_significant_bits = (
      $i_code_point_in_decimal_notation % 64  // 2**6
    );
    $i_first_byte_significant_bits = intdiv(
      (
        $i_code_point_in_decimal_notation
        - $i_last_byte_significant_bits
      ),
      64,  // 2**6
    );
    $s_result = chr(
      $i_first_byte_base_value
      + $i_first_byte_significant_bits
    );
    $s_result .= chr(
      $i_continuation_base_value
      + $i_last_byte_significant_bits
    );
    return $s_result;
```

```
85    }//end if($i_code_point_in_decimal_notation < 2048)
86    // 1110xxxx 10xxxxxx 10xxxxxx
87    if($i_code_point_in_decimal_notation < 65536){  // 2**16
88      $i_first_byte_base_value = 224;
89      $arr_arr_data_per_byte_reverse = [
90        ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
91        ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
92        ['i_bits' => 4, 'i_base_value' => $i_first_byte_base_value],
93      ];
94      $i_rest = $i_code_point_in_decimal_notation;
95      $s_result = '';
96      foreach($arr_arr_data_per_byte_reverse as $arr_data){
97        // _Pragma unroll
98        // J'aurais pu "roller" les 4 cas des ifs.
99        // Mais pour les perfs, il faudrait tout unroller vu le nombre
100       // d'itérations. Et j'aimerais bien comparer du code unrollé
101       // aux 2 niveaux à la main avec celui unrollé par le compilateur
102       // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
103       // l'intérieur d'abord (pour ne pas refaire le même boulot)
104       // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
105       // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
106       $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
107       $i_rest = intdiv(
108         $i_rest - $i_significant_bits,
109         2**$arr_data['i_bits'],
110       );
111       // .= or =.
112       // Below could have been a case of "=.".
113       $s_result = (
114         chr($i_significant_bits + $arr_data['i_base_value'])
115         .$s_result
116       );
117     }//end foreach($arr_arr_data_per_byte_reverse as $arr_data)
118     return $s_result;
119   }//end if($i_code_point_in_decimal_notation < 65536)
120   // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
121   if($i_code_point_in_decimal_notation < 2097152){  // 2**21
122     $i_first_byte_base_value = 240;
123     $arr_arr_data_per_byte_reverse = [
124       ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
125       ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
126       ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
127       ['i_bits' => 3, 'i_base_value' => $i_first_byte_base_value],
128     ];
129     $i_rest = $i_code_point_in_decimal_notation;
130     $s_result = '';
```

```
131     foreach($arr_arr_data_per_byte_reverse as $arr_data){
132       // _Pragma unroll
133       $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
134       $i_rest = intdiv(
135         $i_rest - $i_significant_bits,
136         2**$arr_data['i_bits'],
137       );
138       // .= or =.
139       // Below could have been a case of "=.".
140       $s_result = (
141         chr($i_significant_bits + $arr_data['i_base_value'])
142         .$s_result
143       );
144     }
145     return $s_result;
146   }//end if($i_code_point_in_decimal_notation < 2097152)
147   // phpcs:disable Squiz.Strings.DoubleQuoteUsage.NotRequired
148   throw new Exception(
149     "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
150     ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
151     ." 3 octets au lieu de 4 :face-palm:."
152     ." Du coup, ça peut donner quelques \"surprises\"."
153     ." Sinon, y'a aussi ASCII dans MySQL."
154     ." Et comme ça joue sur la mémoire et les performances,"
155     ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
156     ." :)"
157   );
158   // phpcs:enable Squiz.Strings.DoubleQuoteUsage.NotRequired
159 }//end decimal_code_point_to_UTF8()
160
161
162
163 /**
164 Converts a string of hexadecimal digits to the byte-string corresponding
165 to the UTF-8 encoding of the Unicode code point of same integer value
166 that is denoted by the hexadecimal digits.
167
168 @param string $s_code_point_in_hexadecimal_notation The code point.
169
170 @throws Exception When the code point is outside of Unicode range.
171
172 @return string The corresponding UTF-8 character.
173 */
174 function hexa_code_point_to_UTF8(
175   string $s_code_point_in_hexadecimal_notation
176 ) : string {
```

```
  $s_code_point_in_hexadecimal_notation = strtoupper(
    $s_code_point_in_hexadecimal_notation
  );
  $i_code_point_in_decimal_notation = 0;
  $i_ord_0 = ord('0');
  $i_ord_A = ord('A');
  $i_digit_offset = -$i_ord_0;
  $i_letter_offset = 10 - $i_ord_A;
  for(
    $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
    $i < $i_max;
    ++$i
  ){
    $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
    $i_code_point_in_decimal_notation *= 16;
    if(ctype_digit($s_hexa_digit)){
      $i_code_point_in_decimal_notation += (
        ord($s_hexa_digit) + $i_digit_offset
      );
      continue;
    }
    if(ctype_xdigit($s_hexa_digit)){
      $i_code_point_in_decimal_notation += (
        ord($s_hexa_digit) + $i_letter_offset
      );
      continue;
    }
    throw new Exception('Not an hexadecimal digit.');
  }
  return decimal_code_point_to_UTF8(
    $i_code_point_in_decimal_notation
  );
}//end hexa_code_point_to_UTF8()



/**
This function throws an exception with extended debug informations
if the input string is not valid ASCII.
Otherwise, it returns true.

@param string $s_string The input string.

@throws Exception When the input string is not valid ASCII.

@return bool
```

```
223 */
224 function check_string_is_valid_ASCII(string $s_string) : bool {
225   $i_offset_in_octets_from_string_start = 0;
226   $i_current_line_number = 1;
227   $i_offset_in_octets_from_line_start = -1;
228   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
229     $i_current_octet = ord($s_string[$i]);
230     $i_offset_in_octets_from_string_start = $i;
231     ++$i_offset_in_octets_from_line_start;
232     if($i_current_octet > 127){
233       throw new Exception(
234         'Non-ASCII character found on line '
235         .$i_current_line_number
236         .'; the octet/character '
237         .($i_offset_in_octets_from_line_start + 1)
238         .' has value '
239         .$i_current_octet
240         .' which is not ASCII.'
241         .' (Sequential position without line splitting:'
242         .' This is at octet/character '
243         .($i_offset_in_octets_from_string_start + 1)
244         .'.)',
245       );
246     }
247     // if($s_string[$i] === "\n"){
248     if($i_current_octet === 10){
249       ++$i_current_line_number;
250       $i_offset_in_octets_from_line_start = -1;
251     }
252   }//for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
253   return true;
254 }//end check_string_is_valid_ASCII()
255
256
257
258 /**
259 This function throws an exception with extended debug informations
260 if the content of the file with given file_path is not valid ASCII.
261 Otherwise, it returns true.
262
263 @param string $s_file_path The input file_path.
264
265 @throws Exception When the input string is not valid ASCII.
266
267 @return bool
268 */
```

```
function check_file_is_valid_ASCII(string $s_file_path) : bool {
  $s_string = file_get_contents($s_file_path);
  if($s_string === false){
    throw new Exception('File '.$s_file_path.' not found.');
  }
  return check_string_is_valid_ASCII($s_string);
}//end check_file_is_valid_ASCII()



/**
This function throws an exception with extended debug informations
if the input string is not valid UTF-8.
Otherwise, it returns true.

@param string $s_string The input string.

@throws Exception When the input string is not valid UTF-8.

@return bool
*/
function check_string_is_valid_UTF8(string $s_string) : bool {
  $i_continuation_octet_needed = 0;
  $i_offset_in_octets_from_string_start = 0;
  $i_offset_in_characters_from_string_start = -1;
  $i_character_start_position_from_string_start = 0;
  $i_current_line_number = 1;
  $i_offset_in_octets_from_line_start = -1;
  $i_offset_in_characters_from_line_start = -1;
  $i_character_start_position_from_line_start = -1;
  for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
    $i_current_octet = ord($s_string[$i]);
    $i_offset_in_octets_from_string_start = $i;
    ++$i_offset_in_octets_from_line_start;
    if($i_continuation_octet_needed > 0){
      if($i_current_octet < 128 || $i_current_octet >= 192){
        throw new Exception(
          'Non-UTF8 character found on line '
          .$i_current_line_number
          .'; the octet '
          .($i_offset_in_octets_from_line_start + 1)
          .', part of the character '
          .($i_offset_in_characters_from_line_start + 1)
          .', has value '
          .$i_current_octet
          .' which is not a continuation octet.'
```

```
315           .' This character starts at octet '
316           .($i_character_start_position_from_line_start + 1)
317           .' of the current line.'
318           .' (Sequential positions without line splitting:'
319           .' This is at character '
320           .($i_offset_in_characters_from_string_start + 1)
321           .' and octet '
322           .($i_offset_in_octets_from_string_start + 1)
323           .'.'
324           .' This character starts at octet '
325           .($i_character_start_position_from_string_start + 1)
326           .'.)',
327         );
328       }//end if($i_current_octet < 128 || $i_current_octet >= 192)
329       --$i_continuation_octet_needed;
330       continue;
331     }//end if($i_continuation_octet_needed > 0)
332
333     ++$i_offset_in_characters_from_string_start;
334     ++$i_offset_in_characters_from_line_start;
335     $i_character_start_position_from_string_start = (
336       $i_offset_in_octets_from_string_start
337     );
338     $i_character_start_position_from_line_start = (
339       $i_offset_in_octets_from_line_start
340     );
341     if($i_current_octet < 128){ // 0xxxxxxx ASCII
342       // if($s_string[$i] === "\n"){
343       if($i_current_octet === 10){
344         ++$i_current_line_number;
345         $i_offset_in_octets_from_line_start = -1;
346         $i_offset_in_characters_from_line_start = -1;
347       }
348       continue;
349     }
350     if($i_current_octet >= 128 && $i_current_octet < 192){
351       throw new Exception(
352         'Non-UTF8 character found on line '
353         .$i_current_line_number
354         .'; the octet '
355         .($i_offset_in_octets_from_line_start + 1)
356         .', part of the character '
357         .($i_offset_in_characters_from_line_start + 1)
358         .', has value '
359         .$i_current_octet
360         .' which is a continuation octet.'
```

```
361         .' This character starts at octet '
362         .($i_character_start_position_from_line_start + 1)
363         .' of the current line.'
364         .' (Sequential positions without line splitting:'
365         .' This is at character '
366         .($i_offset_in_characters_from_string_start + 1)
367         .' and octet '
368         .($i_offset_in_octets_from_string_start + 1)
369         .'.'
370         .' This character starts at octet '
371         .($i_character_start_position_from_string_start + 1)
372         .'.)',
373       );
374     }//end if($i_current_octet >= 128 && $i_current_octet < 192)
375     if($i_current_octet >= 192 && $i_current_octet < 224){
376       // 110xxxxx 10xxxxxx
377       $i_continuation_octet_needed = 1;
378       continue;
379     }
380     if($i_current_octet >= 224 && $i_current_octet < 240){
381       // 1110xxxx 10xxxxxx 10xxxxxx
382       $i_continuation_octet_needed = 2;
383       continue;
384     }
385     if($i_current_octet >= 240 && $i_current_octet < 248){
386       // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
387       $i_continuation_octet_needed = 3;
388       continue;
389     }
390     throw new Exception(
391       'Non-UTF8 character found on line '
392       .$i_current_line_number
393       .'; the octet '
394       .($i_offset_in_octets_from_line_start + 1)
395       .', part of the character '
396       .($i_offset_in_characters_from_line_start + 1)
397       .', has value '
398       .$i_current_octet
399       .' which is invalid.'
400       .' This character starts at octet '
401       .($i_character_start_position_from_line_start + 1)
402       .' of the current line.'
403       .' (Sequential positions without line splitting:'
404       .' This is at character '
405       .($i_offset_in_characters_from_string_start + 1)
406       .' and octet '
```

```
      .($i_offset_in_octets_from_string_start + 1)
      .'.'
      .' This character starts at octet '
      .($i_character_start_position_from_string_start + 1)
      .'.)',
    );
  }//end for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
  return true;
}//end check_string_is_valid_UTF8()



/**
This function throws an exception with extended debug informations
if the content of the file with given file_path is not valid ASCII.
Otherwise, it returns true.

@param string $s_file_path The input file_path.

@throws Exception When the input string is not valid UTF-8.

@return bool
*/
function check_file_is_valid_UTF8(string $s_file_path) : bool {
  $s_string = file_get_contents($s_file_path);
  if($s_string === false){
    throw new Exception('File '.$s_file_path.' not found.');
  }
  return check_string_is_valid_UTF8($s_string);
}//end check_file_is_valid_UTF8()



/*
?>
<?php
require_once("./unicode.libr.php");
var_dump(hexa_code_point_to_UTF8("002B"));
var_dump(hexa_code_point_to_UTF8("00E6"));
var_dump(hexa_code_point_to_UTF8("1400"));
var_dump(hexa_code_point_to_UTF8("10111"));
// https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
// Some invisible unicode characters that are a problem:
// 2000-200F 2028-202F 205F 2060-206F
$arr = [
  '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
```

```
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
foreach($arr as $s_hexa){
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  | sed -e 's/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
}
//*/
?>
```

3.68 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
```

```
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# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  # Options:
  #   --add-test-files
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --fixed-point-build
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf -- "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed-point-build* ]]; then
    LFBFL_fixed_point_build="--fixed-point-build"
  fi
  readonly LFBFL_fixed_point_build
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download
  declare -i LFBFL_add_test_files=0
  if [[ "$*" == *--add-test-files* ]]; then
    LFBFL_add_test_files=1
  fi
  readonly LFBFL_add_test_files
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  if [[ LFBFL_add_test_files -eq 1 ]]; then
    mkdir --parents -- --a/
    {
      printf "<?php\nfunction my_function(:;){\n\n}\n"
```

```
    # shellcheck disable=SC2016
    printf '$my_variable = "aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa

```

```
    cat -- --a2.py
    cp -- --a2.py --a/
    {
      # Error 0 is missing shell header.
      printf "my_function(){\n"
      printf "   # Error 1: Indentation is not correct.\n"
      printf "  sed -e 's/a/b/g' \\\\\n"
      printf "      -e 's/a/b/g' \"\$1\" # Indentation is correct.\n"
      printf "  my_variable=\"%%s\\n\"\n"
      # shellcheck disable=SC2016
      printf '  printf "${my_variable}"  # Error 2: shellcheck warning\n'
      local LFBFL_ttp="ttp"
      local LFBFL_URL="h${LFBFL_ttp}://example.com"
      printf '  printf "%s" # Error 3: http -> https\n' "${LFBFL_URL}"
      # shellcheck disable=SC2016
      printf '  printf "${#some_'
      printf 'arr_1}" # Error 4\n'
      # shellcheck disable=SC2016
      printf '  printf "${#'
      printf 'arr_1}"  # Error 5\n'
      printf '  if !'
      # shellcheck disable=SC2016
      printf ' [[ $1 -eq 1 ]]; # Error 6\n'
      printf '  then  # Error 7 : misplaced then\n'
      printf '    my_variable="a"\n'
      printf '  fi\n'
      printf "\n"
      # shellcheck disable=SC2016
      printf '  if [[ $1 -eq 1 ]];\n'
      printf '  then  # Error 8 : another case of misplaced then\n'
      printf '    my_variable="a"\n'
      printf '  fi\n'
      printf '  my_variable="aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
      printf 'aaaaaaaaaaaaaaaaaaaaaaaaaaaaaa"  # Error 9 line too long.\n'
      printf '}\n'
    } > --a.sh
    cp -- --a.sh --a/
  fi
  ./build_and_checks.exec.sh "."\
    "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"\
    "${LFBFL_check_download}"
  if [[ LFBFL_add_test_files -eq 1 ]]; then
    cat -- --a.py
    cat -- --a2.py
    rm --force --recursive -- --a/
```

```
    rm --force -- --a.php --a.py --a2.py --a.sh
  fi
}

upgrade_build_and_checks "$@"
```

3.69 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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#
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# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1=file_path
  # $2=download_URL
  # $3=correct_sha512
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded file)
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf -- "%s wget_sha512 %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
```

```
  readonly LFBFL_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="y"
  fi
  readonly LFBFL_check_download
  declare -i LFBFL_i_error=0

  if [[ ! -f "$1" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"} --output-document="$1"\
      -- "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum -- "$1"\
    | cut --fields=1 --delimiter=' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
    printf --\
      "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
      "$1" "$3" "${LFBFL_present_sha512}"
    LFBFL_i_error=1
  fi
  if [[ -n "${LFBFL_check_download}" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"}\
      --output-document="$1.wget_sha512.temp"\
      -- "$2"
    declare -r LFBFL_present_sha512_2=$(
      sha512sum -- "$1.wget_sha512.temp"\
      | cut --fields=1 --delimiter=' '
    )
    if [[ "${LFBFL_check_download}" != "keep" ]]; then
      rm -- "$1.wget_sha512.temp"
    fi
    if [[ "${LFBFL_present_sha512_2}" != "$3" ]]; then
      printf --\
        "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
        "Online version of $1" "$3" "${LFBFL_present_sha512_2}"
      LFBFL_i_error=1
    fi
  fi
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !